

SCOTT–VOGELIUS–NITSCHKE ON INSCRIBED POLYHEDRA: CONVERGENCE, SHARPNESS AND THE PENALTY THRESHOLD IN THREE DIMENSIONS

JAIDEEP SAI PADHI

ABSTRACT. We analyse the Scott–Vogelius–Nitsche method of Gjerde and Scott in three dimensions, when a smooth no-slip surface Γ is replaced by a triangulated surface Γ_h with vertices on Γ : continuous P_k velocities with discontinuous P_{k-1} pressures on straight-sided tetrahedra, exactly divergence-free, with the wall condition imposed weakly by Nitsche’s method. This is the three-dimensional form of Scott’s zero-gradient prize question, which asks whether the error behaves like $h_\Gamma^{3/2} + h^k$.

For the method as printed, the answer is no as soon as the wall pressure is not constant. The printed Nitsche form has no pressure traction, and the discrete velocity leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$; the constant is exactly 1 in the slip and energy norms. The method converges at rate h/μ in H^1 and $(h/\mu)^{1/2}$ in energy, and both rates are attained. Adding the traction with the wall mean of the discrete pressure removed gives a well-posed, exactly divergence-free method (the device is that of Frachon, Nilsson and Zahedi), and it attains $h_\Gamma^{3/2} + h^k$. On Alfeld refinements, for every $k \geq 3$, the only stability input, a discrete inf-sup constant uniform over the perturbed domains, is proved here from a local result of Guzmán and Neilan and a Lipschitz-perturbation argument of Bernardi, Costabel, Dauge and Girault.

The term $h_\Gamma^{3/2}$ is sharp. We prove $\|\nabla(\tilde{u} - u_h)\| \geq ch_\Gamma^{3/2} \|\mathfrak{h}\| \partial_N u\|_{L^2(\Gamma)} - Ch_\Gamma^2$, where $\mathfrak{h}$ is the second fundamental form, on Alfeld refinements for every $k \geq 3$ and on arbitrary shape-regular meshes for $k \geq 9$. For $k \geq 4$ the local witnesses come from one exact rank computation on a reference macro-element; for $k = 3$ a single macro-element provably does not suffice, and the witnesses live on the star of an interior edge, where the relevant moment has a closed form.

The setting covers general smooth obstacles, nonconvex and with several components. The closest-point projection $\Gamma_h \rightarrow \Gamma$, usually assumed in surface finite elements, is a homeomorphism as soon as the tetrahedral mesh is shape regular and does not reach the far side of the obstacle; we show by examples that neither condition can be dropped. The Nitsche penalty must grow like the ratio ρ of bulk to boundary mesh size: exact certificates on single Alfeld macro-elements, and a continuum witness with half-space constant $2327591/151008 \approx 15.41$, show this under local fineness. For every penalty above the coercivity threshold, the scaled error of the printed method converges in H^1 to the Stokes flow driven by the leak, at rate exactly $h_\Gamma^{1/2}$ on Alfeld refinements with $k \geq 4$, with a lower bound weighted by the second fundamental form; drag and torque errors are identified in the same way. The constants of the lower bounds are explicit. On Worsey–Farin splits the inf-sup condition fails for the full pressure space, as is known. The upper bounds, the leak law, the leak limit and the drag statements extend to steady Navier–Stokes flow under linearised-stability assumptions. A first three-dimensional implementation confirms exact incompressibility, the optimal rate on a manufactured box problem and the leak law with constant tending to 1; at the mesh levels we could run, the sphere tests are still pre-asymptotic for the $h_\Gamma^{3/2}$ rate.

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1. INTRODUCTION

The question. Scott–Vogelius elements, continuous piecewise- P_k velocities with discontinuous piecewise- P_{k-1} pressures, give discrete velocities that are divergence-free pointwise whenever $\operatorname{div} V_h = \Pi_h$ [33, 19]. In three dimensions this holds on Alfeld (barycentric) refinements for $k \geq 3$ [37, 18]; see [26, 11] for surveys. At a curved no-slip wall, exact incompressibility has a price. If the wall is replaced by an inscribed polygon or polyhedron and the no-slip condition is imposed strongly, piecewise-polynomial divergence-free fields inherit gradient constraints at the wall vertices, and the H^1 error can lock at $O(h^{1/2})$ [32, 16]. Gjerde and Scott [16] keep the polygon and impose the wall condition weakly by Nitsche’s method [28], with penalty μ/h and no pressure term. Scott’s zero-gradient prize [32] asks for a proof that this Scott–Vogelius–Nitsche method converges, with the expected error $h_\Gamma^{3/2} + h^k$, and adds: *if this does not hold, why not?*

The two-dimensional question is answered in a companion paper [29]. This paper treats the three-dimensional question: a smooth wall surface Γ , approximated by a closed triangulated surface Γ_h whose vertices lie on Γ , and a tetrahedral Scott–Vogelius mesh of the region bounded by Γ_h and a polyhedral outer boundary.

The methods. The method as printed in [16, (27)–(28)], which we call GS, contains the symmetric Nitsche terms and the penalty μ/h but not the traction term $-\langle p_h, v \cdot n \rangle$. Adding the plain traction makes the square system singular (Proposition 5.15). The corrected method CNS* adds the traction with the wall mean of the discrete pressure removed, $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$, to the momentum row only. It keeps the velocity exactly divergence-free. The device is not ours: the velocity of CNS* coincides with that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi [13], transcribed to our setting [29]. What we add is its analysis on inscribed polyhedra.

Positioning. The optimal-order alternative is to fit the curved boundary. Neilan and Otus proved optimal convergence of Scott–Vogelius elements on isoparametric curved triangulations in two dimensions [27, Theorem 6.1]. Chen and Liu [4] have recently extended this to three dimensions: isoparametric curved tetrahedral meshes, Piola-mapped Scott–Vogelius elements with $k \geq 3$ on Alfeld splits, strongly imposed Dirichlet data and a commuting-interpolant load, with the optimal and pressure-robust rate h^k . Liu, Neilan and Otus [23] reach order h^k in two dimensions on a mesh inside the curved domain, with a mean-free boundary multiplier, a zero-flux constraint and a Taylor boundary correction. Our setting is the cheap one that Scott’s question is about: straight-sided, affinely mapped tetrahedra, an inscribed triangulated wall, no curved elements, multipliers or boundary corrections. Its best possible rate is $h_\Gamma^{3/2} + h^k$, and we show that this rate is attained and cannot be improved. We do not compete with [4] on rate. Our uniform inf-sup argument (Section 4) follows the same route as [4, Lemma 3.9]; see Remark 4.7.

Main results. Notation is fixed in Section 2: h is the bulk mesh size in Scott’s penalty μ/h , h_Γ the largest diameter of a face of Γ_h , $\rho := h/\min_F \operatorname{diam} F$, $\gamma := \mu h_\Gamma/h$, $\bar{p}$ the mean of p over Γ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Constants do not depend on $h, h_\Gamma, \mu, \gamma, \rho$. The solution (u, p) is assumed smooth on $\bar{\Omega}$; at the edges and corners of the outer box this is a hypothesis on the solution, not a consequence of smooth data (Remark 2.6). The domain is a general smooth obstacle problem (Assumption 2.1): the wall Γ may have several components and need not be convex; the exterior of the unit ball in a box is the model case.

1. *Geometry of inscribed surfaces* (Section 3). On each face F the distance to Γ is, up to $O(h_\Gamma^3)$, the quadratic $\frac{1}{2} \sum_{i < j} \mathfrak{h}(e_{ij}, e_{ij}) \mu_i \mu_j$ in the barycentrics of F , with the second fundamental form $\mathfrak{h}$ evaluated on the edge vectors (Lemma 3.1); on the sphere this is exact up to $O(h_\Gamma^4)$. The closest-point projection $\Gamma_h \rightarrow \Gamma$ is a homeomorphism whenever the tetrahedral mesh is

shape regular and the domain Ω_h does not reach the far side of the obstacle, condition (OB) (Theorems 3.4 and 3.6). Neither hypothesis can be dropped (Examples 3.8 and 3.9).

2. *Stability* (Section 4). The continuous inf-sup constant of Ω_h is uniform in h (Lemma 4.3), and on Alfled refinements with $k \geq 3$ so is the discrete one (Lemma 4.6), with an explicit dependence on the constant of a local result of Guzmán and Neilan [18, Theorem 3.1].
3. *The printed method leaks* (Theorems 5.11, 5.12, 5.13 and Corollary 5.14). GS converges, with $\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2}G$ and $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ when $G > 0$, and

$$\tilde{u} - u_h = \frac{h}{\mu} v_\varphi + o((h/\mu)^{1/2}) \text{ in energy}, \quad v_\varphi|_{\Gamma_h} \approx -(p - \bar{p})n,$$

so that $\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}/((h/\mu)G) \rightarrow 1$ and $\|\tilde{u} - u_h\|/((h/\mu)^{1/2}G) \rightarrow 1$.

4. *The leak limit and drag* (Section 7). For every $\gamma \geq \gamma_0$, with no upper bound, $(\mu/h)(\tilde{u} - u_h)$ minus its geometric part converges in H^1 to the Stokes flow U with Dirichlet data $-(p - \bar{p})n$ on Γ , at rate $h_\Gamma^{1/2}$ (Theorem 7.10); on Alfled refinements with $k \geq 4$ the rate is exactly $\frac{1}{2}$, with a lower bound weighted by the second fundamental form and an explicit constant (Theorem 7.13). So the H^1 constant of GS is $\|\nabla U\|/G$, and $\|\nabla(\tilde{u} - u_h)\| \leq C(h/\mu + h_\Gamma^{3/2} + h^k + h^{k+1}h_\Gamma^{-1/2})$ uniformly in γ (Corollary 7.16). The drag, lift and torque of GS have error $-\frac{h}{\mu}K_\psi$ to leading order, with K_ψ given by reciprocity (Theorem 7.24); those of CNS* converge at rate 2 modulo (E_Z), a regularity assumption for the dual Stokes flow at the outer boundary (Theorem 7.27).
5. *The corrected method attains Scott's rate* (Theorem 5.16). CNS* is well posed for $\gamma \geq \gamma_2$, and for bounded γ and $h_\Gamma \geq h^2$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k).$$

On Alfled refinements this holds for every $k \geq 3$ with no further stability hypothesis.

6. *Sharpness* (Theorems 6.7, 6.10, 6.27 and Proposition 6.11). For CNS*, and for GS with any G ,

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \|\mathfrak{h}|\partial_N u\|_{L^2(\Gamma)} - C h_\Gamma^2$$

on Alfled refinements for every $k \geq 3$, and on every shape-regular mesh for $k \geq 9$. For $k \geq 4$ the proof reduces to an exact rank computation on one reference macro-element. For $k = 3$ a single macro-element cannot work (Proposition 6.12); the witnesses live on the star of an interior edge, where the moment of a “vertex move” has a closed form (Theorem 6.19), and a Star Nondegeneracy Theorem (Theorem 6.25) covers saddle and parabolic points.

7. *Penalty* (Section 8). Coercivity holds for $\mu \geq C\rho$. Conversely, N_h is indefinite on the divergence-free space if $\mu \text{diam } F/h$ is below an exactly certified constant $\lambda_T > 32$ of a single near-regular Alfled macro-element (Theorem 8.2); $\lambda_T \geq 20$ holds for every macro-element in an explicit box of shapes around the regular one (Proposition 9.2); tall macro-elements have no single-cell witness, and a continuum witness with half-space constant 2327591/151008 gives $\mu^* \geq c\rho$ for all shapes under local fineness (Theorem 8.3).
8. *Navier–Stokes* (Corollary 10.1). Under the discrete linearised-stability conditions of [29], which hold for small data and, for $h \leq h_0$ (not explicit), on nonsingular solution branches, the error bounds of items 3 and 5 and the leak law carry over to steady Navier–Stokes flow, and so do the leak limit and the drag statements of item 4, with the Oseen leak flow in place of U (Theorem 10.3).
9. *Explicit constants* (Section 9). On Alfled refinements the witness constants are explicit functions of the macro-tetrahedron, from one certified reference norm; for regular macro-elements the constant of the leak lower bound is $5.20 \cdot 10^{-3}$ and that of the sharpness bound $2.37 \cdot 10^{-5}$. The discrete inf-sup constant is not explicit; we say why.

10. *Worsey–Farin splits.* If the wall faces are subdivided, the discrete divergence is continuous along the interior edges of each wall face, so the inf-sup condition fails for the full discontinuous P_{k-1} pressure space (Proposition 4.9), as is known from [17, 10]. The sharpness witnesses exist for $k \geq 4$ (Proposition 6.30), but the upper bounds are not proved there.
11. *Numerics* (Section 11). A three-dimensional Alfeld solver confirms pointwise incompressibility, the optimal rates h^3 and h^4 on a manufactured problem, a stable pair without spurious modes, and the leak law with constant tending to 1 in slip and energy. At the mesh levels that fit our machine the sphere tests are pre-asymptotic for the $h_\Gamma^{3/2}$ rate; we give the commands for the finer levels.

What changes from two dimensions. Three things change. (i) In two dimensions the leading term of the tilt $n(x^*) - n_e$ of a chord is odd about the chord midpoint, which gives a dual-norm cancellation. In three dimensions its face mean is proportional to the offset between centroid and circumcentre (Lemma 3.2), which is nonzero on non-equilateral faces. This turns out to be harmless for the rates: all upper bounds only need a penalty-weighted bound (Lemma 5.3); the face means enter only constants (Remark 7.6). (ii) Vertex terms of the two-dimensional normal-load cancellation become edge terms, which cancel to leading order because each edge of Γ_h lies in exactly two faces (Lemma 5.10). (iii) Stream functions are unavailable. We replace them by a Piola–Hestenes extension of the closed 2-form of the velocity (Lemma 3.12) and by an interpolate-and-correct construction of the discrete leak field (Lemma 5.9). The lower-bound argument is new in three dimensions: the sagitta is a quadratic form in the barycentrics, and the whole lower bound reduces to three edge moments of the normal derivative on each face.

Where a two-dimensional proof carries over with only notational changes we give the three-dimensional replacement of each ingredient and refer to [29] for the unchanged algebra; the three-dimensional arguments are given in full.

Outline. Section 2 fixes the setting. Section 3 contains the geometry and the lift theorem, Section 4 the stability results, Section 5 the error analysis of both methods, Section 6 the lower bounds, Section 7 the leak limit and drag, Section 8 the penalty, Section 9 the explicit constants, Section 10 the Navier–Stokes corollary and Section 11 the computations. Section 12 lists what is not proved. Appendix A lists the code.

2. SETTING, METHODS AND ASSUMPTIONS

2.1. Domain.

Assumption 2.1 (Domain). $\Omega \subset \mathbb{R}^3$ is a bounded connected Lipschitz domain with $\partial\Omega = \Gamma \sqcup \Sigma$. The wall $\Gamma = \Gamma_1 \sqcup \cdots \sqcup \Gamma_m$ is a union of closed, connected, embedded surfaces of class C^{k+3} , each a whole component of $\partial\Omega$. The outer boundary $\Sigma \neq \emptyset$ is a finite union of closed polyhedral surfaces (whole components of $\partial\Omega$) and carries strongly imposed Dirichlet data.

The model case is $\Omega = R \setminus \overline{D}$ with $R = (-L, L)^3$, $L > 1$, and D the open unit ball.

N denotes the unit normal of Γ pointing into Ω and $n := -N$ the outward normal of Ω on Γ . Let d be the signed distance to Γ , positive in Ω , π the closest-point projection, and $\mathfrak{h}_y(X, Y) := -\langle D_X N, Y \rangle$ the second fundamental form with respect to N , so that $\mathfrak{h} > 0$ where Ω is locally convex; for the exterior of the unit ball, $\mathfrak{h} = -I$. Put $\kappa_\infty := \sup_\Gamma |\mathfrak{h}|$. We use two standard facts about the fixed surface Γ .

- (T) There is $r > 0$ with $r\kappa_\infty \leq \frac{1}{2}$ such that $\Psi(y, s) := y + sN(y)$ is a C^{k+2} diffeomorphism of $\Gamma \times (-r, r)$ onto a tube $\mathcal{T}_r$, with $d(\Psi(y, s)) = s$, $\pi(\Psi(y, s)) = y$ and $\nabla d = N \circ \pi$. The volume element is $dx = J dA ds$ with $J = \det(I - sS_y) \in [\frac{1}{4}, \frac{9}{4}]$, S the shape operator. $\mathcal{T}_r$ has positive distance from Σ , and the tubes around different Γ_i are disjoint.

- (Gr) There is $a_0 \leq r/4$ with $\kappa_\infty a_0 \leq \frac{1}{40}$ such that, for each $y \in \Gamma$, in the frame $(y; \tau_1, \tau_2, N(y))$ with coordinates $(\xi, \eta) \in \mathbb{R}^2 \times \mathbb{R}$, the set $\Gamma \cap (-2a_0, 2a_0)^3$ is the graph of $f_y \in C^{k+3}$ with $f_y(0) = 0$, $\nabla f_y(0) = 0$, $D^2 f_y(0) = \mathfrak{h}_y$, $|\nabla f_y(\xi)| \leq 2\kappa_\infty |\xi|$ and $|f_y(\xi)| \leq \kappa_\infty |\xi|^2$, and $\Omega \cap (-2a_0, 2a_0)^3 = \{\eta > f_y(\xi)\}$.

2.2. Inscribed surface and mesh.

Assumption 2.2 (Inscribed surface and mesh).

- (G1) Γ_h is a closed triangulated surface (each edge lies in exactly two faces) whose vertices lie on Γ . Its faces have diameter in $[c_0 h_\Gamma, h_\Gamma]$ and minimal angle at least $\theta_0 > 0$.
 (G3) $h_\Gamma \leq h_*$, a constant depending on Γ , c_0 , θ_0 and the shape regularity below.
 (M0³) Ω_h is a bounded connected open polyhedral set with $\partial\Omega_h = \Sigma \sqcup \Gamma_h$, and $\mathcal{T}_h$ is a conforming, shape-regular tetrahedral mesh of Ω_h (inradius $\geq c_s \text{diam } T$) whose boundary faces on Γ_h are exactly the faces of Γ_h . We write $\mathcal{F}_h^\Gamma$ for this set of faces.
 (OB) For each i , the parallel surface $\Gamma_i^- := \{y - \frac{r}{2}N(y) : y \in \Gamma_i\}$ on the obstacle side is not contained in Ω_h .

(OB) holds whenever Ω_h agrees with Ω away from a thin neighbourhood of Γ , for example for every mesh that only modifies Ω near Γ . In the model case, P_h is a convex polyhedron with triangular faces and vertices on the unit sphere, and $\Omega_h = R \setminus P_h$; then $P_h \supset B(0, 1 - Ch_\Gamma^2) \supset \Gamma^-$, so (OB) holds. The two-dimensional analogue of (G1) and (OB) is discussed in [29]. Section 3 shows that (G1), (M0³) and (OB) imply that $\pi|_{\Gamma_h}$ is a homeomorphism onto Γ ; in surface finite elements this lift property is usually assumed.

Put

$$h := \max_{T \in \mathcal{T}_h} \text{diam } T, \quad h_\Gamma := \max_{F \in \mathcal{F}_h^\Gamma} \text{diam } F, \quad \rho := \frac{h}{\min_{F \in \mathcal{F}_h^\Gamma} \text{diam } F}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the effective wall penalty, $\mu/h = \gamma/h_\Gamma$. By (G1), $\rho/h \leq 1/(c_0 h_\Gamma)$. No quasi-uniformity is assumed.

Definition 2.3 (Alfeld refinement). $\mathcal{T}_h$ is an *Alfeld refinement* if it is obtained from a conforming shape-regular macro mesh $\mathcal{T}_h^M$ of Ω_h by splitting every macro-tetrahedron into four at its barycentre. The faces of the macro-tetrahedra on Γ_h are not subdivided, so each $F \in \mathcal{F}_h^\Gamma$ is a face of exactly one macro-tetrahedron T_F and of one of its four sub-tetrahedra.

(D) $\tilde{f}$ is a smooth bounded extension of f to a fixed neighbourhood of $\bar{\Omega}$, and $h \leq h_*$.

2.3. Spaces and forms. Fix $k \geq 3$. V_h is the space of continuous piecewise- P_k vector fields on $\mathcal{T}_h$, $V_h^R := \{v \in V_h : v|_\Sigma = 0\}$ and $V_h^0 := \{v \in V_h : v|_{\partial\Omega_h} = 0\}$. Π_h is the space of discontinuous piecewise- P_{k-1} functions and $Z_h := \{v \in V_h^R : \text{div } v = 0\}$. With $g := u|_\Sigma$ and g_I its P_k Lagrange interpolant, $W_h := \{z \in V_h : z|_\Sigma = g_I, \text{div } z = 0\}$.

On Γ_h , n is the outward unit normal of Ω_h (on the face F we also write n_F), $\partial_n v := (\nabla v)n$ and $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ pairing. Let

$$a_h(u, v) := \frac{1}{2}(Du, Dv), \quad Du := \nabla u + (\nabla u)^\top, \\ N_h(u, v) := a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle.$$

Definition 2.4 (Method GS [16, (28)], as printed). Find $u_h \in V_h$ with $u_h|_\Sigma = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \text{div } v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \text{div } u_h) = 0 \quad \forall q \in \Pi_h.$$

The transcription conventions (the sign of the pressure, and the test space V_h^R) are those of [29].

Definition 2.5 (Method CNS*). As GS, with momentum row

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v), \quad \bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q.$$

The mean is over all of Γ_h , also when Γ has several components; per-component means are inconsistent (Remark 5.6). The added term appears in the momentum row only, so the velocity stays exactly divergence-free. Its velocity is that of the zero-net-flux formulation of [13, Section 5.1] (see [29] for the transcription; the argument is dimension-free).

(IS3) There is $\beta > 0$, independent of h , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there is $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.

On Alfled refinements with $k \geq 3$, (IS3) is a theorem (Lemma 4.6). On meshes whose wall faces are subdivided, such as Worsey–Farin splits, it is false as formulated (Proposition 4.9). On other meshes it is an assumption; see Remark 4.8.

2.4. Exact solution, extensions and norms. (u, p) is a Stokes solution, $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$, $u|_\Sigma = g$, with $u \in C^{k+1}(\bar{\Omega})$ and $p \in C^k(\bar{\Omega})$. Lemma 3.12 provides extensions $\tilde{u} \in C^{k+1}$ and $\tilde{p} \in C^k$ to a fixed neighbourhood $\mathcal{O} \supset \bar{\Omega}_h$ with $\operatorname{div} \tilde{u} = 0$. The defect

$$F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$$

vanishes on Ω and is supported in $\bar{\Omega}_h \setminus \Omega$.

Remark 2.6 (Edges and corners of the outer box). In the model case $\Sigma = \partial R$ has edges and corners, and near them the smoothness of (u, p) is a hypothesis on the *solution*, not a consequence of smooth data. Along an edge of R the singular exponents include those of the plane Stokes problem in a right angle, the roots of $\sin^2(\lambda\pi/2) = \lambda^2$ with $\operatorname{Re} \lambda > 1$, the first being $\lambda_1 = 2.739593 \pm 1.119025i$ (computed for the companion paper [29]). So for generic smooth, compatible data one can only expect $u \in H^s$ near an edge for $s < 1 + 2.7396$, unless the singular coefficients vanish; the corners of R add vertex singularities, whose exponents we have not computed [7]. The hypothesis $u \in C^{k+1}(\bar{\Omega})$ holds for the manufactured solutions of Section 11. For data-driven problems on meshes that are not graded at the edges of R , the terms h^k in the H^1 bounds, and every other term that uses the smoothness of u near Σ , should be read with h^k replaced by $h^{\min(k, 2.7396) - \epsilon}$. Near Γ nothing changes: the wall is smooth and $\mathcal{T}_r$ has positive distance from Σ . The same caveat applies to the global regularity used in Remark 10.2 (Stokes $H^2 \times H^1$ on Ω , which for the box minus a ball we can only reduce, by a partition of unity and a Bogovskii correction, to the H^2 theorem for convex polyhedra attributed to [7], whose statement we have not checked), to the approximation term E_U^0 of Section 7, and to assumption (E_Z) of Section 7.

For fields with traces on Γ_h ,

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{F \in \mathcal{F}_h^\Gamma} \operatorname{diam} F \|\partial_n v\|_{L^2(F)}^2.$$

We write $\bar{p} := |\Gamma|^{-1} \int_\Gamma p$ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. C and c depend on k , the constants in (G1) and (M0³), Ω , and norms of $(u, p, \tilde{u}, \tilde{p})$; they never depend on $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed. C_p may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

3. GEOMETRY OF INSCRIBED SURFACES

3.1. Faces. For a face $F \in \mathcal{F}_h^\Gamma$ with vertices $z_1, z_2, z_3 \in \Gamma$ we write μ_1, μ_2, μ_3 for its barycentric coordinates, $e_{ij} := z_i - z_j$, $\ell := \operatorname{diam} F$, x_F for its centroid, $y_F := \pi(x_F)$, P_y for the orthogonal projection onto $T_y \Gamma$ and $\mathfrak{h}_F := \mathfrak{h}_{y_F} \circ P_{y_F}$. For $x \in F$ put $x^* := \pi(x)$.

Lemma 3.1 (Faces on a general surface). *Assume (G1) and (G3). Then for every $F \in \mathcal{F}_h^\Gamma$ and $x \in F$:*

- (a) $d(x) = \frac{1}{2} \sum_{i < j} \mathfrak{h}_F(e_{ij}, e_{ij}) \mu_i \mu_j + O(\ell^3)$; in particular $|d(x)| \leq C\ell^2$.
- (b) There is $\sigma_F \in \{\pm 1\}$, independent of $x \in F$, with $|\sigma_F n_F - n(x^*)| \leq C\ell$ (where $n = -N$); hence $|n_F \cdot N(x^*)| \geq 1 - C\ell^2$.
- (c) $\pi|_F$ is a C^2 diffeomorphism onto $\pi(F)$, with area Jacobian $1 + O(\ell^2)$.
- (d) With $w_F := (\mathfrak{h}_F(e_{ij}, e_{ij}))_{i < j} \in \mathbb{R}^3$ and $|\mathfrak{h}|$ the operator norm, $|w_F| \geq c(\theta_0) \ell^2 |\mathfrak{h}_{y_F}| - C\ell^3$.

Proof. Use the frame (Gr) at $y := z_1$. Then $z_j = (\xi_j, f(\xi_j))$ with $|\xi_j| \leq \ell$; the vertical parts are at most $\kappa_\infty \ell^2$, so the projected triangle $\tilde{F} = \text{conv}(\xi_j)$ has minimal angle at least $\theta_0/2$ for h_* small, and F is the graph over $\tilde{F}$ of the linear interpolant $L := I_1 f$, with the same barycentrics.

(a) Write $f = \frac{1}{2} \xi^\top H \xi + R_3$ with $H = \mathfrak{h}_{z_1}$ and $|R_3| \leq C\ell^3$ on $\tilde{F}$. For a quadratic form, $I_1 Q - Q = \sum_{i < j} \mu_i \mu_j Q(\xi_i - \xi_j)$; this is the polarised variance identity $\sum_i \mu_i |\xi_i - o|^2 - |\xi - o|^2 = \sum_{i < j} \mu_i \mu_j |\xi_i - \xi_j|^2$. For $x = (\xi, L(\xi))$ we have $d(x) = (L - f)(\xi) N_3 + O((L - f)^2) = (L - f)(\xi) + O(\ell^4)$, since $N_3 = (1 + |\nabla f|^2)^{-1/2} = 1 + O(\ell^2)$. Finally $\xi_i - \xi_j = P_{z_1} e_{ij}$, and replacing (z_1, P_{z_1}) by (y_F, P_{y_F}) changes $\mathfrak{h}(e, e)$ by $O(\ell)|e|^2$ because $\Gamma \in C^3$.

(b) $n_F \parallel (\nabla L, -1)$ and $N(x^*) \parallel (-\nabla f(\xi^*), 1)$. On $\tilde{F}$, $|\nabla L - \nabla f| \leq C\ell / \sin(\theta_0/2)$ (linear interpolation), and $|\xi^* - \xi| \leq C\ell^2$.

(c) $D\pi(x) = (I - dS)^{-1} P_{x^*}$. Restricted to the plane of F its Jacobian is $|n_F \cdot N(x^*)|(1 + O(d)) = 1 + O(\ell^2)$ by (a) and (b). Injectivity on F : $\pi|_F$ is a C^1 -small perturbation of the vertical projection, which is injective.

(d) The map $H \mapsto (H(u_{ij}, u_{ij}))_{i < j}$ with $u_{ij} := P_{y_F} e_{ij} / |P_{y_F} e_{ij}|$, from symmetric forms on $T_{y_F} \Gamma$ to $\mathbb{R}^3$, is invertible with inverse bounded by $C(\theta_0)$: the three directions are pairwise non-parallel with angles bounded below, and three distinct doubled angles give three non-collinear points on a circle. Also $|P_{y_F} e_{ij}| \geq c\ell$. $\square$

For the model case the sagitta is exactly a circumsphere function, and the face mean of the normal tilt measures the offset of the centroid from the circumcentre.

Lemma 3.2 (Faces inscribed in the unit sphere). *Let F be a face of a convex polyhedron inscribed in the unit sphere, with circumcentre o_F , circumradius r_F , outward (from the centre) unit normal ν_F , and $d_F := (1 - r_F^2)^{1/2}$. Write $\ell_{ij} := |e_{ij}|$, and let x_c be the centroid.*

- (a) For $x \in F$, $1 - |x| = \frac{1}{2}(r_F^2 - |x - o_F|^2) + \delta(x) = \frac{1}{2} \sum_{i < j} \ell_{ij}^2 \mu_i \mu_j + \delta(x)$ with $0 \leq \delta \leq r_F^4/4$.
- (b) $n(x^*) - n_F = -(x - o_F) + O(r_F^2)$, where $n(x^*) = -x^*$ and $n_F = -\nu_F$. The leading term is tangential to F and vanishes at the circumcentre.
- (c) $|F|^{-1} \int_F (n(x^*) - n_F) dA = -(x_c - o_F) + O(r_F^2)$. The leading term vanishes if and only if F is equilateral.
- (d) With e_{12}, e_{23}, e_{31} the cyclically oriented edge vectors, $\sum_{\text{cyc}} \ell_{ij}^2 e_{ij} = \pm 12|F| R_{\pi/2}(x_c - o_F)$, where $R_{\pi/2}$ is the rotation by $\pi/2$ in the plane of F .

Proof. (a) With $s := r_F^2 - |x - o_F|^2 \in [0, r_F^2]$ we have $|x|^2 = d_F^2 + |x - o_F|^2 = 1 - s$, and $1 - \sqrt{1 - s} \in [\frac{s}{2}, \frac{s}{2} + \frac{s^2}{4}]$ for $s \leq \frac{1}{2}$. For $x = \sum \mu_i z_i$, $\sum_i \mu_i |z_i - o_F|^2 - |x - o_F|^2 = \sum_{i < j} \mu_i \mu_j \ell_{ij}^2$, and $|z_i - o_F| = r_F$. (b) $n(x^*) - n_F = -(o_F + y)/|x| + \nu_F = -y/|x| + \nu_F(1 - d_F/|x|)$ with $y := x - o_F \perp \nu_F$, $|x|^{-1} = 1 + O(r_F^2)$ and $0 \leq 1 - d_F/|x| \leq 1 - d_F$. (c) integrates (b); centroid and circumcentre coincide exactly for equilateral triangles. (d) With $a + b + c = 0$ the cyclic edge vectors, $|a|^2 a + |b|^2 b + |c|^2 c = (|a|^2 - |c|^2)a + (|b|^2 - |c|^2)b$; the identity with $x_c - o_F$ is an elementary computation in the plane (we checked it symbolically and on random triangles). $\square$

In two dimensions the circumcentre of a chord is its midpoint, so the tilt of a chord is odd about its midpoint; this is the cancellation used in [29]. By (c) it is lost on every non-equilateral face. We never use it (Lemma 5.3).

3.2. The closest-point projection is a homeomorphism. Lemma 3.1 is a statement about single faces. It does not exclude folds: two faces sharing an edge can have $\sigma_F = -\sigma_{F'}$ and project to the same side of the image of the edge. In surface finite elements the lift $\pi|_{\Gamma_h}$ is therefore assumed to be bijective [8, 9]; in discrete differential geometry, polyhedral surfaces that are normal graphs are assumed as well [21, 24]. We show that in our setting it is a consequence of the volume mesh.

For $y \in \Gamma$ let $\ell_y := \{\Psi(y, s) : |s| \leq r/2\}$ be the normal segment. Let S_1 be the union of the edges of Γ_h and $V := \Gamma \setminus \pi(S_1)$, which is open and dense by Lemma 3.1(c). For $y \in V$, the set $\ell_y \cap \Gamma_h$ is finite and consists of interior points of faces; let $k(y)$ be its cardinality. By Lemma 3.1(a), $\Gamma_h \subset \{|d| \leq C_1 h_\Gamma^2\} \subset \mathcal{T}_{r/4}$, and a normal segment meets a face at most once and transversally (Lemma 3.1(b)).

Lemma 3.3 (Crossing rule). *Let $y \in V$ and $\ell_y \cap \Gamma_h = \{\Psi(y, s_j)\}_{j=1}^k$ with $s_1 < \dots < s_k$, the j -th point interior to the face F_j . On each open interval of $(-\frac{r}{2}, \frac{r}{2}) \setminus \{s_j\}$ the property $\Psi(y, s) \in \Omega_h$ is constant. Crossing s_j upwards goes from outside to inside Ω_h if $\sigma_{F_j} = +1$, and from inside to outside if $\sigma_{F_j} = -1$.*

Proof. ℓ_y does not meet Σ , so the only points of $\partial\Omega_h$ on it are the $\Psi(y, s_j)$. Near an interior point of F_j , Ω_h is the open half-space side into which $-n_{F_j}$ points, and $\sigma_{F_j} = +1$ means that this is the side of increasing s . $\square$

Theorem 3.4 (Orientation implies the lift). *Assume (G1), (G3), that Ω_h is a bounded open polyhedral set with $\partial\Omega_h = \Sigma \sqcup \Gamma_h$ lying on one side of each face, and*

(OR) $\sigma_F = +1$ for every $F \in \mathcal{F}_h^\Gamma$, i.e. $n_F \cdot n(\pi(x)) > 0$ on F .

Then for each i , π maps $\Gamma_h^{(i)} := \Gamma_h \cap \mathcal{T}_r(\Gamma_i)$ homeomorphically onto Γ_i , piecewise as a C^2 diffeomorphism, and $n_F \cdot n(\pi(x)) \geq 1 - Ch_\Gamma^2$ on every face.

Proof. Step 1: $k \leq 1$ on V . By Lemma 3.3 and (OR) every crossing goes from outside to inside. If $k(y) \geq 2$, the status just above s_1 is inside and just below s_2 is outside, which contradicts constancy on (s_1, s_2) .

Step 2: faces. π is injective on each face (Lemma 3.1(c)).

Step 3: edges. Let x_0 be interior to the edge $E = F \cap F'$ and $\gamma := \pi(E)$, a C^2 arc. π maps half-disc neighbourhoods of x_0 in F and in F' onto one-sided neighbourhoods of $\pi(x_0)$ bounded by γ . If they lay on the same side of γ , they would overlap in an open set, which contains a point $y \in V$ with a preimage in the interior of F and one in the interior of F' , so $k(y) \geq 2$. Hence they lie on opposite sides, and π is injective near x_0 in $F \cup F'$, a neighbourhood of x_0 in Γ_h .

Step 4: vertices. Let v be a vertex and $St(v)$ its closed star. On $St(v)$, $|\pi(x) - v| \geq \frac{1}{2}|x - v|$ (segments stay inside faces), and $\text{dist}(v, \partial St(v)) \geq c(\theta_0)c_0 h_\Gamma$. Fix $\varepsilon < \frac{1}{4}c(\theta_0)c_0 h_\Gamma$ such that $B := \{y \in \Gamma : 0 < |y - v| < \varepsilon\}$ is a punctured topological disc, and let $A := \{x \in St(v) : 0 < |\pi(x) - v| < \varepsilon\}$. By Steps 2–3 and invariance of domain, $\pi|_A : A \rightarrow B$ is a local homeomorphism; it is proper, since preimages of compact sets are closed in the compact $St(v)$ and contained in A . A proper local homeomorphism onto a connected, locally compact Hausdorff space is a finite covering [12, Theorem 4.22]. Its number of sheets is at most $k(y) \leq 1$ for $y \in B \cap V$, so it is one, and π is injective near v .

Step 5: global. By Steps 2–4 and invariance of domain, $\pi : \Gamma_h^{(i)} \rightarrow \Gamma_i$ is a local homeomorphism. It is proper ($\Gamma_h^{(i)}$ is compact), so its image is open and closed, hence all of the connected Γ_i , and it is a covering with a constant number of sheets. That number is $k(y) \leq 1$ at any $y \in V \cap \Gamma_i$. A continuous bijection of a compact space onto a Hausdorff space is a homeomorphism. The normal bound is Lemma 3.1(b) with $\sigma_F = +1$. $\square$

Corollary 3.5 (Normal graph). *Under the assumptions of Theorem 3.4, $\eta_h := d \circ (\pi|_{\Gamma_h})^{-1}$ is Lipschitz and piecewise C^2 on Γ , $\Gamma_h = \{\Psi(y, \eta_h(y)) : y \in \Gamma\}$, $\|\eta_h\|_\infty \leq Ch_\Gamma^2$, $\|\nabla_\Gamma \eta_h\|_\infty \leq Ch_\Gamma$, and*

$$\Omega_h \cap \mathcal{T}_{r/2} = \{x \in \mathcal{T}_{r/2} : d(x) > \eta_h(\pi(x))\}.$$

In particular $\Gamma_i^- \cap \Omega_h = \emptyset$ and $\Gamma_i^+ := \{y + \frac{r}{2}N(y)\} \subset \Omega_h$, so (OB) holds. Moreover $|\Omega \Delta \Omega_h| \leq Ch_\Gamma^2$, and both $\Omega \setminus \Omega_h$ and $\Omega_h \setminus \Omega$ lie in $\{|d| \leq Ch_\Gamma^2\}$.

Proof. The bounds on η_h follow from Lemma 3.1: on $\pi(F)$, $\nabla_\Gamma \eta_h = D(\pi|_F)^{-T} \nabla_F d$ and $|\nabla_F d| = |N - (N \cdot n_F)n_F| \leq Ch_\Gamma$. For $y \in V$ the single crossing of ℓ_y is at $s = \eta_h(y)$ and goes from outside to inside, so $\Omega_h \cap \ell_y = \{\Psi(y, s) : \eta_h(y) < s < \frac{r}{2}\}$. The two sets in the display are open in $\mathcal{T}_{r/2}$, disjoint from Γ_h , have Γ_h as relative boundary, and agree on the dense set $\bigcup_{y \in V} \ell_y \setminus \Gamma_h$; two such sets coincide ($U_1 \setminus \overline{U_2}$ is open and misses a dense set). For Γ_i^- : it does not meet $\partial\Omega_h$, and points of $\mathcal{T}_{r/2}$ close to it have $d < \eta_h \circ \pi$, so no point of Γ_i^- lies in the open set Ω_h . The same argument gives $\Gamma_i^+ \subset \Omega_h$. $\square$

Theorem 3.6 (The volume mesh implies the orientation). *Assume (G1), (G3), ($M0^3$) and (OB). Then (OR) holds, and hence the conclusions of Theorem 3.4 and Corollary 3.5.*

Proof. (OB) is global on each component. Γ_i^- is connected and disjoint from $\partial\Omega_h$, so it lies entirely inside or entirely outside Ω_h ; by (OB), $\Gamma_i^- \cap \Omega_h = \emptyset$.

Each face sees the side it faces. Let $T \in \mathcal{T}_h$ be the tetrahedron with face F , c_T its incentre and $\varrho_T \geq c_s \text{diam } T \geq c_s c_0 h_\Gamma$ its inradius. Then $c_T = t - \varrho_T n_F$, with $t \in F$ the point where the insphere touches F , and $c_T \in \mathcal{T}_{r/4}$. By Lemma 3.1(b), $n_F = -\sigma_F N(\pi t) + O(h_\Gamma)$, and with $\nabla d = N \circ \pi$,

$$d(c_T) = d(t) - \varrho_T \int_0^1 N(\pi(t - \tau \varrho_T n_F)) \cdot n_F d\tau = \sigma_F \varrho_T + O(h_\Gamma^2) + O(\varrho_T h_\Gamma).$$

So $\sigma_F d(c_T) \geq c_s c_0 h_\Gamma - Ch_\Gamma^2 > C_1 h_\Gamma^2$ for $h_\Gamma \leq h_*$. With $y := \pi(c_T)$, the normal segment from c_T to $\Psi(y, \sigma_F \frac{r}{2})$ misses Σ and, since $|s| > C_1 h_\Gamma^2$ along it, misses Γ_h . It starts in $\text{int } T \subset \Omega_h$, so $\Psi(y, \sigma_F \frac{r}{2}) \in \Omega_h$.

Conclusion. If $\sigma_F = -1$ for some F , then $\Psi(y, -\frac{r}{2}) \in \Gamma_i^- \cap \Omega_h = \emptyset$, a contradiction. $\square$

Remark 3.7. (1) Only the tetrahedra touching Γ_h are used, through $\varrho_T \geq c \text{diam } F$. (2) Without (OB) the same proof shows that on each Γ_i either all $\sigma_F = +1$, or all $\sigma_F = -1$ (a mirror statement), or both occur, and then every generic crossing number is even; Example 3.9 is of the third kind. (3) The manifold condition in (G1) is not needed for Theorem 3.4: an edge in three or more faces, or a vertex whose star is not a disc, would produce $k(y) \geq 2$. (4) The structure of the proof (local homeomorphism plus properness gives a covering; one sheet from a single fibre) is that of Amenta, Choi, Dey and Leekha [1, Theorem 15], who assume that adjacent triangles meet at an angle greater than $\pi/2$. Here the fibre count comes from the in/out crossing rule of $\partial\Omega_h$, and the orientation from the tetrahedra touching Γ_h ; no angle condition between neighbouring faces is needed.

Both hypotheses are necessary. In the next two examples $\Omega = R \setminus \overline{D}$ with D the unit ball, $N(y) = y$ and $r = \frac{1}{2}$.

Example 3.8 ((G1) and (OB) do not suffice: an inscribed sliver). Let P_h be a convex polyhedron inscribed in the unit sphere satisfying (G1) with $h_\Gamma = h$, and let F be a face with $\text{diam } F = h$, so that its circumradius satisfies $r_F \geq h/2$. Consider the square Q of side $h/3$ in the plane of F whose centre lies at distance $h/20$ from o_F , and let q_1, q_2, q_3 be the points of the spherical cap over F whose orthogonal projections onto the plane of F are three vertices of Q . The plane through q_1, q_2, q_3 meets the sphere in a circle; choose q_4 on the cap, within h^2 of the point over the

fourth vertex of Q , and not on that circle. (The square is placed off-centre, and q_4 is perturbed, because four points of the sphere over a square centred on the axis are concyclic, hence coplanar.) All q_j project to within $\frac{\sqrt{2}}{6}h + \frac{h}{20} + h^2 < \frac{h}{2} \leq r_F$ of o_F , so they lie strictly beyond the plane of F . Let $P' := \text{conv}\{q_j\}$, a genuine tetrahedron of thickness $O(h^2)$, and put $\Omega_h := R \setminus (P_h \cup P')$, $\Gamma_h := \partial P_h \sqcup \partial P'$.

- $P' \cap P_h = \emptyset$, since P_h lies in the closed half-space bounded by the plane of F and P' in the open complementary one.
- The faces of P' are, up to a relative perturbation of order h , right isosceles triangles with legs $h/3$; their minimal angles tend to 45° and their diameters to $\frac{\sqrt{2}}{3}h$. So (G1) holds with $c_0 \leq 0.47$, and Γ_h is embedded. Two faces of P' face away from the centre and two face towards it.
- (OB) holds: $\Gamma^- = \{|x| = \frac{3}{4}\} \subset P_h$ for small h .
- $\pi|_{\Gamma_h}$ is not injective: a generic point of $\pi(P')$ has three preimages, one on ∂P_h and one on each side of P' . The faces of P' facing the centre have $\sigma = -1$; the thin lens between F and P' belongs to Ω_h .

By Theorem 3.6, this Ω_h admits no shape-regular conforming tetrahedral mesh with the faces of Γ_h as boundary faces, once h is small: the lens has thickness $O(h^2)$ and is bounded by faces of diameter at least $c_0 h$. So a volume hypothesis such as (M0³) is needed if (OR) or the lift itself is not assumed. We have not constructed an example with connected Γ_h .

Example 3.9 ((G1) and (M0³) do not suffice: a filled obstacle). Let P' be as in Example 3.8, $\Omega_h := R \setminus P'$ and $\Gamma_h := \partial P'$. Then Γ_h is connected and homeomorphic to Γ , (G1) and (G3) hold, $\partial\Omega_h = \partial R \sqcup \Gamma_h$ and Ω_h is connected; but $\pi|_{\Gamma_h}$ covers a patch of diameter $O(h)$ twice and misses the rest of Γ , and (OB) fails. A shape-regular mesh exists: with o the centre of Q and $p_\pm := o \pm hN(o)$, the bipyramid $\text{conv}(P' \cup \{p_+, p_-\}) \setminus \text{int } P'$ is the union of four shape-regular tetrahedra $\text{conv}(G, p_\pm)$, G the faces of P' . Completing this to a shape-regular conforming mesh of the box minus a convex polyhedron of fixed shape at scale h is routine; we do not write it out. So even “ Γ_h is a triangulation of Γ ” does not give the lift; a global condition tying Ω_h to Ω , such as (OB), is needed.

3.3. Consequences. From now on (G1), (G3), (M0³) and (OB) hold. By Theorem 3.6, $\sigma_F = +1$ for every face, $\pi|_{\Gamma_h}$ is a homeomorphism and Corollary 3.5 applies. Since Ω_h is connected by (M0³), $\Omega_h \setminus \overline{\mathcal{T}_{r/2}} = \Omega \setminus \overline{\mathcal{T}_{r/2}}$: a component C of $\mathbb{R}^3 \setminus (\Sigma \cup \overline{\mathcal{T}_{r/2}})$ misses $\partial\Omega_h \cup \partial\Omega$, so it lies in or outside each set; if $\overline{C}$ meets $\partial\mathcal{T}_{r/2}$, it lies in both or in neither by Corollary 3.5, and if not, then $\partial C \subset \Sigma$ and C would be a relatively closed and open proper subset of the connected sets Ω_h and Ω , so it lies in neither.

Lemma 3.10 (One boundary face per tetrahedron). *For $h_\Gamma \leq h_*$, no tetrahedron of $\mathcal{T}_h$ has two faces in Γ_h . The same holds for the macro-tetrahedra of an Alfeld refinement.*

Proof. Two such faces would share an edge E of Γ_h , and the dihedral angle of the tetrahedron at E would equal the dihedral angle of Ω_h at E , which is $\pi + O(h_\Gamma)$ because the normals of the two faces agree with n up to $O(h_\Gamma)$ (Theorem 3.4). Shape regularity bounds the dihedral angles of tetrahedra by $\pi - \delta$. $\square$

Lemma 3.11 (Uniform Korn, trace and Friedrichs inequalities). *There is a bi-Lipschitz homeomorphism $\Phi_h : \overline{\Omega} \rightarrow \overline{\Omega}_h$, the identity outside $\mathcal{T}_{r/4}$, mapping Γ onto Γ_h , with*

$$\|D\Phi_h - I\|_{L^\infty} + \|D\Phi_h^{-1} - I\|_{L^\infty} \leq Ch_\Gamma.$$

Consequently there are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^3$ with $v|_\Sigma = 0$ satisfies

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v).$$

Proof. Let $\chi \in C^\infty(\mathbb{R})$ with $\chi = 1$ near 0, $\chi = 0$ on $[r/4, \infty)$, and put $\Phi_h(\Psi(y, s)) := \Psi(y, s + \chi(s)\eta_h(y))$ for $0 \leq s < r/2$, and $\Phi_h := \text{id}$ elsewhere on $\bar{\Omega}$. Since $|\chi'\eta_h| \leq Ch_\Gamma^2 < 1$, $s \mapsto s + \chi(s)\eta_h(y)$ is increasing and maps $[0, r/2)$ onto $[\eta_h(y), r/2)$; by Corollary 3.5 and the preceding paragraph, Φ_h maps $\bar{\Omega}$ onto $\bar{\Omega}_h$ and Γ onto Γ_h . In (y, s) coordinates, $D\hat{\Phi}_h = I + \begin{pmatrix} 0 & 0 \\ \chi \nabla_{\Gamma\eta_h} \chi' \eta_h & \chi' \eta_h \end{pmatrix}$, and $D\Psi(y, s')D\Psi(y, s)^{-1} = I + O(|s - s'|)$; with Corollary 3.5 this gives the bound. For $w := v \circ \Phi_h$, $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$. On the fixed Lipschitz domain Ω , w vanishes on Σ , a set of positive surface measure, so Korn's first inequality, the trace inequality and the Friedrichs inequality hold; changing variables back and absorbing $Ch_\Gamma \|\nabla v\|$ for $h_\Gamma \leq h_*$ gives the claim. $\square$

Lemma 3.12 (Divergence-free extension). *Let $u \in C^{k+1}(\bar{\Omega})$ be divergence free and $p \in C^k(\bar{\Omega})$, with $\Gamma \in C^{k+3}$. There are $\tilde{u} \in C^{k+1}(\mathcal{O})$ and $\tilde{p} \in C^k(\mathcal{O})$ on $\mathcal{O} := \Omega \cup \mathcal{T}_{r/2}$, with $\tilde{u} = u$, $\tilde{p} = p$ on Ω and $\text{div } \tilde{u} = 0$ on $\mathcal{O}$. Moreover $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ for $v \in V_h^R$.*

Proof. Let $\omega := \iota_u \text{vol}$, a closed C^{k+1} 2-form on $\bar{\Omega}$ ($d\omega = (\text{div } u)\text{vol}$). On a collar $\Gamma_i \times [0, r)$ the pull-back $\beta := \Psi^* \omega$ is closed and of class C^{k+1} up to $s = 0$, since $\Psi \in C^{k+2}$. In local coordinates (σ_1, σ_2) on Γ_i , $\beta = a d\sigma_1 \wedge d\sigma_2 + b_1 d\sigma_2 \wedge ds + b_2 ds \wedge d\sigma_1$. For $\lambda > 0$ let $R_\lambda(y, s) := (y, -\lambda s)$; then $R_\lambda^* \beta$ has coefficients $a(\sigma, -\lambda s)$ and $-\lambda b_\alpha(\sigma, -\lambda s)$. Choose distinct $\lambda_j \in (0, 1]$, $j = 1, \dots, k+3$, and c_j with $\sum_j c_j (-\lambda_j)^m = 1$ for $m = 0, \dots, k+2$ (a Vandermonde system), and set $\tilde{\beta} := \sum_j c_j R_{\lambda_j}^* \beta$ for $s \in (-r/2, 0)$ [20]. The coefficients of $\tilde{\beta}$ and β and all their derivatives of order at most $k+1$ agree at $s = 0$ (for a this uses $m \leq k+1$, for b_α the shifted conditions $m+1 \leq k+2$), so the glued form is C^{k+1} on $\Gamma_i \times (-r/2, r)$. It is closed on each side, because pull-backs commute with d , hence closed everywhere. Pushing forward by Ψ^{-1} and taking the vector field $\tilde{u}$ with $\iota_{\tilde{u}} \text{vol}$ equal to the result gives $\tilde{u} \in C^{k+1}$ with $\text{div } \tilde{u} = 0$. For $\tilde{p}$ use the scalar Hestenes reflection.

Since $\tilde{u}|_\Gamma = 0$ and $|d| \leq Ch_\Gamma^2$ on Γ_h , $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$. F vanishes on Ω and is supported in $S_h := \overline{\Omega_h} \setminus \Omega = \{\eta_h \circ \pi \leq d \leq 0\}$, a layer of width at most Ch_Γ^2 in the normal direction. A one-dimensional Poincaré inequality along normal segments, with the Jacobian J of (T) , gives $\|v\|_{L^2(S_h)} \leq Ch_\Gamma (\|v\|_{\Gamma_h} + h_\Gamma \|\nabla v\|)$, and $|(F, v)| \leq \|F\|_\infty |S_h|^{1/2} \|v\|_{L^2(S_h)}$ with $|S_h| \leq Ch_\Gamma^2$; conclude with Lemma 3.11. $\square$

The extension needs neither a vector potential nor the zero-flux condition on Γ , and it works verbatim in two dimensions. Only $\tilde{u} \in C^2$ near Γ is used in the geometric lemmas.

Lemma 3.13 (Wall structure). *On Γ : $\nabla u = \partial_n u \otimes n$, $\partial_n u \cdot n = 0$ and $(\nabla u)^T n = 0$. In particular $W := \partial_N u|_\Gamma$, the wall shear, is a tangential field.*

Proof. The tangential derivatives of u vanish on Γ , and $0 = \text{div } u = n \cdot \partial_n u$. $\square$

4. STABILITY

Lemma 4.1 (Inverse traces). *There is C_I , depending only on k and shape regularity, such that for $v \in V_h$*

$$\sum_{F \in \mathcal{F}_h^\Gamma} \text{diam } F \|\partial_n v\|_{L^2(F)}^2 \leq C_I^2 \|\nabla v\|^2, \quad \text{hence} \quad \|\partial_n v\|_{\Gamma_h} \leq C_I (\rho/h)^{1/2} \|\nabla v\|,$$

and for $q \in \Pi_h$, $\|q\|_{\Gamma_h}^2 \leq C_I^2 \sum_F (\text{diam } F)^{-1} \|q\|_{T_F}^2 \leq C_I^2 (\rho/h) \|q\|^2$, where T_F is the tetrahedron with face F .

Proof. Scaling on each T_F ; the T_F are distinct by Lemma 3.10, and $(\text{diam } F)^{-1} \leq \rho/h$. $\square$

Lemma 4.2 (Coercivity and continuity). *Let $\mu_0 := 4\kappa C_I^2 \rho$. The condition $\gamma \geq \gamma_0 := 4\kappa C_I^2 / c_0$ implies $\mu \geq \mu_0$. If $\mu \geq \mu_0$, then $N_h(v, v) \geq c_1 \|v\|^2$ for all $v \in V_h^R$ and $|N_h(w, v)| \leq M \|w\| \|v\|$, with $c_1 = 1/\max(2\kappa(1 + C_I^2), 2)$ and $M = 3 + (\kappa C_I^2)^{-1/2}$.*

Proof. As in two dimensions [29]: $2|\langle \partial_n v, v \rangle| \leq \frac{1}{2}a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2$ by Lemma 4.1, Young's inequality and Lemma 3.11; add $\sum_F \text{diam } F \|\partial_n v\|_F^2 \leq C_I^2 \|\nabla v\|^2$. For continuity, each ∂_n -term is at most $(\rho/\mu)^{1/2} \|w\| \|v\|$. $\square$

4.1. The continuous inf-sup constant. Let $C_B(U)$ denote the Bogovskii constant of a bounded connected Lipschitz domain U : the smallest C such that every $q \in L_0^2(U)$ is the divergence of some $v \in H_0^1(U)^3$ with $\|\nabla v\| \leq C\|q\|$. By the closed range theorem, $C_B(U)^{-1}$ is the continuous inf-sup constant of U .

Lemma 4.3 (Uniform continuous inf-sup constant). *For $h_\Gamma \leq h_*$, $C_B(\Omega_h) \leq 2C_B(\Omega)$.*

This is an instance of the continuity of the inf-sup constant under Lipschitz-close perturbations, due to Bernardi, Costabel, Dauge and Girault [2, Theorem 4.4]: if F is a diffeomorphism with $\|DF - I\|_\infty, \|DF^{-1} - I\|_\infty \leq \varepsilon$, the inf-sup constants of U and $F(U)$ differ by at most $c(d)\varepsilon$. Our map Φ_h of Lemma 3.11 is only bi-Lipschitz and piecewise smooth, so we include the short argument, which uses nothing more.

Proof. Let $\Phi := \Phi_h$, $\varepsilon := \|D\Phi - I\|_\infty + \|D\Phi^{-1} - I\|_\infty \leq Ch_\Gamma$ and $J := \det D\Phi$, so $|J - 1| \leq C\varepsilon$. Given $q \in L_0^2(\Omega_h)$, put $\hat{q} := J(q \circ \Phi)$. Then $\int_\Omega \hat{q} = \int_{\Omega_h} q = 0$ and $(1 - C\varepsilon)\|q\| \leq \|\hat{q}\| \leq (1 + C\varepsilon)\|q\|$. Let $\hat{w} \in H_0^1(\Omega)^3$ with $\text{div } \hat{w} = \hat{q}$ and $\|\nabla \hat{w}\| \leq C_B(\Omega)\|\hat{q}\|$, and $w := \hat{w} \circ \Phi^{-1} \in H_0^1(\Omega_h)^3$. By the chain rule $(\nabla w) \circ \Phi = \nabla \hat{w} (D\Phi)^{-1}$, so

$$(\text{div } w) \circ \Phi = \text{div } \hat{w} + \text{tr}(\nabla \hat{w} ((D\Phi)^{-1} - I)),$$

and, changing variables,

$$\begin{aligned} (q, \text{div } w)_{\Omega_h} &= (\hat{q}, \text{div } \hat{w})_\Omega + \int_\Omega \hat{q} \text{tr}(\nabla \hat{w} ((D\Phi)^{-1} - I)) \\ &\geq \|\hat{q}\|^2 - \sqrt{3}\varepsilon \|\hat{q}\| \|\nabla \hat{w}\| \geq \|\hat{q}\|^2 (1 - \sqrt{3}\varepsilon C_B(\Omega)). \end{aligned}$$

Also $\|\nabla w\| \leq (1 + C\varepsilon)\|\nabla \hat{w}\| \leq (1 + C\varepsilon)C_B(\Omega)\|\hat{q}\|$. Hence $\sup_w (q, \text{div } w)/\|\nabla w\| \geq (1 - C\varepsilon)\|q\|/C_B(\Omega)$ for every $q \in L_0^2(\Omega_h)$, which is the inf-sup condition on Ω_h with constant $(1 - C\varepsilon)/C_B(\Omega)$. For $h_\Gamma \leq h_*$ this is at least $1/(2C_B(\Omega))$. $\square$

Remark 4.4. A second route covers Ω_h by finitely many pieces that are star-shaped with respect to balls, uniformly in h , and applies the corresponding estimates for Bogovskii's operator [15, 6]. We carried this out for the model case and for general obstacles, with explicit cones and graph charts, but Lemma 4.3 makes it unnecessary.

4.2. The discrete inf-sup constant on Alfeld refinements. For a macro-tetrahedron K with Alfeld split K^r let $\dot{P}_k^c(K^r)$ be the continuous piecewise- P_k vector fields on K^r vanishing on ∂K , and $\dot{P}_{k-1}(K^r)$ the piecewise- P_{k-1} functions on K^r with mean zero on K . We use the following local result.

Fact 4.5 (Guzmán and Neilan [18, Theorem 3.1]). *Let $k \geq 1$ and $d = 3$. For every macro-tetrahedron K and every $p \in \dot{P}_{k-1}(K^r)$ there is $v \in \dot{P}_k^c(K^r)$ with $\text{div } v = p$ on K and $\|v\|_{H^1(K)} \leq C_{\text{loc}}\|p\|_{L^2(K)}$, where C_{loc} depends only on k and the shape regularity of K^r .*

We read the wording of this theorem, and the statements of [18, Proposition 4.1, Proposition 6.1, Corollary 6.2], through a text-extraction tool rather than directly from the typeset PDF; the wording should be rechecked against the published version before submission. The global stability result of [18] (Proposition 6.1 and Corollary 6.2, $k \geq d$) combines Fact 4.5 with the stability of the Bernardi–Raugel pair [18, Proposition 4.1], [3], for which no dependence of the constant on the domain is stated there. For a family of domains Ω_h that dependence is exactly what matters, and the following lemma supplies it. Zhang [37] proves stability on Alfeld refinements for $d = 3$, $k \geq 3$; we do not use his proof.

Lemma 4.6 ((IS3) on Alfeld refinements, uniformly in h). *Let $\mathcal{T}_h$ be the Alfeld refinement of a conforming macro mesh $\mathcal{T}_h^M$ of Ω_h with shape-regularity constant σ , and $k \geq 3$. For every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there is $v \in V_h^0$ with $\operatorname{div} v = q$ and*

$$\|\nabla v\| \leq [C_{\text{loc}} + C_{\Pi} C_B(\Omega_h) (1 + \sqrt{3} C_{\text{loc}})] \|q\|,$$

where $C_{\Pi} = C_{\Pi}(\sigma)$ is the constant of the Fortin operator in step 1. By Lemma 4.3, (IS3) holds with $\beta^{-1} \leq C_{\text{loc}} + 2C_{\Pi} C_B(\Omega_h) (1 + \sqrt{3} C_{\text{loc}})$, independent of h .

Proof. (1) *A local Fortin operator for the Bernardi–Raugel pair.* Let I_{SZ} be the Scott–Zhang P_1 interpolant on $\mathcal{T}_h^M$ that preserves homogeneous boundary values [34]. For $u \in H_0^1(\Omega_h)^3$, $|I_{SZ}u|_{1,K} \leq C|u|_{1,\omega_K}$ and $\|u - I_{SZ}u\|_K \leq Ch_K|u|_{1,\omega_K}$, with $C = C(\sigma)$ and ω_K the patch of K . For every interior face F of $\mathcal{T}_h^M$ let b_F be its cubic face bubble; $b_F n_F$ is piecewise P_3 on the Alfeld split and continuous, so it lies in V_h^0 because $k \geq 3$. Set

$$\Pi u := I_{SZ}u + \sum_{F \text{ interior}} \alpha_F b_F n_F, \quad \alpha_F := \frac{\int_F (u - I_{SZ}u) \cdot n_F}{\int_F b_F}.$$

Then $\int_F \Pi u \cdot n_F = \int_F u \cdot n_F$ on interior faces, and both vanish on boundary faces. With $|\alpha_F| \leq C|F|^{-1/2}\|u - I_{SZ}u\|_{L^2(F)}$ and the scaled trace inequality, $|\Pi u|_{1,K} \leq C(\sigma)|u|_{1,\omega'_K}$, hence $\|\nabla \Pi u\| \leq C_{\Pi}\|\nabla u\|$ by finite overlap. No constant of the domain enters: every estimate is on a patch, and $u \in H_0^1$ is used only through the preserved zero trace.

- (2) *Macro means.* Let $\bar{q}$ be the L^2 projection of q onto macro-piecewise constants, so $\bar{q} \in L_0^2(\Omega_h)$ and $\|\bar{q}\| \leq \|q\|$. Take $u \in H_0^1(\Omega_h)^3$ with $\operatorname{div} u = \bar{q}$ and $\|\nabla u\| \leq C_B(\Omega_h)\|\bar{q}\|$, and $v_1 := \Pi u \in V_h^0$. By the divergence theorem and the flux property, $\int_K \operatorname{div} v_1 = \int_K q$ for every $K \in \mathcal{T}_h^M$, and $\|\nabla v_1\| \leq C_{\Pi} C_B(\Omega_h)\|q\|$.
- (3) *Local fluctuations.* $r := q - \operatorname{div} v_1$ is piecewise P_{k-1} on each K^r and has zero mean on every K . Fact 4.5 gives $w_K \in \dot{P}_k^c(K^r)$ with $\operatorname{div} w_K = r|_K$ and $\|\nabla w_K\|_K \leq C_{\text{loc}}\|r\|_K$; extended by zero, $w := \sum_K w_K \in V_h^0$.
- (4) *Sum.* $v := v_1 + w$ satisfies $\operatorname{div} v = q$, and with $\|r\| \leq \|q\| + \sqrt{3}\|\nabla v_1\|$ the bound follows. $\square$

Remark 4.7 (Where the domain could enter). Only through $C_B(\Omega_h)$ in step 2. The local problem in step 3 is posed on one macro-element with zero data on its whole boundary, so boundary vertices of Ω_h play no role, and on Alfeld refinements there is no analogue of the two-dimensional singular-vertex condition. Subdivided wall faces do produce one (Proposition 4.9). The same combination (uniform continuous inf-sup via [2], Bernardi–Raugel face fluxes, and [18, Theorem 3.1]) is used by Chen and Liu [4, Lemma 3.9] for polyhedral approximations of curved domains.

Remark 4.8 ((IS3) on other meshes). For the Scott–Vogelius pair with discontinuous P_{k-1} pressures in three dimensions, the known stability results are for split meshes. On Alfeld refinements they hold for $k \geq 3$ [37, 18]; see [11, Table 2.1]. On Freudenthal meshes stability holds for $k \geq 6$ [38]; such meshes cannot be fitted to an inscribed surface. According to [11], Neilan [25, Proposition 6.5] extends Zhang’s $k \geq 6$ result to more general meshes satisfying a condition that he states as a conjecture. We know of no unconditional result on general unsplit tetrahedral meshes, and on such meshes (IS3) remains an assumption.

Worsey–Farin splits are different. The stable Worsey–Farin Stokes pairs of Guzmán, Lischke and Neilan [17] and of Fabien, Guzmán, Neilan and Zytoon [10] do *not* use the full space of discontinuous piecewise- P_{k-1} pressures: the pressure space carries conditions at the singular edges of the split. With boundary conditions, [10] impose $\theta_e(q) = q^{(1)} - q^{(2)} = 0$ on the boundary singular edges (and an alternating-sum condition on interior ones). So (IS3) *as formulated* fails on Worsey–Farin splits (Proposition 4.9); this is known, and it is the three-dimensional analogue of the two-dimensional singular boundary vertex. None of our upper bounds, which all use (IS3)

for the full space Π_h , applies to Worsey–Farin meshes as they stand; they would have to be redone with a constrained pressure space, together with a uniform inf-sup constant for that pair on the perturbed domains Ω_h (as Lemma 4.6 provides for Alfeld) and a lift lemma onto the constrained trace space of Proposition 4.9.

Proposition 4.9 ((IS3) fails on subdivided wall faces). *Let a wall face $F \in \mathcal{F}_h^\Gamma$ of a macro-element be subdivided into coplanar sub-faces, and let E be an edge of that subdivision interior to F . For $v \in V_h^0$, $\operatorname{div} v$ is continuous across the interior face that meets F along E , at every point of E . Consequently $\operatorname{div} V_h^0 \subsetneq \Pi_h \cap L_0^2$ for discontinuous P_{k-1} pressures, and (IS3) is false. On one barycentric Worsey–Farin macro-element (interior point at the barycentre, face points at the face barycentres, 12 sub-tetrahedra) with $v = 0$ on the whole boundary, $\operatorname{codim} \operatorname{div} V_h^0 = 12k - 4$ exactly for $k = 2, 3, 4$ (ranks 27/47, 87/119, 195/239), which is $4 \times (3k - 1)$: k conditions on each of the three edges from the face point to the vertices of a face, one of them redundant at the face point. [COMPUTER-ASSISTED, EXACT] for the count.*

Proof. Let s_1, s_2 be the sub-tetrahedra sharing the interior face through E , both with a face in the plane of F , and $x \in E$. Then $\nabla v|_{s_j}(x)t = 0$ for t in the plane of F , since $v = 0$ there, and $\nabla v|_{s_1}(x)d = \nabla v|_{s_2}(x)d$ for a direction d in the common face transversal to F , by continuity across that face. These directions span $\mathbb{R}^3$, so $\nabla v|_{s_1}(x) = \nabla v|_{s_2}(x)$ and $\operatorname{div} v$ is continuous at x . A discontinuous piecewise- P_{k-1} function with a jump along E is therefore not in $\operatorname{div} V_h^0$. $\square$

Computation: `notes/v7/td/wf_witness.py 2|3|4`; output `wf_ranks.log`.

The velocities of GS and CNS* with the full space Π_h are still pointwise divergence-free when they exist, but the pressure is not unique. Moreover the trace space of Z_h on a subdivided face is a proper subspace of the continuous piecewise- P_k zero-flux traces: along E the jump of $\operatorname{div}_F v_F - (d_F \cdot \nabla_F)(v \cdot n_F)/d_n$ must vanish, where $d = d_F + d_n n_F$ is the direction of the interior face. So Lemma 7.2 does not transfer either.

Lemma 4.10 (Divergence range). *Under (IS3), $\operatorname{div} V_h^R = \Pi_h$ and $W_h \neq \emptyset$, and the velocities of GS and CNS* are pointwise divergence-free.*

Proof. $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ is (IS3). For $F \in \mathcal{F}_h^\Gamma$, the cubic face bubble b_F of the tetrahedron T_F gives $b_F n_F \in V_h^R$ ($k \geq 3$), and $\int_{\Omega_h} \operatorname{div}(b_F n_F) = \int_F b_F \neq 0$; this adds the constants. The rest is as in two dimensions [29]: for any $\mathcal{G} \in V_h$ with trace g_I on Σ there is $v \in V_h^R$ with $\operatorname{div} v = \operatorname{div} \mathcal{G}$, so $\mathcal{G} - v \in W_h$, and the continuity row forces $\operatorname{div} u_h = 0$. $\square$

5. ERROR ANALYSIS

Throughout this section (IS3) holds; on Alfeld refinements with $k \geq 3$ it is Lemma 4.6.

5.1. Consistency.

Lemma 5.1 (Error equation). *For $v \in V_h^R$,*

$$N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v), \quad \mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v).$$

Proof. Integrate $a_h(\tilde{u}, v) = (D(\tilde{u}), \nabla v)$ by parts on Ω_h ($v = 0$ on Σ), use $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$, $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$, and $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$. $\square$

Proposition 5.2 (The transposed-gradient term). *On $F \in \mathcal{F}_h^\Gamma$, $(\nabla \tilde{u})^\top n_F = n(x^*)(\partial_n u(x^*) \cdot (n_F - n(x^*))) + O(h_\Gamma^2) = O(h_\Gamma)$. Hence, for $\gamma \geq \gamma_0$ and $v \in V_h^R$,*

$$|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq Ch_\Gamma \|v\|_{L^1(\Gamma_h)} \leq Ch_\Gamma (h/\mu)^{1/2} \|v\| = C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|.$$

Proof. By Lemma 3.13, $(\nabla u)^\top n_F = n(\partial_n u \cdot n_F) = n(\partial_n u \cdot (n_F - n))$ on Γ . Taylor-expand $\nabla \tilde{u}$ from x^* to x (distance $O(h_\Gamma^2)$) and use $|n_F - n(x^*)| \leq Ch_\Gamma$ (Theorem 3.4). Then use $|\Gamma_h| \leq C$ and $\|v\|_{\Gamma_h}^2 \leq (h/\mu)\|v\|^2$. $\square$

In the $\|\nabla v\|$ -dual norm the transposed term is generically only of order h_Γ in three dimensions: on the sphere, testing with the interpolant of $\chi\psi n$ gives $\sum_F |F|\psi(x_F^*) \partial_n u(x_F^*) \cdot (x_c - o_F) + O(h_\Gamma^2)$ by Lemma 3.2(c), and the two-dimensional odd-symmetry cancellation is not available. The energy-dual bound above is all that the error analysis uses.

Lemma 5.3 (Geometric consistency). *For $\gamma \geq \gamma_0$: $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$ for $v \in V_h^R$, and $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$ for $v \in V_h^0$.*

Proof. The transposed term is Proposition 5.2. By Lemmas 3.12 and 4.1, $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 C_I (c_0 h_\Gamma)^{-1/2} \|\nabla v\|$; $\frac{\mu}{h} |\langle \tilde{u}, v \rangle| \leq (\mu/h)^{1/2} Ch_\Gamma^2 \|v\| = C\gamma^{1/2} h_\Gamma^{3/2} \|v\|$; and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$. On V_h^0 the first and third terms vanish. $\square$

Lemma 5.4 (Fluxes). *For $v \in Z_h$, $\sum_i \int_{\Gamma_{h,i}} v \cdot n = 0$, where $\Gamma_{h,i} := \pi^{-1}(\Gamma_i) \cap \Gamma_h$. If $m \geq 2$, the map $v \mapsto (\int_{\Gamma_{h,i}} v \cdot n)_i$ is onto $\{x \in \mathbb{R}^m : \sum_i x_i = 0\}$ for small h .*

Proof. The first statement is $\int_{\partial\Omega_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$ with $v = 0$ on Σ . For the second, apply Lemma 5.9 below with $q = \alpha_i$ constant on Γ_i , $\sum \alpha_i |\Gamma_i| = 0$: its field has flux $-|\Gamma_i| \alpha_i + O(h_\Gamma^2 + h^k) |\alpha|$ through $\Gamma_{h,i}$, by Lemma 3.1(c). $\square$

Corollary 5.5 (GS on Z_h). *For $v \in Z_h$ and every constant c , $N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v)$ with $\mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle$.*

Remark 5.6 (The constant is global). By Lemma 5.4, $\int_{\Gamma_{h,i}} v \cdot n$ is free on Z_h for each component separately when $m \geq 2$; only the total vanishes. So the constant in $\mathcal{R}$ can only be a single global one, and the relevant size of the wall pressure is $G = \|p - \bar{p}\|_{L^2(\Gamma)}$ with the global mean $\bar{p}$, not the sum of per-component variances. Correspondingly, the correction in CNS* uses the global mean $\bar{p}_\Gamma$; per-component means would make the method inconsistent. The two-dimensional statements and proofs are in [29]; they are dimension-free.

5.2. Approximation and the leak field. Let

$$\varepsilon_h^{(3)} := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{k+1} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

Lemma 5.7 (Approximation in W_h). *$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h^{(3)}$. For bounded γ the second term is at most Ch^k if $h_\Gamma \geq h^2$. If the outer data have zero net discrete flux, $\int_\Sigma g_I \cdot \nu = 0$, it reduces to $C(1 + \gamma^{1/2})h_\Gamma^{k+1/2}$. Every $z \in W_h$ satisfies $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |\int_\Sigma g_I \cdot \nu| / |\Gamma_h|^{1/2}$.*

Proof. Let $w_h := I_h \tilde{u}$ be the Lagrange interpolant. Boundary tetrahedra have diameter at most Ch_Γ , so $\|\nabla(\tilde{u} - w_h)\| \leq Ch^k$, $\|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$ and $\|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$, whence $\|\tilde{u} - w_h\| \leq C(1 + (\gamma h_\Gamma)^{1/2}) h^k$ and $\|\operatorname{div} w_h\| \leq Ch^k$. *Net flux.* Since $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_\Sigma g \cdot \nu = 0$, $m := \int_{\Omega_h} \operatorname{div} w_h = \int_\Sigma (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n$, and $|m| \leq Ch^{k+1}$ by pointwise interpolation estimates. (Lagrange interpolation on triangles has no extra quadrature exactness in general, unlike the one-dimensional Newton–Cotes case of [29].) *Bubble.* $b := \sum_{F \in \mathcal{F}_h} b_F n_F$ satisfies $\int_{\Gamma_h} b \cdot n = c_b |\Gamma_h|$ with $c_b > 0$, $\|\nabla b\| + \|\operatorname{div} b\| \leq Ch_\Gamma^{-1/2}$ (there are $O(h_\Gamma^{-2})$ bubbles, each with H^1 -seminorm squared $O(h_\Gamma)$), $\|b\|_{\Gamma_h} \leq C$ and $\sum_F \operatorname{diam} F \|\partial_n b\|_F^2 \leq Ch_\Gamma^{-1}$. With $a := m/(c_b |\Gamma_h|)$, $\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2})$. *Correction.* $q := \operatorname{div}(w_h - ab) \in \Pi_h \cap L_0^2$ with $\|q\| \leq C\varepsilon_h^{(3)}$; (IS3) gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|v\| \leq (1 + C_I^2)^{1/2} \beta^{-1} \|q\|$. Then $z := w_h - ab - v \in W_h$. The lower bound holds because $\int_{\Gamma_h} z \cdot n = -\int_\Sigma g_I \cdot \nu$ for $z \in W_h$, while $\int_{\Gamma_h} \tilde{u} \cdot n = 0$. $\square$

Lemma 5.8 (A divergence-free field with normal trace q). *Let $q \in C^{k+1}(\Gamma)$ with $\int_\Gamma q = 0$. There is $w \in H^{k+1}(\Omega_h \cup \Omega)^3$, of class C^{k+1} on $\mathcal{T}_{r/16}$, with $\operatorname{div} w = 0$, $w = 0$ near Σ , and $w|_\Gamma = qN = -qn$.*

Proof. Let $\bar{q}_i$ be the mean of q on Γ_i and $\phi_i \in C^{k+2,\alpha}(\Gamma_i)$ a solution of $\Delta_\Gamma \phi_i = q - \bar{q}_i$ (Schauder theory on the closed C^{k+3} surface). Let $\chi \in C^\infty(\mathbb{R})$ with $\chi = 1$ on $(-\infty, r/8]$ and $\chi = 0$ on $[r/4, \infty)$. On $\Gamma \times (-r, r)$ with volume $dA ds$ put $V := (-\chi'(s)\nabla_\Gamma \phi_i, \chi(s)q)$, whose product divergence is $-\chi'\Delta_\Gamma \phi_i + \chi'q = \chi'\bar{q}_i$. Let w_0 be its Piola push-forward by Ψ , i.e. $\iota_{w_0} \operatorname{vol} = (\Psi^{-1})^* \iota_V (dA \wedge ds)$. Since $\Psi^* \operatorname{vol} = J dA \wedge ds$, $\operatorname{div} w_0 = J^{-1} \chi'(s) \bar{q}_i$; at $s = 0$, $D\Psi$ is the identity on $T\Gamma$ plus $\partial_s \mapsto N$, $J = 1$ and $\chi' = 0$, so $w_0 = qN$ on Γ . Regularity: $\nabla_\Gamma \phi_i \in C^{k+1,\alpha}$ and $D\Psi, J \in C^{k+1}$, so $w_0 \in C^{k+1}$. $g := \operatorname{div} w_0 \in C^k$ is supported in $\{r/8 \leq d \leq r/4\} \subset \Omega$, and $\int g = \sum_i \bar{q}_i |\Gamma_i| (\chi(r/4) - \chi(0)) = -\int_\Gamma q = 0$. Let $U \subset \{x \in \Omega : d(x) > r/16, \operatorname{dist}(x, \Sigma) > \delta\}$ be a connected Lipschitz domain containing $\operatorname{supp} g$ in its interior (it exists because Ω is connected and deformation-retracts onto such sets). A Bogovskii field $w_1 \in H_0^{k+1}(U)^3$ with $\operatorname{div} w_1 = -g$ exists [6, 15]. Put $w := w_0 + w_1$ (extended by zero); $w = w_0$ on $\mathcal{T}_{r/16}$. If $m = 1$ or q has zero mean on each component, $w_1 = 0$. $\square$

For the model case and $q = (p - \bar{p})|_\Gamma$ this is the explicit field $w = \chi(r)r^{-2}q(\omega)e_r - \chi'(r)r^{-1}\nabla_{S^2}\phi_0(\omega)$ in spherical coordinates, with $\Delta_{S^2}\phi_0 = q$.

From now on $q := (p - \bar{p})|_\Gamma$, which requires $p|_\Gamma \in C^{k+1}$, and w is the field of Lemma 5.8.

Lemma 5.9 (Discrete leak field). *Let b and c_b be as in the proof of Lemma 5.7, and*

$$v_\varphi := I_h w - ab - v_0, \quad a := \frac{\int_{\Gamma_h} I_h w \cdot n}{c_b |\Gamma_h|},$$

with $v_0 \in V_h^0$ given by (IS3) with $\operatorname{div} v_0 = \operatorname{div}(I_h w - ab)$. Then $v_\varphi \in Z_h$ and

$$\begin{aligned} \|v_\varphi - w\|_{L^\infty(\Gamma_h)} &\leq Ch_\Gamma^{k+1}, & \|\nabla(v_\varphi - w)\| &\leq Ch^k, \\ \|\partial_n(v_\varphi - w)\|_{\Gamma_h} &\leq Ch^k h_\Gamma^{-1/2}, & \sum_F \operatorname{diam} F \|\partial_n(v_\varphi - w)\|_F^2 &\leq Ch^{2k}. \end{aligned}$$

Proof. Since $\operatorname{div} w = 0$ on Ω_h and $w = 0$ near Σ , $\int_{\Gamma_h} w \cdot n = 0$. So $|a| \leq C \|I_h w - w\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$ (boundary tetrahedra have diameter $\leq Ch_\Gamma$ and $w \in C^{k+1}$ there), and $\operatorname{div}(I_h w - ab) \in \Pi_h \cap L_0^2$ with norm at most $\sqrt{3} \|\nabla(I_h w - w)\| + \sqrt{3} |a| \|\nabla b\| \leq Ch^k$. Hence $\|\nabla v_0\| \leq Ch^k$; as $v_0 = 0$ on Γ_h , Lemma 4.1 gives $\|\partial_n v_0\|_{\Gamma_h} \leq Ch^k h_\Gamma^{-1/2}$ and $\sum_F \operatorname{diam} F \|\partial_n v_0\|_F^2 \leq Ch^{2k}$. The terms $I_h w - w$ and ab contribute $O(h_\Gamma^{k+1})$ pointwise on Γ_h , $O(h^k)$ in H^1 and $O(h_\Gamma^k)$ pointwise for ∂_n . $\square$

These bounds replace the $W^{1,\infty}$ bound for the curl of a C^1 stream-function interpolant used in two dimensions [29], in every use below.

Lemma 5.10 (Normal-load cancellation). *Let $S \in W^{1,\infty}$ on a neighbourhood of Γ_h with $|S - (S \cdot n_F)n_F| \leq Ch_\Gamma$ on every $F \in \mathcal{F}_h^\Gamma$. (For instance $S = w$, since $w = qN + O(d)$ and $|n_F - n| \leq Ch_\Gamma$.) Then for $v \in Z_h$*

$$|\langle S, \partial_n v \rangle| \leq C (\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C \|\nabla v\|,$$

with C depending on $\|S\|_{W^{1,\infty}}$, k and the mesh constants.

Proof. Tangential part. $|\langle S - (S \cdot n_F)n_F, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$ by Lemma 4.1. *Normal part.* On F , v is a polynomial on T_F with $\operatorname{div} v = 0$, and n_F is constant; so $\partial_n v \cdot n_F = -\operatorname{div}_F v_F$, where v_F is the part of v tangential to F and div_F the in-plane divergence. With $\sigma_F := S \cdot n_F$ and $\nu_{\partial F}$ the in-plane outward conormal,

$$\int_F \sigma_F \partial_n v \cdot n_F = \int_F \nabla_F \sigma_F \cdot v_F - \int_{\partial F} \sigma_F v \cdot \nu_{\partial F},$$

and the first term is at most $C\|v\|_{L^1(F)}$. *Edge terms.* Each edge E of Γ_h lies in exactly two faces F, F' , and v is continuous across E , so the two contributions combine to $\int_E v \cdot (\sigma_F \nu_{E,F} + \sigma_{F'} \nu_{E,F'})$. With t_E a unit tangent, $\nu_{E,F} = t_E \times n_F$ and $\nu_{E,F'} = -t_E \times n_{F'}$, so $\nu_{E,F} + \nu_{E,F'} = t_E \times (n_F - n_{F'}) = O(h_\Gamma)$; and $\sigma_F - \sigma_{F'} = S \cdot (n_F - n_{F'}) = O(h_\Gamma)$. So the coefficient is $O(h_\Gamma)$. *Summing.* $\|v\|_{L^1(\partial F)} \leq C(\text{diam } F)^{-1} \|v\|_{L^1(F)}$ for $v|_F \in P_k$, so $\sum_E h_\Gamma \|v\|_{L^1(E)} \leq C\|v\|_{L^1(\Gamma_h)} \leq C\|v\|_{\Gamma_h}$. Finish with Lemma 3.11. $\square$

If boundary faces of $\mathcal{T}_h$ subdivide the faces of Γ_h (for example on Worsey–Farin splits), the extra edges lie in a plane, where $n_F = n_{F'}$ and the conormals are opposite, so their contributions cancel exactly. On such meshes, however, (IS3) as formulated fails (Proposition 4.9), so the results of this section do not apply to them as they stand.

5.3. The printed method. Throughout this subsection $\gamma \geq \gamma_0$. By Theorem 3.4, Lemma 3.1(c) and $\tilde{p}(x) = p(x^*) + O(h_\Gamma^2)$ on Γ_h ,

$$\|\tilde{p} - \bar{p}\|_{\Gamma_h} \leq G(1 + Ch_\Gamma^2) + Ch_\Gamma^2, \quad \|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h_\Gamma^{k+1}), \quad \|v_\varphi\|^2 = \frac{\mu}{h} \|v_\varphi\|_{\Gamma_h}^2 + O(1). \quad (1)$$

Theorem 5.11 (GS: energy norm, exact rate $1/2$). $\|\tilde{u} - u_h\| \leq C[(h_\Gamma/\gamma)^{1/2}G + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$. If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$,

$$\|\tilde{u} - u_h\| \geq (2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C(1 + \gamma^{1/2})h_\Gamma^{3/2}.$$

Here $(h_\Gamma/\gamma)^{1/2} = (h/\mu)^{1/2}$.

Proof. Upper bound. For $z \in W_h$, $e_h := z - u_h \in Z_h$, and by Corollary 5.5 with $c = \bar{p}$ and Lemma 4.2, $c_1\|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h)$. By (1), $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|e_h\|$; apply Lemmas 5.3 and 5.7. *Lower bound.* On F , $v_\varphi \cdot n_F = w(x^*) \cdot n_F + O(h_\Gamma^2) = -q(x^*)n(x^*) \cdot n_F + O(h_\Gamma^2)$ and $n(x^*) \cdot n_F = 1 + O(h_\Gamma^2)$; with Lemma 3.1(c), $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$. By continuity $M\|\tilde{u} - u_h\|\|v_\varphi\| \geq |\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|$, and $|\mathcal{G}(v_\varphi)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v_\varphi\|$. By (1), $|\mathcal{R}(v_\varphi)|/\|v_\varphi\| \geq \frac{1}{2}(h/\mu)^{1/2}G$ for $h \leq h_0(G)$. $\square$

Theorem 5.12 (GS: H^1 rate 1). With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h}{\mu} + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}].$$

Proof. Split. For $z \in W_h$ write $z - u_h = \delta_R + \delta_G$ with $\delta_R, \delta_G \in Z_h$ solving $N_h(\delta_R, v) = \mathcal{R}(v)$ and $N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v)$ for all $v \in Z_h$. Then $\|\nabla \delta_G\| \leq \|\delta_G\| \leq c_1^{-1}(C(1 + \gamma^{1/2})h_\Gamma^{3/2} + ME_h)$. *Ansatz.* Put $r_h := \delta_R - (h/\mu)v_\varphi \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$m(v) := -\langle (\tilde{p} - \bar{p})n + v_\varphi, v \rangle, \quad \ell(v) := -\frac{h}{\mu}[a_h(v_\varphi, v) - \langle \partial_n v_\varphi, v \rangle - \langle v_\varphi, \partial_n v \rangle].$$

Bound on m . On F , $(\tilde{p} - \bar{p})n_F + v_\varphi = q(x^*)(n_F - n(x^*)) + O(h_\Gamma^2) = O(h_\Gamma)$, with nonzero face means in general (Lemma 3.2(c)). So we use only the energy-dual bound $|m(v)| \leq C_p h_\Gamma \|v\|_{L^1(\Gamma_h)} \leq C_p h_\Gamma (h/\mu)^{1/2} \|v\|$. *Bound on ℓ .* $|a_h(v_\varphi, v)| \leq C\|\nabla v\|$; $|\langle \partial_n v_\varphi, v \rangle| \leq (C + Ch^k h_\Gamma^{-1/2})\|v\|_{\Gamma_h} \leq (C_p + Ch^k h_\Gamma^{-1/2})\|\nabla v\|$; and by Lemma 5.10 with $S = w$ and Lemma 5.9, $|\langle v_\varphi, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + Ch_\Gamma^{k+1} \|\partial_n v\|_{L^1(\Gamma_h)} \leq C_p \|\nabla v\|$. So $|\ell(v)| \leq [C_p(h/\mu) + C(h/\mu)h^k h_\Gamma^{-1/2}]\|\nabla v\|$, and $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1}h^k h_\Gamma^{1/2} \leq Ch^k$. *Close.* Testing with r_h , $c_1\|r_h\| \leq C_p(h_\Gamma(h/\mu)^{1/2} + h/\mu) + Ch^k$, and $h_\Gamma(h/\mu)^{1/2} \leq \frac{1}{2}(h_\Gamma^2 + h/\mu)$ gives $\|r_h\| \leq C_p(h/\mu + h_\Gamma^2) + C\varepsilon_h^{(3)}$. Hence $\|\nabla \delta_R\| \leq (h/\mu)\|\nabla v_\varphi\| + \|r_h\| \leq C_p(h/\mu + h_\Gamma^2) + C\varepsilon_h^{(3)}$, and $h_\Gamma^2 \leq h_\Gamma^{3/2}$. $\square$

The rate is 1 and not $1/2$ because $\operatorname{div} v = 0$ turns the normal part of the missing traction into a tangential derivative (Lemma 5.10), as in two dimensions.

Theorem 5.13 (GS: boundary slip and H^1 lower bound). *Let $G > 0$ and $X := C_p(h_\Gamma/\gamma) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}$. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h_\Gamma/\gamma \leq \delta G, \quad h_\Gamma \leq h_0(G).$$

All three hold for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

- (i) $c(h_\Gamma/\gamma)G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2}\|\tilde{u} - u_h\|$;
- (ii) $\|\nabla(\tilde{u} - u_h)\| \geq c(h_\Gamma/\gamma)G - Ch^{k+1/2}$.

Proof. Let $e_u := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 5.5 gives

$$\frac{\mu}{h}\langle e_u, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e_u, v) + \langle \partial_n e_u, v \rangle + \langle e_u, \partial_n v \rangle.$$

(1) $\mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k)$. (2) $|\mathcal{G}(v)| \leq C(h_\Gamma + \gamma h_\Gamma^3 + h_\Gamma^{3/2})$: the transposed term is $O(h_\Gamma)$ pointwise (Proposition 5.2); in the penalty term, $\tilde{u} = dW(x^*) + O(d^2)$ is tangential to Γ while $v = -qn(x^*) + O(h_\Gamma^2)$ is normal, so $\tilde{u} \cdot v = O(h_\Gamma^4)$. By the hypotheses, $|\mathcal{G}(v)| \leq \delta G^2$. (3) $|a_h(e_u, v)| \leq C\|\nabla e_u\| \leq CX \leq C\delta G^2$ by Theorem 5.12. (4) By Theorem 5.11, $|\langle e_u, \partial_n v \rangle| \leq C_p\|e_u\|_{\Gamma_h} \leq C_p(h/\mu)^{1/2}\|e_u\| \leq C_p[(h/\mu)G + X] \leq 2C_p\delta G^2$. (5) With Lemma 4.1, $|\langle \partial_n e_u, v \rangle| \leq (\|\partial_n(\tilde{u} - z)\|_{\Gamma_h} + Ch_\Gamma^{-1/2}\|\nabla(z - u_h)\|)\|v\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}XG \leq C\delta G^2$. For δ small, $|\langle e_u, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$; divide by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$. For (ii) let $E \in H^1(\Omega_h)^3$ lift $(g - g_I)|_\Sigma$, vanishing near Γ_h , with $\|E\|_{H^1} \leq Ch^{k+1/2}$; then $\|e_u\|_{\Gamma_h} = \|e_u - E\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e_u\| + \|\nabla E\|)$. $\square$

Corollary 5.14 (Leak law). *Let $G > 0$ and $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h^{(3)}/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$. Then*

$$\tilde{u} - u_h = \frac{h}{\mu}v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

To leading order the printed method replaces no-slip by the leak condition $(\mu/h)u_h \cdot n \approx p - \bar{p}$, with the global mean $\bar{p}$.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - z) + \delta_G + \delta_R$ and $\delta_R = (h/\mu)v_\varphi + r_h$ as above, so $\vartheta_h = (\tilde{u} - z) + \delta_G + r_h$. Relative to $(h/\mu)^{1/2}G$: $\|r_h\| \leq C_p(h/\mu + h_\Gamma^2) + C\varepsilon_h^{(3)}$ contributes $O((h/\mu)^{1/2} + (\gamma h_\Gamma^3)^{1/2} + \varepsilon_h^{(3)}(\gamma/h_\Gamma)^{1/2}) \rightarrow 0$, and $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$ contributes $O((\gamma^{1/2} + \gamma)h_\Gamma + \varepsilon_h^{(3)}(\gamma/h_\Gamma)^{1/2}) \rightarrow 0$. The Γ_h norms carry an extra factor $(h/\mu)^{1/2}$. The limits follow from (1). $\square$

For fixed γ and bounded ρ , Theorems 5.12 and 5.13 give $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$. Section 7 identifies the constant: $\|\nabla(\tilde{u} - u_h)\|\mu/(hG) \rightarrow \|\nabla U\|/G$, with U the Stokes flow driven by the leak as Dirichlet data (Corollary 7.16), and the leak limit holds for every $\gamma \geq \gamma_0$ (Theorem 7.10).

5.4. The corrected method.

Proposition 5.15 (The plain traction is singular). *If $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ is replaced by $\langle p_h, v \cdot n \rangle$, the square system $(V_h^R \times \Pi_h)$ tested by $V_h^R \times \Pi_h$ has the kernel vector $(0, 1)$ and is singular.*

Proof. For $v \in V_h^R$, $-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle = -\int_{\partial\Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0$, since $v = 0$ on Σ . $\square$

The L_0^2 variant, and the relation to [13], are discussed in [29]; the arguments there are dimension-free.

Theorem 5.16 (CNS*: well posed, with Scott's rate). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution and*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}].$$

Consequently, for $\gamma \in [\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h^2$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k),$$

and with outer data of zero discrete net flux the proviso $h_\Gamma \geq h^2$ is not needed. On Alfled refinements this holds for every $k \geq 3$.

Proof. Consistency. Let $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$. For a constant c , $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$. With $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, Lemma 5.1 gives $N_h(\tilde{u}, v) + B^*(p^*, v) = (\tilde{f}, v) + \mathcal{G}(v)$, so with $e_u := \tilde{u} - u_h$, $e_p := p^* - p_h$,

$$N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v) \quad \forall v \in V_h^R. \quad (2)$$

Split. $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ (π_h the L^2 projection onto Π_h) and $\xi \in \Pi_h$; $(\eta, \operatorname{div} v) = 0$ for $v \in V_h^R$, and $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$. *Pressure.* For $v \in V_h^0$, $(\xi - \bar{\xi}, \operatorname{div} v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v)$, where $\bar{\xi}$ is the mean over Ω_h . With $|\langle e_u, \partial_n v \rangle| \leq (\rho/\mu)^{1/2} C_I \|e_u\| \|\nabla v\|$ and Lemma 5.3, (IS3) gives $\beta \|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$. *Velocity.* For $z \in W_h$, $e_h := z - u_h \in Z_h$. Testing (2) with e_h , and using $(e_p, \operatorname{div} e_h) = 0$ and $\int_{\Gamma_h} e_h \cdot n = 0$,

$$c_1 \|e_h\|^2 \leq ME_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|,$$

with $|\langle \eta, e_h \cdot n \rangle| \leq Ch_\Gamma^k (h/\mu)^{1/2} \|e_h\|$ and, by Lemma 4.1 and $\rho/\mu \leq (c_0 \gamma)^{-1}$, $|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|$. *Absorb.* Insert the pressure bound and $\|e_u\| \leq E_h + \|e_h\|$; for $\gamma \geq \gamma_2$ with $C_I (c_0 \gamma_2)^{-1/2} C \beta^{-1} \leq c_1/2$ the pressure term is absorbed. *Uniqueness.* For zero data, $z = 0$, so $u_h = 0$; then p_h is constant, and (2) reads $-c \int_{\Gamma_h} v \cdot n = 0$ for all v , so $c = 0$ by the face bubble of Lemma 4.10. The system is square. *H^1 bound.* For bounded γ the flux term of $\varepsilon_h^{(3)}$ is $Ch^{k+1} h_\Gamma^{-1/2} \leq Ch^k$ iff $h_\Gamma \geq h^2$; with zero net outer flux it is $C(1 + \gamma^{1/2}) h_\Gamma^{k+1/2}$ (Lemma 5.7). The L^2 part uses the lift E of Theorem 5.13 and Lemma 3.11. On Alfled refinements (IS3) is Lemma 4.6. $\square$

Remark 5.17 (Flux-corrected outer data). In the model case one may replace g_I by $\tilde{g}_I := g_I - \frac{m_R}{3|R|} x$ with $m_R := \int_{\partial R} g_I \cdot \nu$; since $\int_{\partial R} x \cdot \nu = 3|R|$, this has zero net flux. In two dimensions the corresponding H^1 bound holds uniformly in γ [29]. In three dimensions Corollary 7.16 gives a γ -uniform H^1 bound for GS with g_I itself (the flux term appears as $h^{k+1} h_\Gamma^{-1/2}$); for CNS* we have not carried the uniformity over.

6. SHARPNESS OF $h_\Gamma^{3/2}$

Theorem 5.16 gives $\|\nabla(\tilde{u} - u_h)\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$ for CNS*, and Theorem 5.11 the same for GS when $G = 0$. We prove the reverse inequality whenever $|\mathfrak{h}| \partial_N u \not\equiv 0$ on Γ . The lower bound does not use (IS3), except to guarantee that u_h exists.

6.1. An abstract lower bound. Let $Y_h := Z_h \cap V_h^0$ and

$$Y_h^1 := \left\{ v \in Y_h : \int_F \partial_n v \, dA = 0 \quad \forall F \in \mathcal{F}_h^\Gamma \right\}.$$

For $F \in \mathcal{F}_h^\Gamma$ let T_F be the tetrahedron of $\mathcal{T}_h$ with face F ; the T_F are distinct (Lemma 3.10).

Lemma 6.1. *If $v \in Y_h$ and $F \in \mathcal{F}_h^\Gamma$, then on F (trace from T_F) $\nabla v = \partial_n v \otimes n_F$ and $\partial_n v \cdot n_F = 0$.*

Proof. $v|_{T_F}$ is a polynomial vanishing on the plane of F , so its tangential derivatives vanish there, and $0 = \operatorname{div} v = \partial_n v \cdot n_F$. $\square$

Lemma 6.2 (Error identity on Y_h). *Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$) or of GS ($\gamma \geq \gamma_0$, any G), and $e_u := \tilde{u} - u_h$. For $v \in Y_h$, $a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$.*

Proof. By (2), resp. Corollary 5.5, $N_h(e_u, v) = \mathcal{G}(v)$ on Y_h : the pressure terms vanish because $\operatorname{div} v = 0$ and $v = 0$ on Γ_h . Since $v = 0$ on Γ_h , $N_h(e_u, v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle$, and the transposed and penalty parts of $\mathcal{G}(v)$ vanish. $\square$

Lemma 6.3 (Trace control on Y_h^1). *Let C_{PT} be the constant in the scaled Poincaré trace inequality $\|w - \bar{w}_T\|_{L^2(F)} \leq C_{PT}(\operatorname{diam} T)^{1/2} \|\nabla w\|_T$ for a face F of a tetrahedron T , where $\bar{w}_T$ is the mean of w over T . Then $|\langle w, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla w\| \|\nabla v\|$ for $w \in H^1(\Omega_h)^3$ and $v \in Y_h^1$.*

Proof. $\int_F w \cdot \partial_n v = \int_F (w - \bar{w}_{T_F}) \cdot \partial_n v$; use Cauchy–Schwarz on each F , the local form of Lemma 4.1, and Cauchy–Schwarz over the distinct T_F . $\square$

Corollary 6.4 (Abstract lower bound). *With $\Lambda_0 := 2 + C_{PT} C_I$,*

$$\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} \frac{|\langle \tilde{u}, \partial_n v \rangle + (F, v)|}{\|\nabla v\|}.$$

Proof. $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$, and Lemmas 6.2 and 6.3. $\square$

Neither the pressure nor the penalty enters: the identity is the discrete momentum equation on Y_h , which $\tilde{u}$ violates through Nitsche’s symmetric term acting on the nonzero trace of $\tilde{u}$ on Γ_h .

6.2. The leading term: three edge moments per face. Let $W := \partial_N u|_\Gamma$, a tangential field (Lemma 3.13), and $W_F := W(y_F)$. For $F \in \mathcal{F}_h^\Gamma$ put $w_{ij}^F := \mathfrak{h}_F(e_{ij}, e_{ij})$, $Q_F := \frac{1}{2} \sum_{i < j} w_{ij}^F \mu_i \mu_j$ (so $d = Q_F + O(\ell^3)$ on F by Lemma 3.1(a)), and for $v \in Y_h$

$$\mathcal{M}_F(v) := \int_F Q_F \partial_n v \, dA = \frac{1}{2} \sum_{i < j} w_{ij}^F \int_F \mu_i \mu_j \partial_n v \, dA, \quad \ell_W(v) := \sum_{F \in \mathcal{F}_h^\Gamma} W_F \cdot \mathcal{M}_F(v).$$

By Lemma 6.1, $\mathcal{M}_F(v)$ is parallel to F . In the model case $w_{ij}^F = -\ell_{ij}^2$ (Lemma 3.2).

Lemma 6.5 (Leading term). *For $v \in Y_h$, $|\langle \tilde{u}, \partial_n v \rangle - \ell_W(v)| \leq Ch_\Gamma^{5/2} \|\nabla v\|$ and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.*

Proof. On Γ_h , $x = x^* + d(x)N(x^*)$ and $\tilde{u}(x^*) = 0$, so $\tilde{u}(x) = d(x)W(x^*) + O(d^2)$. With $|d - Q_F| \leq Ch_\Gamma^3$, $|W(x^*) - W_F| \leq Ch_\Gamma$ and $|Q_F| \leq Ch_\Gamma^2$, $\tilde{u} = Q_F W_F + R$ on F with $|R| \leq Ch_\Gamma^3$. Hence $\int_F \tilde{u} \cdot \partial_n v = W_F \cdot \mathcal{M}_F(v) + O(h_\Gamma^3) \|\partial_n v\|_{L^1(F)}$, and $\sum_F \|\partial_n v\|_{L^1(F)} \leq \sum_F |F|^{1/2} C_I (\operatorname{diam} F)^{-1/2} \|\nabla v\|_{T_F} \leq Ch_\Gamma^{-1/2} \|\nabla v\|$ since $\#\mathcal{F}_h^\Gamma \leq Ch_\Gamma^{-2}$. The bound on (F, v) is Lemma 3.12. $\square$

6.3. The local condition and the lower bound.

Assumption 6.6 ((M3³)). There are $\kappa_0 > 0$ and K_{ov}, C_s, C_1 such that for every $F \in \mathcal{F}_h^\Gamma$ and every unit $t \in T_{y_F} \Gamma$ there is $v_{F,t} \in Y_h^1$ with

- (i) $\|\nabla v_{F,t}\| = 1$;
- (ii) $\operatorname{supp} v_{F,t} \subset \omega_F$, a union of tetrahedra within distance $C_s h_\Gamma$ of F , each tetrahedron lying in at most K_{ov} of the sets ω_F ;
- (iii) $t \cdot \sum_{F' \subset \omega_F \cap \Gamma_h} \mathcal{M}_{F'}(v_{F,t}) \geq \kappa_0 |\mathfrak{h}_{y_F}| (\operatorname{diam} F)^{5/2} - C_1 (\operatorname{diam} F)^{7/2}$.

Condition (iii) is scale invariant: under $x \mapsto \lambda x$ (with v renormalised) $\|\nabla v\|$ scales like $\lambda^{1/2}$, w^F like λ^2 and $\mathcal{M}_F$ like λ^3 . Since (iii) is linear in v , only the sign of v is at our disposal; statements about $-\mathcal{M}_F$ are equivalent. The vector t is tangent to Γ at y_F , not to F ; this is the form in which the witnesses are assembled.

Theorem 6.7 (H^1 lower bound). *Assume the setting of Section 2, (M^3) , and $|\mathfrak{h}|W \not\equiv 0$. Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$) or of GS ($\gamma \geq \gamma_0$, any G). Then for $h_\Gamma \leq h_1$*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \| |\mathfrak{h}| W \|_{L^2(\Gamma)} - C h_\Gamma^2,$$

with $c = c(\kappa_0, K_{\text{ov}}, \Lambda_0, c_0, \theta_0)$ and C independent of γ, μ, ρ, h , the pressure and $\varepsilon_h^{(3)}$. Here h_1 depends also on $C_s, C_1, \|W\|_{W^{1,\infty}}$ and $\| |\mathfrak{h}| W \|_{L^2}$. In the model case $|\mathfrak{h}| = 1$.

Proof. For each F with $W_F \neq 0$ let $t_F := W_F/|W_F| \in T_{y_F}\Gamma$ and $v_F := |\mathfrak{h}_{y_F}| |W_F| v_{F,t_F}$ (else $v_F := 0$), and $v := \sum_F v_F \in Y_h^1$. By linearity,

$$\ell_W(v) = \sum_F \left[W_F \cdot \sum_{F'} \mathcal{M}_{F'}(v_F) + \sum_{F'} (W_{F'} - W_F) \cdot \mathcal{M}_{F'}(v_F) \right],$$

the inner sums running over the (boundedly many) faces $F' \subset \omega_F$. Since $|\mathcal{M}_{F'}(v)| \leq C h_\Gamma^2 |F'|^{1/2} \|\partial_n v\|_{L^2(F')} \leq C h_\Gamma^{5/2} \|\nabla v\|$ and $|W_{F'} - W_F| \leq C C_s h_\Gamma \|W\|_{W^{1,\infty}}$, (M^3) (iii) gives

$$\ell_W(v) \geq \kappa_0 c_0^{5/2} h_\Gamma^{5/2} \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2 - C h_\Gamma^{7/2} \sum_F |\mathfrak{h}_{y_F}| |W_F| (|W_F| + \|W\|_{W^{1,\infty}}),$$

and $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2$. The projections $\pi(F)$ tile Γ with area Jacobian $1 + O(h_\Gamma^2)$ (Theorem 3.4, Lemma 3.1(c)), and $|\mathfrak{h}|, |W|$ are Lipschitz, so $S := \sum_F |F| |\mathfrak{h}_{y_F}|^2 |W_F|^2 = \| |\mathfrak{h}| W \|_{L^2(\Gamma)}^2 + O(h_\Gamma)$; with $c_A h_\Gamma^2 \leq |F| \leq h_\Gamma^2$ (c_A from c_0, θ_0) we get $h_\Gamma^{-2} S \leq \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2 \leq (c_A h_\Gamma^2)^{-1} S$, and the error sum is at most $C \# \mathcal{F}_h^\Gamma \leq C h_\Gamma^{-2}$. Hence

$$\frac{\ell_W(v)}{\|\nabla v\|} \geq \kappa_0 c_0^{5/2} \left(\frac{c_A}{K_{\text{ov}}} \right)^{1/2} h_\Gamma^{3/2} S^{1/2} - C \frac{h_\Gamma^{5/2}}{S^{1/2}},$$

and for $h_\Gamma \leq h_1$ the second term is at most half the first and $S \geq \frac{3}{4} \| |\mathfrak{h}| W \|^2$. Conclude with Corollary 6.4 and Lemma 6.5. $\square$

On flat parts of Γ the faces lie in Γ and the chordal bump vanishes, so the weight $|\mathfrak{h}|$ cannot be removed.

6.4. Witnesses by affine transport. Let T be a tetrahedron with a face F and apex p_0 , $\hat{T} = \text{conv}(\hat{p}_0, \dots, \hat{p}_3)$ with $\hat{p}_0 = e_3$, $\hat{p}_1 = 0$, $\hat{p}_2 = e_1$, $\hat{p}_3 = e_2$ and $\hat{F} = \hat{T} \cap \{\hat{x}_3 = 0\}$, and $x = A\hat{x} + a$ the affine map with $\hat{F} \mapsto F$, $\det A > 0$. Let $\nu = -n_F$ be the inward unit normal of T on F and h_T the height of T over F . For $\hat{v}$ vanishing on $\hat{F}$ let $v := (\det A)^{-1} A \hat{v} \circ A^{-1}$ (Piola) and $\hat{g} := \partial_{\hat{\nu}} \hat{v}|_{\hat{F}}$.

Lemma 6.8 (Piola transport). (a) *The Piola map preserves continuity, zero boundary values and piecewise- P_k structure on affinely corresponding subdivisions, and $\text{div } v = (\det A)^{-1} (\widehat{\text{div}} \hat{v}) \circ A^{-1}$; it maps the Alfeld split of $\hat{T}$ onto that of T .* (b) *For every polynomial weight ϕ in the barycentrics of F , $\int_F \phi \partial_\nu v dA = h_T^{-2} A \int_{\hat{F}} \phi \hat{g} d\hat{A}$.* (c) $\|\nabla v\|_T \leq (\det A)^{-1/2} \|A\| \|A^{-1}\| \|\hat{\nabla} \hat{v}\|_{\hat{T}}$.

Proof. (a) is standard (A is constant and maps barycentre to barycentre). (b) On $\hat{F}$, $\hat{\nabla} \hat{v} = \hat{g} \otimes e_3$, so $\partial_\nu v = (\det A)^{-1} A \hat{g} (A^{-\top} e_3 \cdot \nu)$ with $A^{-\top} e_3 = |A^{-\top} e_3| \nu$. By Nanson's formula $dA = \det A |A^{-\top} e_3| d\hat{A}$, so the factor is $|A^{-\top} e_3|^2 = |\nabla \lambda_0|^2 = h_T^{-2}$. (c) $\nabla v = (\det A)^{-1} A (\hat{\nabla} \hat{v}) A^{-1}$. $\square$

Since $\partial_n = -\partial_\nu$ on F , $\mathcal{M}_F(v) = -\frac{1}{2} \sum w_{ij} \int_F \mu_i \mu_j \partial_\nu v$; the sign is absorbed by $v \mapsto -v$.

Proposition 6.9 (Joint criterion). *Let $\hat{\mathcal{Y}}$ be a space of continuous piecewise-polynomial divergence-free fields on a subdivision of $\hat{T}$ vanishing on $\partial\hat{T}$, $\hat{\mathcal{Y}}^1 := \{\hat{v} \in \hat{\mathcal{Y}} : \int_{\hat{F}} \hat{g} = 0\}$, and suppose the joint edge-moment map*

$$\hat{E} : \hat{\mathcal{Y}}^1 \rightarrow (\mathbb{R}^2)^3, \quad \hat{v} \mapsto \left(\int_{\hat{F}} \mu_1 \mu_2 \hat{g}, \int_{\hat{F}} \mu_1 \mu_3 \hat{g}, \int_{\hat{F}} \mu_2 \mu_3 \hat{g} \right)$$

is onto, with right inverse $\hat{R}$ (Euclidean norm on $\mathbb{R}^6$ to $\|\hat{\nabla} \cdot\|_{L^2(\hat{T})}$). Then for every tetrahedron T with face F , every real weight vector $w \neq 0$ and every unit $t \parallel F$ there is v in the Piola image, supported in T , with $\|\nabla v\| = 1$ and

$$t \cdot \frac{1}{2} \sum_{i < j} w_{ij} \int_F \mu_i \mu_j \partial_\nu v \geq \frac{|w| (\det A)^{1/2}}{2h_T^2 \|A\| \|A^{-1}\|^2 \|\hat{R}\|} \geq c(\text{shape}) \frac{|w| (\text{diam } F)^{1/2}}{\|\hat{R}\|}.$$

Proof. $s := A^{-1}t$ is parallel to $\hat{F}$. Take $\hat{v} := \hat{R}((w_{ij}s/|w|^2)_{i < j})$, so that $\sum_{i < j} w_{ij} \hat{E}_{ij}(\hat{v}) = s$ and $\|\hat{\nabla} \hat{v}\| \leq \|\hat{R}\| \|A^{-1}\|/|w|$. By Lemma 6.8(b), $\frac{1}{2} \sum w_{ij} \int_F \mu_i \mu_j \partial_\nu v = \frac{1}{2} h_T^{-2} A s = \frac{1}{2} h_T^{-2} t$; normalise with (c). Shape regularity gives $\|A\| \lesssim \text{diam } T$, $\|A^{-1}\| \lesssim (\text{diam } T)^{-1}$, $\det A \gtrsim (\text{diam } T)^3$ and $h_T \lesssim \text{diam } T \lesssim \text{diam } F$. $\square$

No compactness over shapes is used: $\hat{\mathcal{Y}}^1$ and $\hat{E}$ are fixed, and the shape enters only through A and w .

Theorem 6.10 ((M3³) on Alfeld refinements, $k \geq 4$). *Let $\mathcal{T}_h$ be an Alfeld refinement and $k \geq 4$. Then (M3³) holds with $\omega_F = T_F$ (the macro-tetrahedron containing F), $K_{\text{ov}} = 1$ and $\kappa_0 = c(\text{shape regularity}, \theta_0)/\|\hat{R}\|$. [COMPUTER-ASSISTED, EXACT]*

Proof. Let $\hat{\mathcal{Y}}_k$ be the continuous piecewise- P_k divergence-free fields on the Alfeld split of $\hat{T}$ vanishing on $\partial\hat{T}$. An exact rational computation in the Bernstein–Bézier basis (35 interior domain points, 105 unknowns, 80 divergence equations) gives, for $k = 4$: the divergence has rank 79 (onto the mean-free piecewise P_3), $\dim \hat{\mathcal{Y}}_4 = 26$, and the eight functionals ($\int_{\hat{F}} \hat{g}$ and the three edge moments, two tangential components each) have rank 8 on $\hat{\mathcal{Y}}_4$. So $\hat{E}$ is onto on $\hat{\mathcal{Y}}_4^1$ ($\dim = 24$), and $\hat{\mathcal{Y}}_4 \subset \hat{\mathcal{Y}}_k$. By Lemma 6.8(a),(b) the Piola image on T_F , extended by zero, lies in Y_h^1 : it is continuous and piecewise P_k on the Alfeld refinement, divergence-free, zero on $\partial T_F \supset F$, has $\int_F \partial_n v = 0$, and $T_F \cap \Gamma_h = F$ (Lemma 3.10), so no other face constraint or moment is affected. For unit $t \in T_{y_F} \Gamma$ apply Proposition 6.9 to $t' := P_F t / |P_F t|$, P_F the projection onto the plane of F ; $|P_F t| \geq \frac{1}{2}$ because the angle between $T_{y_F} \Gamma$ and F is $O(h_\Gamma)$, and $t \cdot \mathcal{M}_F = |P_F t| t' \cdot \mathcal{M}_F$ because $\mathcal{M}_F \parallel F$. Finally $|w^F| \geq c\ell^2 |\mathfrak{h}_{y_F}| - C\ell^3$ (Lemma 3.1(d)). $\square$

Computation: `notes/v6/td.sharp/alfeld.witness.py 4`; output `alfeld.k4.log`.

The rank computation was repeated independently (`notes/v6/referee6/aw.py`), and the Piola identity of Lemma 6.8(b) was checked symbolically on a random affine map. In floating point, the singular values of $\hat{E}$ on $\hat{\mathcal{Y}}_4^1$ in the H^1 -seminorm are 0.240, 0.175, 0.123, 0.069, 0.0211, 0.0109, so $\|\hat{R}\| = 91.7$ (`k4.constant.py`); the certified value is $\|\hat{R}\| = 91.678155469$, and κ_0 is explicit (Proposition 9.1; $\kappa_0 = 2.11 \cdot 10^{-3}$ for a regular macro-tetrahedron).

Proposition 6.11 ((M3³) on any mesh, $k \geq 9$). *Let $K = T_F$, λ_0 its barycentric coordinate vanishing on F , $B := (\lambda_1 \lambda_2 \lambda_3)^2$, $\tau \parallel F$ a unit vector and $w \neq 0$ real weights. With $Q_w := \sum_{i < j} w_{ij} \mu_i \mu_j$ written in $\lambda_1, \lambda_2, \lambda_3$, $\bar{Q}_w := \int_F B Q_w / \int_F B$ and $\psi := \lambda_0^2 B (Q_w - \bar{Q}_w)(\nu \times \tau)$, the field $v := \text{curl } \psi \in P_9^3$, extended by zero, lies in Y_h^1 and*

$$\frac{1}{2} \sum_{i < j} w_{ij} \int_F \mu_i \mu_j \partial_\nu v = -|\nabla \lambda_0|^2 \tau \int_F B (Q_w - \bar{Q}_w)^2, \quad \frac{1}{|F|} \int_F B (Q_w - \bar{Q}_w)^2 \geq \frac{|w|^2}{8316000}.$$

Hence (M^3) holds for $k \geq 9$ on every shape-regular mesh, with $\omega_F = T_F$ and $K_{\text{ov}} = 1$.

Proof. ψ vanishes to second order on ∂K , so v vanishes on ∂K , is continuous after extension by zero, and is divergence free. With $f := \lambda_0^2 B(Q_w - \bar{Q}_w)$, $\text{curl}(f\Phi) = \nabla f \times \Phi + f \text{curl} \Phi$, and on F only the term with $2\lambda_0 \nabla \lambda_0$ survives one normal derivative: $\partial_\nu v|_F = 2|\nabla \lambda_0|^2 B(Q_w - \bar{Q}_w) \nu \times (\nu \times \tau) = -2|\nabla \lambda_0|^2 B(Q_w - \bar{Q}_w) \tau$, which has mean zero. The identity is $\int B(Q_w - \bar{Q}_w) Q_w = \int B(Q_w - \bar{Q}_w)^2$. The lower bound is $w^\top \mathcal{V} w$, where $\mathcal{V} = \frac{1}{1188000} I - \frac{1}{2772000} (J - I)$ is the exact B -weighted covariance matrix of the three edge bubbles on F (J the all-ones matrix), with eigenvalues $1/831600$ (twice) and $1/8316000$. By scaling, $\|\nabla v\| \leq C|w|(\text{diam } K)^{-1/2}$ and the moment is at least $c|w|^2|F|/h_K^2$, so the ratio is at least $c|w|(\text{diam } F)^{1/2}$. Transfer to $t \in T_{y_F} \Gamma$ as in Theorem 6.10. $\square$

Computation: `notes/v6/td_sharp/bubble_variance.py`; output `bubble_variance.log`.

Proposition 6.12 ($k = 3$: one macro-element is not enough). *On the Alfeld split with $k = 3$, $\dim \hat{\mathcal{Y}}_3 = 6$, $\dim \hat{\mathcal{Y}}_3^1 = 4$, and the image of $\hat{E}$ on $\hat{\mathcal{Y}}_3^1$ is the line spanned by $(\hat{p}_1 - \hat{p}_2, \hat{p}_3 - \hat{p}_1, \hat{p}_2 - \hat{p}_3)$. Consequently, for a field supported in one macro-tetrahedron T_F , the physical moment $\mathcal{M}_F(v)$ is a multiple of $\sum_{\text{cyc}} w_{ij} e_{ij}$; on the sphere this is a multiple of $R_{\pi/2}(x_c - o_F)$ (Lemma 3.2(d)), which vanishes on equilateral faces, and it spans one direction only. So (M^3) cannot be verified with $\omega_F = T_F$ for $k = 3$. [COMPUTER-ASSISTED, EXACT]*

Computation: `notes/v6/td_sharp/alfeld_witness.py` 3; output `alfeld_k3.log`.

6.5. Degree three: edge stars. Throughout this subsection $\mathcal{T}_h$ is an Alfeld refinement with minimal angle θ of the macro mesh, and $k = 3$ (larger k follow, since the witnesses are also piecewise P_k and the moments do not depend on k).

Edge stars. Let z be a vertex of Γ_h and q a macro vertex with $[z, q]$ a macro edge and $q \notin \Gamma_h$. For the apex q_F of T_F this holds for each vertex z of F : q_F has distance at least $c \text{diam } F$ from the plane of F , on the side of Ω_h , while Γ_h lies within $C(\text{diam } F)^2$ of that plane near F (as in the proof of Theorem 3.6). The relative interior of $[z, q]$ lies in Ω_h , so the *edge star* $R = R(z, q)$, the union of the macro-tetrahedra containing $[z, q]$, has a link that is a cycle $x_0, \dots, x_{m-1}$ with $3 \leq m \leq m_{\max}(\theta)$:

$$T_j = \text{conv}(q, z, x_j, x_{j+1}), \quad f_j = \text{conv}(q, z, x_j) \text{ (interior faces)}, \quad F_j := \text{conv}(z, x_j, x_{j+1}).$$

The faces of the T_j on Γ_h are among the F_j ; let $I := \{j : F_j \in \mathcal{F}_h^\Gamma\}$. Let $Y(R)$ be the continuous piecewise- P_3 divergence-free fields on the Alfeld refinement of R vanishing on ∂R , and $Y^1(R) := \{v \in Y(R) : \int_{F_i} \partial_\nu v = 0, i \in I\}$. Extension by zero maps $Y^1(R)$ into Y_h^1 .

Traces. Fix $i \in I$, write $F = F_i = \text{conv}(z, a, b)$, $T = T_i$ with barycentre B , and $H := \text{dist}(q, \text{plane of } F)$. The Alfeld sub-tetrahedra of T at z are $s_1 = \text{conv}(B, z, a, b) \supset F$, $s_2 = \text{conv}(B, q, z, b)$ and $s_3 = \text{conv}(B, q, z, a)$. Since $v|_{s_1}$ vanishes on F , $\nabla v|_{s_1} = \hat{g} \otimes \nabla \lambda_B^{s_1}$ on F with $\hat{g}(x) := \partial_{B-x} v(x) \in P_2(F; \mathbb{R}^3)$, and $\partial_\nu v = \frac{4}{H} \hat{g}$. In this subsection we work with

$$\hat{\mathcal{M}}_F(v) := \frac{4}{H} \int_F Q_F \hat{g} dA = -\mathcal{M}_F(v),$$

which is harmless because only the sign of the witnesses changes. With m_{xy} the midpoint of $[x, y]$ and n_S a normal of the plane of S , define

$$U_F := \{\hat{g} \in P_2(F; \mathbb{R}^3) : \hat{g} \cdot n_F \equiv 0, \hat{g}(a) = \hat{g}(b) = 0, \hat{g}(m_{ab}) \cdot n_{qab} = 0\} \quad (\dim 7),$$

$$S_F := \{\hat{g} \in U_F : \hat{g}(z) = 0, \hat{g}(m_{za}) \cdot n_{f_a} = 0, \hat{g}(m_{zb}) \cdot n_{f_b} = 0\} \quad (\dim 3),$$

with $f_a = \text{conv}(q, z, a)$, $f_b = \text{conv}(q, z, b)$. The map $\pi_F : U_F \rightarrow T_F \times \mathbb{R} \times \mathbb{R}$, $\hat{g} \mapsto (\hat{g}(z), \hat{g}(m_{za}) \cdot n_{f_a}, \hat{g}(m_{zb}) \cdot n_{f_b})$, is onto with kernel S_F (here T_F also denotes the tangent plane of F).

Lemma 6.13 (Necessary relations). *Let $v \in Y(R)$ and $\delta := \partial_{q-z}v(z)$, which is common to all T_j . Then:*

- (a) $\hat{g}_i \in U_{F_i}$ for every $i \in I$.
- (b) On f_j , in its barycentrics, $v|_{f_j} = \lambda_q \lambda_z (p_q \lambda_q + \delta \lambda_z + p_j \lambda_x)$ with $p_q \in \mathbb{R}^3$ common to all j . The fluxes $A_j \cdot (\frac{\delta + p_q}{30} + \frac{p_j}{60})$, $A_j := \frac{1}{2}(q - z) \times (x_j - z)$, are equal for all j . Along $[z, x_j]$, $h_j := \partial_{q-x}(v|_{f_j})$ satisfies $h_j(m_{zx_j}) = \frac{1}{4}(\delta + p_j)$.
- (c) $\delta_i := \partial_{B-z}v|_{T_i}(z)$ satisfies $\delta_i \cdot \nabla \lambda_B^{s_1} = 0$ and $\delta \cdot \nabla \lambda_q^s + \delta_i \cdot \nabla \lambda_B^s = 0$ for $s = s_2, s_3$, and $\hat{g}_i(z) = \delta_i$.
- (d) $\hat{g}_i \cdot \nabla \lambda_B^{s_3} + h_i \cdot \nabla \lambda_q^{s_3} = 0$ on $[z, a]$, and likewise on $[z, b]$ with s_2 and h_{i+1} .

Proof. (a) At a , the three edges of T at a lie in ∂R , so in each sub-tetrahedron at a , $\nabla v(a) = \delta' \otimes \nabla \lambda_B$ with a common δ' ; $\operatorname{div} v = 0$ makes δ' orthogonal to three independent normals, so $\hat{g}(a) = 0$, and likewise $\hat{g}(b) = 0$. On $[a, b]$, in $\operatorname{conv}(B, q, a, b)$, $\nabla v = \hat{g} \otimes \nabla \lambda_B$ and $\operatorname{div} v = 0$ gives $\hat{g} \cdot n_{qab} = 0$; tangency is $\operatorname{div} v|_{s_1} = \hat{g} \cdot \nabla \lambda_B^{s_1} = 0$. (b) $v|_{f_j} \in P_3$ vanishes on $[z, x_j]$ and $[q, x_j]$, and its restriction to $[z, q]$ is shared. With $\int_T \lambda_q^2 \lambda_z = |T|/30$ and $\int_T \lambda_q \lambda_z \lambda_x = |T|/60$, and $\int_{\partial T_j} v \cdot n = 0$ with $\partial T_j \setminus \partial R = f_j \cup f_{j+1}$, the fluxes agree. (c), (d): at z in s_2 , v vanishes on $[z, b]$, so $\nabla v(z) = \delta \otimes \nabla \lambda_q + \delta_i \otimes \nabla \lambda_B$ with zero trace; similarly in s_3 , s_1 and along $[z, a]$. $\square$

Lemma 6.14 (Single macro-element). *For one Alfeld macro-tetrahedron T with $v = 0$ on ∂T , the trace space $\{\hat{g}\}$ is exactly S_F , and $\hat{g} \mapsto \int_F \hat{g}$ has rank 2 on S_F . [COMPUTER-ASSISTED]*

Proof. $\subseteq$: Lemma 6.13 with $\delta = 0$ and $h \equiv 0$; then $\delta_i = 0$. $\supseteq$: the trace rank on $Y(T)$ is at least 3 (a rank computation modulo a prime, which is a lower bound for the rank over $\mathbb{Q}$; the rank 39 of the divergence is its a priori maximum). The Piola map multiplies $\hat{g}$ by a constant matrix and preserves S_F , so the statement is shape independent. The mean statement is exact on the reference triangle. $\square$

Computation: `notes/v6/td.k3/trace1.py`, `ref.K.py`; output `trace1.log`, `ref.K.log`.

Lemma 6.15 (Realisability). *Given $\delta, p_q, p_0, \dots, p_{m-1} \in \mathbb{R}^3$ with equal fluxes as in Lemma 6.13(b), and $s_i \in S_{F_i}$ ($i \in I$), there is $v \in Y(R)$ with these interior-face data such that, for each $i \in I$, $\hat{g}_i$ differs from a fixed element of U_{F_i} with π_{F_i} -value determined by (c), (d) by exactly s_i .*

Proof. Let v_0 be the continuous piecewise- P_3 field whose Bernstein coefficients on the domain points of the faces f_j are those of the prescribed $v|_{f_j}$ ($\delta/3$ at $\frac{q+2z}{3}$, $p_q/3$ at $\frac{2q+z}{3}$, $p_j/6$ at $\frac{q+z+x_j}{3}$), and zero elsewhere. Then $v_0 = 0$ on ∂R and $\int_{T_j} \operatorname{div} v_0 = 0$ by the equal fluxes. On each Alfeld macro-element, div maps the piecewise- P_3 fields vanishing on ∂T_j onto the mean-free piecewise P_2 (Fact 4.5; exact rank 39 on the reference element), which gives w_j with $\operatorname{div} w_j = -\operatorname{div} v_0$ on T_j and $w_j = 0$ on ∂T_j . $v_1 := v_0 + \sum_j w_j \in Y(R)$ has the prescribed data, and adding single-macro fields (Lemma 6.14) adjusts the S_{F_i} -components independently. $\square$

Theorem 6.16 (Ring Trace Theorem). *The joint trace space $\{(\hat{g}_i)_{i \in I} : v \in Y(R)\}$ consists exactly of the tuples with $\hat{g}_i \in U_{F_i}$ and $\pi_{F_i}(\hat{g}_i)$ given by Lemma 6.13(c), (d) for some data $(\delta, p_q, (p_j))$ with all fluxes equal.*

Proof. Lemma 6.13 and Lemma 6.15. $\square$

An exact implementation of this reduced model agrees with the full finite element computation in dimension and moment rank on eight configurations, including the rank-0 octahedral fan below (`notes/v6/td.k3/validate_reduced.log`).

For $i \in I$ let n_i be a unit normal of F_i , $P_i := I - n_i \otimes n_i$, $H_i := |(q - z) \cdot n_i|$, and

$$r_i := (a_i - z) + (b_i - z) + P_i(q - z), \quad G_i \delta := P_i \delta - \frac{n_i \cdot \delta}{(q - z) \cdot n_i} r_i, \quad c_i := \frac{|F_i| w_{ab}^{F_i}}{H_i},$$

where $w_{ab}^{F_i}$ is the weight of the edge of F_i opposite z .

Lemma 6.17 (Jet map). *In Lemma 6.13(c), $\hat{g}_i(z) = \delta_i = \frac{1}{4}G_i\delta$.*

Proof. The solution is unique, so it suffices to check $\delta = q - z$ and $\delta \in T_{F_i}$. For $\delta = q - z$: $\nabla \lambda_q^s \cdot (q - z) = 1$, and $y = -(B - z) + t(q - z)$ satisfies $\nabla \lambda_B^s \cdot y = -1$; tangency forces $t = \frac{1}{4}$ (heights $H/4$ and H), so $\delta_i = -\frac{1}{4}((a - z) + (b - z)) = \frac{1}{4}G_i(q - z)$. For $\delta \in T_{F_i}$: $\varphi := \frac{1}{4}\lambda_B^s + \lambda_q^s$ is affine, vanishes at z and at b (resp. a), and equals $\frac{1}{4}$ at B and 1 at q , so $\varphi = \text{dist}_\pm(\cdot, F)/H$ and $\nabla \varphi \perp T_F$; hence $\delta_i = \frac{1}{4}\delta$. $\square$

Lemma 6.18 (Face moment of the vertex representative). *Let $e \in T_F$. The set $\{\hat{g} \in U_F : \pi_F(\hat{g}) = (e, 0, 0), \int_F \hat{g} = 0\}$ is a line parallel to $S_F \cap \{\int \hat{g} = 0\}$, and it contains an element $r_F(e)$, linear in e , with $\int_F Q_F r_F(e) dA = -\frac{|F| w_{ab}^F}{360} e$. [COMPUTER-ASSISTED, EXACT]*

Proof. On the reference triangle (exact): a representative has moment Ke with $K = \frac{1}{720} \begin{pmatrix} -w_{ab} & 2(w_{ab} - w_{za}) \\ 0 & -3w_{ab} + 2w_{zb} \end{pmatrix}$ in the basis $(a - z, b - z)$, and the line direction has moment $D = \frac{1}{720}(w_{za} - w_{ab}, w_{ab} - w_{zb})$. Since $K \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 2D = -\frac{w_{ab}}{720} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $K \equiv -\frac{w_{ab}}{720}I$ modulo D ; fix r_F accordingly. Transport: $\int_F Q \hat{g} dA = 2|F|A \int_{\text{ref}} Q \hat{e}$ for $\hat{g} = A\hat{e}$. $\square$

Computation: `notes/v6/td_k3/ref_K.py`; independent re-derivation
`notes/v6/referee8/ref8_lemmaK.py`; output `ref_K.log`, `ref8_lemmaK.log`.

Theorem 6.19 (Vertex witness). *For every edge star $R(z, q)$ with $I \neq \emptyset$, any geometry and any real weights, and every $\delta \in \mathbb{R}^3$, there is $v_\delta \in Y^1(R)$ with*

$$\sum_{i \in I} \hat{\mathcal{M}}_{F_i}(v_\delta) = -\frac{1}{360} \sum_{i \in I} c_i G_i \delta, \quad \|\nabla v_\delta\|_R \leq C_E(\theta) |\delta| (\text{diam } R)^{1/2}.$$

Proof. Take $p_q = -\delta/2$ and $p_j = -\delta$ for all j . All fluxes are $A_j \cdot (\frac{\delta}{60} - \frac{\delta}{60}) = 0$ and $h_j(m_{zx_j}) = 0$, so both ν -components of π_{F_i} vanish (Lemma 6.13(d)). By Lemmas 6.15 and 6.17 there is $v \in Y(R)$ with $\pi_{F_i}(\hat{g}_i) = (\frac{1}{4}G_i\delta, 0, 0)$ for all $i \in I$. Subtracting single-macro fields (Lemma 6.14) makes $\hat{g}_i = r_{F_i}(\frac{1}{4}G_i\delta)$, which has mean zero, and Lemma 6.18 gives the moment. The energy bound: the Bernstein data are $O(|\delta|)$, and the Alfeld right inverse and the single-macro right inverses are fixed reference maps transported affinely, so their norms depend only on θ ($m \leq m_{\max}(\theta)$); scaling gives $(\text{diam } R)^{1/2}$. $\square$

Computation: `notes/v6/td_k3/vertex_formula_general.py`, `fe_vertex_check.py`;
`notes/v6/referee8/ref8_vertex_fe.py`; output `vertex_formula_general.log`,
`fe_vertex_check.log`, `ref8_vertex_fe_eqw.log`.

In the exact reduced model the formula holds in all 30 test cases. Where the weights are generic, the test “modulo the span of the single-macro directions” can be vacuous; the non-vacuous checks are the equal-weight cases and an independent full finite element computation on three non-flat stars inscribed in the sphere with distinct per-face equal weights, where the finite element moment agrees with the closed form to 10^{-15} .

Flat stars and the octahedral degeneracy. If all F_i lie in one plane T , then $G_i|_T = I$ and the tangential moment of v_δ is $-\frac{1}{360}(\sum_i c_i)\delta$, a multiple of the identity. For the octahedral fan ($z = e_3$, $q = 2e_3$, link $\pm e_1, \pm e_2$, all four faces in Γ_h , equal weights), $\sum_i c_i P_T G_i|_T = 0$ exactly and the full finite element rank is 0; its faces are tilted by $\arccos(1/\sqrt{3})$ against T . On shape-regular inscribed meshes with small h_Γ such tilts do not occur.

6.5.1. *The sphere: an explicit criterion.* For a vertex z of Γ_h with unit normal n_z of Γ and an edge star $R = R(z, q)$ put

$$\tau_z := \max_{i \in I} \angle(n_{F_i}, n_z), \quad \rho_z := \max_{i \in I} |r_i|/H_i.$$

On the unit sphere $\angle(n_F, z) = \arcsin r_F$ for every vertex z of F , so $\tau_z \leq C(\theta)h_\Gamma$, while $\rho_z \leq C(\theta)$.

Theorem 6.20 ((M3³) for $k = 3$ on the sphere, explicit). *In the model case, let $\mathcal{T}_h$ be an Alfeld refinement and $k \geq 3$. Suppose every $F \in \mathcal{F}_h^\Gamma$ has a vertex z with $\tau_z(\tau_z + \rho_z) \leq \frac{1}{2}$ for $R(z, q_F)$; this holds for $h_\Gamma \leq h_*(\theta)$. Then (M3³) holds for $h_\Gamma \leq h_*(\theta)$ with $\omega_F = R(z, q_F)$, $K_{\text{ov}}, C_s \leq C(\theta)$ and $\kappa_0 = c(\theta)/C_E(\theta)$.*

Proof. For unit $t' \in n_z^\perp$: $|n_i \cdot t'| \leq \sin \tau_z$ and $|P_i t'|^2 \geq 1 - \sin^2 \tau_z$, so

$$t' \cdot G_i t' = |P_i t'|^2 - \frac{(n_i \cdot t')(r_i \cdot t')}{(q - z) \cdot n_i} \geq 1 - \tau_z(\tau_z + \rho_z) \geq \frac{1}{2}.$$

Given a unit $t \in T_{y_F} \Gamma$, let t' be its normalised projection onto $n_z^\perp$; $|t - t'| \leq Ch_\Gamma$, so $t \cdot G_i t' \geq \frac{1}{2} - Ch_\Gamma(1 + \rho_z) \geq \frac{1}{4}$ for $h_\Gamma \leq h_*(\theta)$. On the sphere all c_i have the same sign ($w_{ab} = -\ell_{ab}^2$); take $\delta := \pm t'$ in Theorem 6.19 with the sign that makes $t \cdot \sum_i \hat{\mathcal{M}}_{F_i}(v_\delta) = \frac{1}{360} \sum_i |c_i| t \cdot G_i t' \geq \frac{1}{1440} |c_F| \geq c(\theta)(\text{diam } F)^3$, using $\ell_{ab} \geq c \text{diam } F$, $|F| \geq c(\text{diam } F)^2$ and $H \leq C \text{diam } F$. Divide by $\|\nabla v_\delta\| \leq C_E(\text{diam } F)^{1/2}$. A macro-tetrahedron lies in $R(z, q_F)$ only if $[z, q_F]$ is one of its six edges, so $K_{\text{ov}} \leq C(\theta)$. $\square$

The only non-explicit constant is $C_E(\theta)$, a finite combination of right-inverse norms of fixed reference maps. On icosphere meshes with a radial prism layer (Alfeld-refined), the criterion holds from the second or third level on, and the matrix $\sum_i c_i P_T G_i|_T$ is nonsingular at every level (notes/v6/td_k3/icosphere_check.log). [NUMERICAL]

6.5.2. *General surfaces.* On a general surface the weights w_{ab} of the vertex move can cancel, $\sum_i c_i = 0$, at saddle points. Two further families of moves in the Ring Trace Theorem repair this. For a face $F = F_i = \text{conv}(z, a, b)$ write $e_a = a - z$, $e_b = b - z$, $P_a = \delta + p_a$, $P_b = \delta + p_b$ (the spoke data), $u_a = \nabla \lambda_q^{s_3}$, $u_b = \nabla \lambda_q^{s_2}$, $w = (w_{za}, w_{zb}, w_{ab})$ its weights, and

$$D_F := (w_{za} - w_{ab})e_a + (w_{ab} - w_{zb})e_b,$$

the single-macro direction of Proposition 6.12.

Lemma 6.21 (Per-face moment). *For every $v \in Y^1(R)$ with interior-face data $(\delta, p_q, (p_j))$ and every $i \in I$,*

$$\hat{\mathcal{M}}_{F_i}(v) \equiv \frac{8|F_i|}{H_i} \left[-\frac{w_{ab}}{2880} G_i \delta - \frac{w_{za} - w_{zb}}{2880} \left((P_a \cdot u_a) e_b - (P_b \cdot u_b) e_a \right) \right] \pmod{\mathbb{R} D_{F_i}},$$

and every value of the right-hand side, plus any multiple of D_{F_i} chosen independently for each i , is attained. [COMPUTER-ASSISTED, EXACT]

Proof. On the reference triangle $z = 0$, $a = e_1$, $b = e_2$, for $\hat{g} \in U_F \cap \{\int \hat{g} = 0\}$ with $\hat{g}(z) = e$, $\tilde{\alpha} := \hat{g}(m_{za}) \cdot \nabla \mu_b$ and $\tilde{\beta} := \hat{g}(m_{zb}) \cdot \nabla \mu_a$, an exact computation gives $\int_{\text{ref}} Q \hat{g} \equiv -\frac{w_{ab}}{720} e + \frac{w_{za} - w_{zb}}{180} (\tilde{\alpha} \hat{e}_2 - \tilde{\beta} \hat{e}_1)$ modulo D . Under $\hat{g} = A \hat{g}_{\text{ref}}$ the quantities $\tilde{\alpha}, \tilde{\beta}$ are affine invariants, so $\int_F Q \hat{g} \equiv 2|F| \left[-\frac{w_{ab}}{720} \hat{g}(z) + \frac{w_{za} - w_{zb}}{180} (\tilde{\alpha} e_b - \tilde{\beta} e_a) \right]$ modulo $\mathbb{R} D_F$. By Lemma 6.17, $\hat{g}(z) = \frac{1}{4} G_i \delta$; by Lemma 6.13(d) with $h_i(m_{za}) = \frac{1}{4} P_a$, and since the tangential part of $\nabla \lambda_B^{s_3}$ equals $4 \nabla \mu_b$, $\tilde{\alpha} = -\frac{1}{16} P_a \cdot u_a$, and likewise $\tilde{\beta} = -\frac{1}{16} P_b \cdot u_b$. Multiply by $4/H_i$. Realisability is Theorem 6.16, and Lemma 6.14 leaves exactly the D_{F_i} -freedom on each face. $\square$

Computation: `notes/v6/td_k3b/ref_nu.py`, `face_formula.py`; output `ref_nu.log`, `face_formula.log` (29/29 faces; rerun with a fresh seed in `notes/v6/referee9`).

The check compares face by face modulo the single line $\mathbb{R}D_{F_i}$, so it is never vacuous.

A star is Γ -flat if its Γ -faces lie in one plane T . Put $z = 0$, $T = \{x_3 = 0\}$, $q = (q', H)$ with $H > 0$; let J be the rotation by $\frac{\pi}{2}$ in T , and orient so that $s_i \wedge s_{i+1} = 2|F_i| > 0$ for $i \in I$, where $s_j := x_j$ are the spokes. Let $\mathfrak{h}$ be a symmetric form on T with weights $w_{ij}^F = \mathfrak{h}(e_{ij}, e_{ij})$; put $\omega_j := \mathfrak{h}(s_j, s_j)$, $w_i := w_{ab}^{F_i}$, $\hat{\tau}_i := \omega_i - \omega_{i+1}$, write $P_j = (\pi_j, \nu_j) \in T \times \mathbb{R}$, and $\sigma_j := Js_j \cdot (\pi_j - \nu_j q'/H)$.

Lemma 6.22 (Flat reduction). *On a Γ -flat star:*

(a) $P_a \cdot u_a = -\sigma_a/(2|F|) + \nu_a/H$ and $P_b \cdot u_b = \sigma_b/(2|F|) + \nu_b/H$, hence, modulo $\text{span}\{D_{F_i}\}$,

$$\sum_{i \in I} \hat{\mathcal{M}}_{F_i} \equiv \frac{1}{360H} \sum_{i \in I} \left[-|F_i|w_i G_i \delta + \hat{\tau}_i \left(\frac{1}{2}(\sigma_i s_{i+1} + \sigma_{i+1} s_i) - \frac{|F_i|}{H}(\nu_i s_{i+1} - \nu_{i+1} s_i) \right) \right].$$

(b) The flux condition for spoke j reads $A_j \cdot P_j = \frac{H}{2}\sigma_j$. The admissible data are exactly: $\nu_j \in \mathbb{R}$ free for each Γ -spoke, and $\sigma_j = \gamma - s_j \wedge \eta$ with $(\gamma, \eta) \in \mathbb{R} \times \mathbb{R}^2$ free; the data δ , (ν_j) , (γ, η) and the D -coefficients are mutually independent.

Proof. (a) In the flat case $u_a = (-\nabla\mu_b, (1 + \nabla\mu_b \cdot q')/H)$, $u_b = (-\nabla\mu_a, (1 + \nabla\mu_a \cdot q')/H)$, $\nabla\mu_b = Js_a/(2|F|)$ and $\nabla\mu_a = -Js_b/(2|F|)$; substitute into Lemma 6.21. (b) $A_j = \frac{1}{2}q \times x_j = \frac{1}{2}(HJs_j, q' \wedge s_j)$ gives $A_j \cdot P_j = \frac{H}{2}\sigma_j$. Lemma 6.13(b) with $p_j = P_j - \delta$ gives $A_j \cdot P_j = 60\Phi - A_j \cdot \psi$ with $\psi := \delta + 2p_q$ and Φ the common flux, both free; so $\sigma_j = \frac{120\Phi}{H} - s_j \wedge (\psi' - \psi_n q'/H)$. Given ν_j and σ_j , the component $Js_j \cdot \pi_j$ is fixed; the component along s_j is invisible. $\square$

Computation: `notes/v6/td_k3b/flat_algebra.py`; output `flat_algebra.log`.

Definition 6.23. A Γ -face $F = \text{conv}(z, a, b)$ is *degenerate at z* if $w_{ab} = 0$ and $w_{za} = w_{zb}$.

Lemma 6.24. F is degenerate at z iff $\mathfrak{h} = 0$ or $\mathfrak{h} = \pm\nu \otimes \nu$ with $\nu \perp b - a$. In particular no face is degenerate at any vertex if $\det \mathfrak{h} \neq 0$, and if $\mathfrak{h} \neq 0$, F is nondegenerate at the vertex opposite any edge e with $\mathfrak{h}(e, e) \neq 0$; such an edge exists.

Proof. With $e = b - a$, $\mathfrak{h}(e_b, e_b) = \mathfrak{h}(e_a, e_a) + 2\mathfrak{h}(e_a, e) + \mathfrak{h}(e, e)$, so degeneracy means $\mathfrak{h}(e, e) = \mathfrak{h}(e_a, e) = 0$. If $\det \mathfrak{h} \neq 0$, the $\mathfrak{h}$ -orthogonal complement of a null vector e is $\mathbb{R}e$, forcing $e_a \parallel e$. If $\mathfrak{h} = \pm\nu \otimes \nu$ both conditions reduce to $\nu \cdot e = 0$. A quadratic form vanishing on three pairwise non-parallel vectors is zero. $\square$

Theorem 6.25 (Star Nondegeneracy). *Let R be a Γ -flat edge star, $\mathfrak{h} \neq 0$ a symmetric form on T with weights $\mathfrak{h}(e, e)$, and assume the Γ -spokes are pairwise distinct as directions. If some Γ -face of R is nondegenerate at z , then $v \mapsto P_T \sum_{i \in I} \hat{\mathcal{M}}_{F_i}(v)$ maps $Y^1(R)$ onto T .*

Proof. Suppose $0 \neq \lambda \in T$ annihilates the image; put $\ell_j := \lambda \cdot s_j$, $\mu := J\lambda$ and $\tau_i := |F_i|\hat{\tau}_i$, and write $\mathbf{1}_i = [i \in I]$. By Lemma 6.22 the families are independent, so λ annihilates each:

$$\begin{aligned} (\delta_T) \quad & \sum_{i \in I} |F_i|w_i = 0 \quad (\delta \in T, G_i|_T = I); \quad (D) \quad \lambda \cdot D_{F_i} = 0 \quad \forall i \in I; \\ (\nu_j) \quad & \tau_{j-1}\ell_{j-1}\mathbf{1}_{j-1} = \tau_j\ell_{j+1}\mathbf{1}_j \quad \text{for every } \Gamma\text{-spoke } j; \\ (\sigma) \quad & \sum_{i \in I} \hat{\tau}_i [\gamma(\ell_i + \ell_{i+1}) + L(s_i)\ell_{i+1} + L(s_{i+1})\ell_i] = 0 \quad \forall \gamma \in \mathbb{R}, L \in T^*. \end{aligned}$$

Step 1: $\tau \equiv 0$. Let $\varpi_i := \tau_i \ell_i \ell_{i+1}$. Multiplying (ν_j) by ℓ_j shows that ϖ is constant on each run of consecutive Γ -faces and zero at the ends of a non-cyclic run. If I is the whole cycle, (σ) with $\gamma = 0$, $L = \lambda \cdot$ gives $2 \sum_i \varpi_i / |F_i| = 0$. So $\varpi \equiv 0$, and $\tau_i = 0$ whenever $\ell_i \ell_{i+1} \neq 0$. Let Z be the set of Γ -spokes with $\ell_p = 0$, i.e. $s_p \parallel \mu$; $|Z| \leq 2$, two such spokes point in opposite directions, and no face has both spokes in Z . For $p \in Z$, (ν_p) defines $X_p := \tau_{p-1}\ell_{p-1}\mathbf{1}_{p-1} = \tau_p\ell_{p+1}\mathbf{1}_p$, which vanishes unless both adjacent faces are Γ -faces, and the only possibly nonzero τ 's are

$\tau_{p-1} = X_p/\ell_{p-1}$ and $\tau_p = X_p/\ell_{p+1}$. With $k_p := |F_{p-1}|^{-1} + |F_p|^{-1}$, (σ) with $L = 0$ and with $L = \mu \cdot$ gives $\sum_{p \in Z} k_p X_p = 0$ and $\sum_{p \in Z} (\mu \cdot s_p) k_p X_p = 0$. The numbers $\mu \cdot s_p$ are nonzero and of opposite signs if there are two, so $X_p = 0$. *Step 2:* $w_i = 0$ for all $i \in I$. Now $\omega_i = \omega_{i+1} =: \Omega_i$, and (D) reads $(\Omega_i - w_i)(\ell_i - \ell_{i+1}) = 0$. Suppose $w_i \neq 0$. If $w_i = \Omega_i$, then $\mathfrak{h}$ takes the same nonzero value c on s_i , s_{i+1} and $s_{i+1} - s_i$, so $\mathfrak{h} = c \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$ in that basis is definite; then all w_i have the sign of c , contradicting (δ_T) . Otherwise $\ell_i = \ell_{i+1}$, i.e. $s_{i+1} - s_i \parallel \mu$, and $w_i = |s_{i+1} - s_i|^2 \mathfrak{h}(\hat{\mu}, \hat{\mu})$. So all nonzero w_i have the sign of $\mathfrak{h}(\hat{\mu}, \hat{\mu})$, and (δ_T) forces them to vanish. Hence every Γ -face is degenerate at z , a contradiction. $\square$

An exact search over all real $\mathfrak{h} \neq 0$ (eliminating $\mathfrak{h}$ through 3×3 minors) on 7 structured and 3034 random Γ -flat stars found degeneracy only when every Γ -face is degenerate, and full finite element ranks confirm the predictions on seven stars, including a saddle with $\sum_i w_{ab} = 0$ (`notes/v6/td_k3b/degen*.log`, `fe_cross.log`). [NUMERICAL, EXACT ARITHMETIC]

For $F \in \mathcal{F}_h^\Gamma$ let z_F be the vertex of F opposite an edge e_F that maximises $|\mathfrak{h}_{y_F}(P_{y_F}e, P_{y_F}e)|$ over the edges of F , and $\omega_F := R(z_F, q_F)$.

Theorem 6.26 (Uniform witness, $k = 3$). *On Alfeld refinements there are $\kappa_0 = \kappa_0(\theta, \theta_0) > 0$, C_1 and h_* such that for $h_\Gamma \leq h_*$, (M^3) holds with $\omega_F = R(z_F, q_F)$ and $K_{\text{ov}}, C_s \leq C(\theta)$.*

Proof. Let $\ell := \text{diam } F$ and $\mathfrak{h}_0 := \mathfrak{h}_{y_F} \circ P_{y_F}$. (1) *Weights.* Every Γ -face F' of ω_F lies within $C\ell$ of y_F , so $w_{ij}^{F'} = \mathfrak{h}_0(e_{ij}, e_{ij}) + O(\ell^3)$ (Lemma 3.1(a), $\Gamma \in C^3$). Moments are linear in the weights with $|\mathcal{M}_{F'}(v)| \leq C|w|\ell^{1/2}\|\nabla v\|$, so replacing the weights costs $C\ell^{7/2}\|\nabla v\|$, the C_1 term. If $\mathfrak{h}_0 = 0$ there is nothing to prove; otherwise write $\mathfrak{h}_0 = |\mathfrak{h}_0|\hat{\mathfrak{h}}$ with $|\hat{\mathfrak{h}}| = 1$. (2) *The parameter map.* For admissible data $x = (\delta, p_q, (p_j), (s_i))$, Lemmas 6.15 and 6.14 give $v(x) \in Y^1(R)$ with $\|\nabla v(x)\| \leq C_E(\theta)|x|\ell^{1/2}$, where C_E is uniform for configurations in a compact set of normalised shapes. Let $\Pi(\mathcal{C}; \hat{\mathfrak{h}})$ be the linear map from admissible data to $\ell^{-3}P_t \sum \mathcal{M}_{F'}(v(x)) \in \mathbb{R}^2$, for the normalised configuration $\mathcal{C}$ and weights $\hat{\mathfrak{h}}(e, e)$. By Lemma 6.21, which holds for every geometry, Π is an explicit rational function of $\mathcal{C}$, linear in $\hat{\mathfrak{h}}$, on an admissible subspace of constant dimension (each flux equation contains its own p_j with $A_j \neq 0$). So $s(\mathcal{C}, \hat{\mathfrak{h}}) := \sigma_2(\Pi)$, the second singular value of a map into $\mathbb{R}^2$, is continuous. (3) *Compactness.* Let $\mathcal{K}$ be the set of normalised Γ -flat configurations with $m \leq m_{\max}(\theta)$, containing the face F_{i_0} of diameter 1, macro-tetrahedra of minimal angle at least $\theta/2$, Γ -spokes separated by angles at least $\theta_0/2$ and consistently oriented, $|\hat{\mathfrak{h}}| = 1$ and $|\hat{\mathfrak{h}}(\hat{e}_F, \hat{e}_F)| \geq c_1(\theta_0)$. The last condition can be imposed because $\mathfrak{h} \mapsto (\mathfrak{h}(e, e))_e$ is invertible with inverse bounded by $C(\theta_0)\ell^{-2}$ (Lemma 3.1(d)), so the maximising edge has $|\hat{\mathfrak{h}}(\hat{e}_F, \hat{e}_F)| \geq c_1$. The spoke separation holds because the Γ -spokes at z are distinct edges of the embedded surface Γ_h around z (Theorem 3.4) and (G1) bounds their angles below. $\mathcal{K}$ is compact. On $\mathcal{K}$, F_{i_0} is nondegenerate at z (Lemma 6.24), so $s > 0$ by Theorem 6.25, and $s \geq s_0(\theta, \theta_0) > 0$. (4) *Actual meshes.* Projecting the Γ -vertices of ω_F onto $T_{y_F}\Gamma$ moves them by $O(\ell^2)$, i.e. $O(\ell)$ after normalisation, and $\mathfrak{h}_0(Pe, Pe) = \mathfrak{h}_0(e, e)$; the distinction between $t \in T_{y_F}\Gamma$ and the plane of the projected faces is also $O(\ell)$. By uniform continuity of Π on a compact neighbourhood of $\mathcal{K}$, $\sigma_2(\Pi_{\text{actual}}) \geq s_0/2$ for $\ell \leq h_*$. For a unit t choose $x \propto \Pi^\top t$ normalised; then $t \cdot \sum \mathcal{M} \geq \frac{s_0}{2C_E} |\mathfrak{h}_{y_F}| \ell^{5/2} - C\ell^{7/2}$. $\square$

The constant κ_0 comes from compactness and is not explicit; the explicit criterion of Theorem 6.20 is available on the sphere.

Theorem 6.27 (Lower bound for $k \geq 3$ on Alfeld refinements). *Under the assumptions of Section 2, on an Alfeld refinement with $k \geq 3$, if $|\mathfrak{h}| \partial_N u \not\equiv 0$ and $h_\Gamma \leq h_1$, then for CNS* ($\gamma \geq \gamma_2$) and for GS ($\gamma \geq \gamma_0$)*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \left(\|\mathfrak{h}| \partial_N u\|_{L^2(\Gamma)} - C h_\Gamma^2 \right),$$

with $c = c(\theta, \theta_0, \Lambda_0)$. It covers saddle points, parabolic points and flat regions.

Proof. Theorem 6.26 and Theorem 6.7; for $k > 3$ use $Y_3 \subset Y_k$. $\square$

At a parabolic point the choice of vertex matters: for $\mathfrak{h} = (x + y)^2$ a fixed vertex can give a degenerate star (`degen.log`, “flat square, 1”). This affects semidefinite regions such as cylinders.

Corollary 6.28 (Sharpness of Scott’s rate). *On Alfeld refinements with $k \geq 3$, or on any mesh with $k \geq 9$ satisfying (IS3), if $|\mathfrak{h}| \partial_N u \neq 0$, $\gamma \in [\gamma_2, \gamma_{\max}]$ and $\varepsilon_h^{(3)} \leq Ch_\Gamma^{3/2}$, then for CNS**

$$ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq Ch_\Gamma^{3/2}.$$

The same holds for GS when $G = 0$.

Proof. Theorems 5.16, 5.11 and 6.7 with Theorems 6.10, 6.27 or Proposition 6.11. $\square$

Remark 6.29 (Worsey–Farin splits). On Worsey–Farin splits each face of Γ_h is subdivided into three coplanar sub-faces of $\mathcal{T}_h$ (so (M0³) is used with this modification), and Y_h^1 then requires $\int \partial_n v = 0$ on each sub-face; the moments $\mathcal{M}_F$ are still taken over the whole face F of Γ_h . The lower bound of Theorem 6.7 uses only the discrete momentum equation on Y_h , so it holds for every solution of GS or CNS* that exists; but existence and the upper bounds are not available there, because (IS3) as formulated fails (Proposition 4.9). Proposition 6.11 still applies for $k \geq 9$, and Proposition 6.30 below gives single-macro witnesses for $k \geq 4$ on barycentric Worsey–Farin splits (interior point at the barycentre, face points at the face barycentres; this split is affine-invariant, so the Piola arguments of Section 6.4 apply, unlike the incentre-based split of [17]). For $k = 3$ a single Worsey–Farin macro-element has no witness, and Worsey–Farin edge stars give rank 2 in all our tests (ranks modulo a prime; `notes/v6/td_k3/wf*.log`) [NUMERICAL]; the trace analysis of Section 6.5 has not been redone for split faces.

Proposition 6.30 (Sharpness witnesses on barycentric Worsey–Farin splits). *Let $\hat{\mathcal{Y}}_k$ be the continuous piecewise- P_k divergence-free fields on the barycentric Worsey–Farin split of $\hat{T}$ vanishing on $\partial\hat{T}$, and $\hat{\mathcal{Y}}_k^1$ the subspace with zero mean of $\hat{g}$ on each of the three sub-faces of $\hat{F}$.*

- (i) $k = 3$: $\dim \hat{\mathcal{Y}}_3 = 18$, $\dim \hat{\mathcal{Y}}_3^1 = 12$, and both the quadratic edge-moment map $\hat{E}$ and the first-moment map vanish identically on $\hat{\mathcal{Y}}_3^1$. There is no single-macro witness of degree 3.
- (ii) $k = 4$: $\dim \hat{\mathcal{Y}}_4^1 = 72$, $\hat{E}$ is onto with $\|\hat{R}_{\text{WF}}\| = 54.001475715$ and the first-moment map is onto with right-inverse norm 9.709257103 (both certified). Hence (M3³) holds on barycentric Worsey–Farin refinements for every $k \geq 4$, with κ_0 as in Proposition 9.1 and $\|\hat{R}\|$ replaced by 54.0015, and Theorem 6.7 gives the lower bound $ch_\Gamma^{3/2} \|\mathfrak{h}|\partial_N u\|_{L^2(\Gamma)} - Ch_\Gamma^2$ for every solution of GS or CNS* that exists.

[COMPUTER-ASSISTED, EXACT]

Computation: `notes/v7/td/wf_witness.py 3|4`; output `wf_ranks.log`.

For $k \geq 4$ this is a sharpness witness: together with an upper bound it would give the exact rate $h_\Gamma^{3/2}$ on barycentric Worsey–Farin splits. The upper bound needs the reformulated stability input of Remark 4.8, which we do not have.

7. THE LEAK LIMIT AND DRAG IN THREE DIMENSIONS

This section identifies the H^1 constant of the printed method, uniformly in the penalty, and treats drag, lift and torque. Throughout, (IS3) holds (on Alfeld refinements with $k \geq 3$ it is Lemma 4.6), $k \geq 3$ and $\gamma \geq \gamma_0$. Put $q := (p - \bar{p})|_\Gamma$ and $\lambda := \mu/h = \gamma/h_\Gamma$. For $F \in \mathcal{F}_h^\Gamma$ let x_c be its centroid, $x_F^* := \pi(x_c)$, $q_F := q(x_F^*)$, P_F the orthogonal projection onto the plane of F , and

$S_F := P_F S_{x_F^*} P_F$, where S is the shape operator ($D_X n = SX$, $\langle SX, Y \rangle = \mathfrak{h}(X, Y)$). $|\cdot|_F$ is the Frobenius norm and

$$G_{\mathfrak{h}} := \|q|\mathfrak{h}|_F\|_{L^2(\Gamma)} \quad (\text{on the unit sphere } |\mathfrak{h}|_F = \sqrt{2}, G_{\mathfrak{h}} = \sqrt{2}G).$$

Definition 7.1 (Leak flow). $(U, P) \in H^1(\Omega)^3 \times L^2(\Omega)$ is the Stokes flow with $-\Delta U + \nabla P = 0$, $\operatorname{div} U = 0$ in Ω , $U = -qn$ on Γ and $U = 0$ on Σ .

It exists because $\int_{\Gamma} U \cdot n = -\int_{\Gamma} q = 0$ (the global mean; per-component fluxes need not vanish, Remark 5.6). Since $q \in C^k(\Gamma)$ with $k \geq 3$, boundary Schauder theory for the Stokes system at the C^{k+3} surface Γ gives $(U, P) \in C^{2,\alpha} \times C^{1,\alpha}$ on a collar of Γ ; no regularity at Σ is used. Lemma 3.12, applied on the collar, gives divergence-free extensions $\tilde{U} \in C^2$, $\tilde{P} \in C^1$ to $\Omega \cup \mathcal{T}_{r/2}$, and $F_U := -\Delta \tilde{U} + \nabla \tilde{P}$ vanishes on Ω , is bounded and supported in $\overline{\Omega_h} \setminus \Omega$, so $|(F_U, v)| \leq Ch_{\Gamma}^2 \|\nabla v\|$ for $v \in V_h^R$. The two-dimensional counterpart of every result below is in [29].

Let $\mathcal{T}_h^{\text{tr}} := \{t \in C^0(\Gamma_h)^3 : t|_F \in P_k(F)^3 \ \forall F \in \mathcal{F}_h^{\Gamma}, \int_{\Gamma_h} t \cdot n = 0\}$ with $L^2(\Gamma_h)$ -orthogonal projection Π_T , let $\delta_R \in Z_h$ solve $N_h(\delta_R, v) = \mathcal{R}(v)$ for $v \in Z_h$ (as in the proof of Theorem 5.12), and $X := \lambda \delta_R$. Recall $Y_h = Z_h \cap V_h^0$ and Y_h^1 from Section 6.

7.1. Tools.

Lemma 7.2 (Lift of zero-flux traces). *Every $t \in \mathcal{T}_h^{\text{tr}}$ is the trace of some $L \in Z_h$ with $L|_{\Sigma} = 0$ and $\|\nabla L\| \leq Ch_{\Gamma}^{-1/2} \|t\|_{\Gamma_h}$, $C = C'(1 + \sqrt{3}\beta^{-1})$. Consequently $\{v|_{\Gamma_h} : v \in Z_h\} = \mathcal{T}_h^{\text{tr}}$.*

Proof. Let $L_0 \in V_h$ carry the nodal values of t at the Lagrange nodes on Γ_h and vanish at all other nodes; its support, the union of the tetrahedra meeting Γ_h , misses Σ for $h \leq h_*$. A node $z \in F$ has $|t(z)| \leq C|F|^{-1/2} \|t\|_{L^2(F)}$, and $\|\nabla \phi_z\|_T^2 \leq C \operatorname{diam} T$; each node lies in boundedly many tetrahedra, so $\|\nabla L_0\|^2 \leq C \sum_F (\operatorname{diam} F)^{-1} \|t\|_F^2 \leq C(c_0 h_{\Gamma})^{-1} \|t\|_{\Gamma_h}^2$. Then $\operatorname{div} L_0 \in \Pi_h$ and $\int_{\Omega_h} \operatorname{div} L_0 = \int_{\Gamma_h} t \cdot n = 0$, and (IS3) gives $v \in V_h^0$ with $\operatorname{div} v = \operatorname{div} L_0$, $\|\nabla v\| \leq \sqrt{3}\beta^{-1} \|\nabla L_0\|$. Put $L := L_0 - v$. Conversely, $v \in Z_h$ has a continuous piecewise- P_k trace with $\int_{\Gamma_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$. $\square$

Lemma 7.3 (Projection identity). *Let $g := -(\tilde{p} - \bar{p})n_F$ on each $F \in \mathcal{F}_h^{\Gamma}$ and $t^* := \Pi_T g$. Then $\mathcal{R}(v) = \langle g, v \rangle = \langle t^*, v \rangle$ and $N_h(X, v) = \lambda \langle t^*, v \rangle$ for all $v \in Z_h$.*

Proof. $\mathcal{R}(v) = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle = \langle g, v \rangle$, and $v|_{\Gamma_h} \in \mathcal{T}_h^{\text{tr}}$ by Lemma 7.2, so $\langle g - \Pi_T g, v \rangle = 0$. $\square$

Lemma 7.4 (Normal data are invisible to ∂_n on Z_h). *Let σ be Lipschitz on a neighbourhood of Γ_h and $g_{\sigma} := \sigma n_F$ on each F . For $v \in Z_h$, $|\langle g_{\sigma}, \partial_n v \rangle| \leq C \|\sigma\|_{W^{1,\infty}} \|v\|_{\Gamma_h}$; for $v \in Y_h$ it vanishes.*

Proof. This is the normal part of the proof of Lemma 5.10: $\partial_n v \cdot n_F = -\operatorname{div}_F v_F$, integration by parts on F , and edge terms with coefficient $|t_E \times (n_F - n_{F'})| \leq Ch_{\Gamma}$. There is no tangential part, because g_{σ} is exactly normal on every face, so no $h_{\Gamma}^{1/2} \|\nabla v\|$ term appears. If $v \in Y_h$, $v = 0$ on Γ_h and $\partial_n v \cdot n_F = \operatorname{div} v = 0$ (Lemma 6.1). $\square$

Lemma 7.5 (Tangential defect of the leak flow on a face). *For $x \in F \in \mathcal{F}_h^{\Gamma}$, with $\tau_F := P_F(n(x_F^*) - n_F)$, $|\tau_F| \leq Ch_{\Gamma}$,*

$$P_F \tilde{U}(x) = -q_F(\tau_F + S_F(x - x_c)) + O(h_{\Gamma}^2), \quad \zeta := \tilde{U} - g = O(h_{\Gamma}) \text{ on } \Gamma_h, \quad \|\zeta\|_{\Gamma_h} \leq Ch_{\Gamma}.$$

Proof. $|x - x^*| \leq Ch_{\Gamma}^2$ (Lemma 3.1(a)), so $\tilde{U}(x) = -q(x^*)n(x^*) + O(h_{\Gamma}^2)$. Expand $n(x^*) = n(x_F^*) + S_{x_F^*}(x^* - x_F^*) + O(h_{\Gamma}^2)$ with $x^* - x_F^* = P_F(x - x_c) + O(h_{\Gamma}^2)$, and use $P_F n_F = 0$ and $q(x^*) = q_F + O(h_{\Gamma})$. For ζ : $\zeta = -q_F(n(x^*) - n_F) + O(h_{\Gamma}^2)$ and $|n(x^*) - n_F| \leq Ch_{\Gamma}$ (Lemma 3.1(b)). $\square$

Remark 7.6 (The face-mean defect). The tangential defect on F is affine, $-q_F S_F(x - x_c)$, plus the constant $-q_F \tau_F = O(h_\Gamma)$. On the unit sphere $S = I$ on tangent planes and $\tau_F = -(x_c - o_F) + O(h_\Gamma^2)$, with o_F the circumcentre of F (Lemma 3.2(b),(c)); the sign is fixed by n pointing into the obstacle. So the defect is the radial field $-q(x - o_F)$ about the circumcentre, and its face mean is proportional to the centroid–circumcentre offset, nonzero on every non-equilateral face. In two dimensions the circumcentre of a chord is its midpoint, the face mean is $O(h_\Gamma^2)$, and the defect is an odd sawtooth. The constant does not change any rate below: the witnesses of Theorem 7.13 have $\int_F \partial_n v = 0$, and Lemma 7.4 does not see normal data. It does enter the constants (Corollary 7.18) and the size of $\mathcal{G}(w_\phi)$ in Lemma 7.25. It is the same offset that makes the transposed term of Proposition 5.2 of order h_Γ in the $\|\nabla v\|$ -dual norm.

7.2. The leak limit, uniformly in the penalty. Let χ be a cutoff with $\chi = 1$ on $\mathcal{T}_{r/4}$ and $\chi = 0$ off $\mathcal{T}_{r/2}$, I_h the P_k Lagrange interpolant, I_{SZ} the P_k Scott–Zhang interpolant preserving the zero trace on Σ , and $I_h^\sharp \tilde{U} := I_h(\chi \tilde{U}) + I_{SZ}((1 - \chi)\tilde{U})$. For $h \leq h_*$, $I_h^\sharp \tilde{U} = I_h \tilde{U}$ on the tetrahedra meeting Γ_h and $I_h^\sharp \tilde{U} = 0$ on Σ . Put

$$E_U^0 := h + \|\nabla((1 - \chi)\tilde{U} - I_{SZ}((1 - \chi)\tilde{U}))\|_{\Omega_h}.$$

$E_U^0 \rightarrow 0$ always (density and stability of I_{SZ}), and $E_U^0 \leq C(h + h^s \|U\|_{H^{1+s}(\Omega)})$ if $U \in H^{1+s}(\Omega)$, $0 < s \leq 1$. Only this term sees the regularity of U at the edges and corners of Σ (Remark 2.6).

Lemma 7.7. *There is $z_U \in Z_h$ with $\|\nabla(\tilde{U} - z_U)\| \leq C E_U^0$ and $\|\tilde{U} - z_U\|_{L^\infty(\Gamma_h)} \leq C h_\Gamma^2$.*

Proof. $w := I_h^\sharp \tilde{U}$ vanishes on Σ , and $m := \int_{\Gamma_h} w \cdot n = \int_{\Gamma_h} (w - \tilde{U}) \cdot n$ satisfies $|m| \leq C h_\Gamma^2$. With b and c_b from the proof of Lemma 5.7, $a := m/(c_b |\Gamma_h|)$ and $\|\nabla(ab)\| \leq C h_\Gamma^{3/2}$; (IS3) gives $v \in V_h^0$ with $\operatorname{div} v = \operatorname{div}(w - ab)$ and $\|\nabla v\| \leq \sqrt{3} \beta^{-1} (\|\nabla(w - \tilde{U})\| + \|\nabla(ab)\|)$. Put $z_U := w - ab - v$. On Γ_h , $z_U = I_h \tilde{U} - ab$. $\square$

Lemma 7.8. $\|t^* - g\|_{\Gamma_h} \leq C h_\Gamma$ and $\|t^* - \tilde{U}\|_{\Gamma_h} \leq C h_\Gamma$.

Proof. $z_U|_{\Gamma_h} \in \mathcal{T}_h^{\text{tr}}$, so $\|\Pi_T g - g\| \leq \|z_U - g\| \leq \|z_U - \tilde{U}\| + \|\zeta\| \leq C h_\Gamma$; and $t^* - \tilde{U} = (t^* - g) - \zeta$. $\square$

Let $X^D \in Z_h$ be the discrete Stokes–Dirichlet field: $X^D|_{\Gamma_h} = t^*$ and $a_h(X^D, v) = 0$ for $v \in Y_h$. It exists and is unique: the affine set $\{z \in Z_h : z|_{\Gamma_h} = t^*\}$ is nonempty by Lemma 7.2, its direction is Y_h , and $a_h(y, y) = \|\nabla y\|^2$ on Y_h .

Lemma 7.9. $\|\nabla(X^D - \tilde{U})\| \leq C(h_\Gamma^{1/2} + E_U^0)$.

Proof. $W := X^D - z_U$ has trace $t^* - z_U \in \mathcal{T}_h^{\text{tr}}$ of norm $\leq C h_\Gamma$; its lift L has $\|\nabla L\| \leq C h_\Gamma^{1/2}$, and $y := W - L \in Y_h$. Green’s formula gives $a_h(\tilde{U}, y) = (F_U, y)$, so $\|\nabla y\|^2 = a_h(\tilde{U} - z_U, y) - (F_U, y) - a_h(L, y)$ and $\|\nabla y\| \leq 2\|\nabla(\tilde{U} - z_U)\| + C h_\Gamma^2 + 2\|\nabla L\|$. $\square$

Theorem 7.10 (Three-dimensional leak limit, uniform in the penalty). *For every $\gamma \geq \gamma_0$, with no upper bound on γ , and $D := X - X^D \in Z_h$,*

$$\|D\| \leq C \left[\left(\frac{h_\Gamma}{\gamma} \right)^{1/2} (1 + h_\Gamma^{-1/2} E_U^0) + h_\Gamma^{1/2} \right], \quad \|\nabla(X - \tilde{U})\|_{\Omega_h} \leq C(h_\Gamma^{1/2} + E_U^0).$$

In particular $\|\nabla \delta_R\| \leq C h/\mu$ uniformly in $\gamma \geq \gamma_0$, and $(\mu/h)\delta_R \rightarrow \tilde{U}$ in H^1 .

Proof. By Lemma 7.3 and $X^D|_{\Gamma_h} = t^*$ the λ -terms cancel exactly: $N_h(D, v) = -a_h(X^D, v) + \langle \partial_n X^D, v \rangle + \langle t^*, \partial_n v \rangle =: \ell(v)$ on Z_h . (i) Let $L_v \in Z_h$ lift $v|_{\Gamma_h}$. Since $v - L_v \in Y_h$, $a_h(X^D, v) = a_h(\tilde{U}, L_v) + a_h(X^D - \tilde{U}, L_v)$. Green’s formula ($\operatorname{div} L_v = 0$, $L_v|_\Sigma = 0$, $\int_{\Gamma_h} L_v \cdot n = 0$) gives $a_h(\tilde{U}, L_v) = (F_U, L_v) + \langle (D(\tilde{U}) - (\tilde{P} - c)I)n, v \rangle$, which is at most $C\|v\|_{\Gamma_h}$; and

$|a_h(X^D - \tilde{U}, L_v)| \leq C(1 + h_\Gamma^{-1/2} E_U^0) \|v\|_{\Gamma_h}$ by Lemmas 7.9 and 7.2. (ii) $\|\partial_n X^D\|_{\Gamma_h} \leq \|\partial_n I_h^\# \tilde{U}\|_{\Gamma_h} + C_I(c_0 h_\Gamma)^{-1/2} \|\nabla(X^D - I_h^\# \tilde{U})\| \leq C(1 + h_\Gamma^{-1/2} E_U^0)$ (Lemmas 4.1, 7.9; $|\nabla I_h \tilde{U}| \leq C$ on the tetrahedra at Γ_h). (iii) $\langle t^*, \partial_n v \rangle = \langle g, \partial_n v \rangle + \langle t^* - g, \partial_n v \rangle$: Lemma 7.4 with $\sigma = -(\tilde{p} - \bar{p})$ bounds the first by $C\|v\|_{\Gamma_h}$, and Lemma 7.8 with $\|\partial_n v\|_{\Gamma_h} \leq (c_0 h_\Gamma)^{-1/2} \|v\|$ the second by $Ch_\Gamma^{1/2} \|v\|$. With $\|v\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2} \|v\|$ and coercivity (Lemma 4.2; c_1 does not depend on $\mu \geq \mu_0$), take $v = D$. Then $\|\nabla(X - \tilde{U})\| \leq \|D\| + \|\nabla(X^D - \tilde{U})\|$ and $(h_\Gamma/\gamma)^{1/2} \leq \gamma_0^{-1/2} h_\Gamma^{1/2}$. $\square$

7.3. A matching lower bound for every penalty.

Lemma 7.11 (Identity on Y_h). *For every $\mu > 0$ for which δ_R exists and every $v \in Y_h$, with $W := X - \tilde{U}$, $a_h(W, v) - \langle W, \partial_n v \rangle = \langle \tilde{U}, \partial_n v \rangle - (F_U, v)$.*

Proof. $N_h(\delta_R, v) = \mathcal{R}(v) = 0$ and $N_h(\delta_R, v) = a_h(\delta_R, v) - \langle \delta_R, \partial_n v \rangle$ for $v \in Y_h$; and $a_h(\tilde{U}, v) = (F_U, v)$. $\square$

Lemma 7.12 (Single-macro leak witness). *Let $\mathcal{T}_h$ be an Alfeld refinement, $k \geq 4$, $F \in \mathcal{F}_h^\Gamma$, $T = T_F$ its macro-tetrahedron with reference map $x = A\hat{x} + a$ as in Section 6.4, $A_F := A|_{\mathbb{R}^2 \times \{0\}}$, $\ell := \text{diam } F$ and h_T the height of T over F . For every symmetric S on the plane of F there is $v \in Y_h^1$, supported in T , with $\|\nabla v\| = 1$ and*

$$S : \mathcal{M}_F^1(v) \geq \kappa_L(T) |S|_F \ell^{3/2}, \quad \mathcal{M}_F^1(v) := \int_F \partial_n v \otimes (x - x_c) dA,$$

$$\kappa_L(T) := \frac{\sigma_{\min}(A_F)^2 (\det A)^{1/2}}{h_T^2 \|A\| \|A^{-1}\| \|\hat{R}_1\| \ell^{3/2}},$$

where $\|\hat{R}_1\| = 3.645838299$ (certified; the interval has width below the last digit) is the norm of the right inverse of the first-moment map $\hat{E}^1 : \hat{v} \mapsto \int_{\hat{F}} \hat{g} \otimes (\hat{x} - \hat{x}_c) \in \mathbb{R}^{2 \times 2}$ on $\hat{\mathcal{Y}}_4^1$, from the Frobenius norm to $\|\hat{\nabla} \cdot\|_{L^2(\hat{T})}$. [COMPUTER-ASSISTED, EXACT]

Proof. The exact computation (Bernstein–Bézier, over $\mathbb{Q}$) gives $\text{rank } \hat{E}^1 = 4$ on $\hat{\mathcal{Y}}_4^1$ (dim 24) and brackets $\lambda_{\min}(\hat{E}^1 \hat{G}^{-1} \hat{E}^{1\top}) = 7.523244811512 \cdot 10^{-2}$ by exact $LDL^\top$ tests. Let $\hat{M} := A_F^\top S A_F / |A_F^\top S A_F|_F$ and $\hat{v} := \hat{R}_1 \hat{M}$, so $\|\hat{\nabla} \hat{v}\| \leq \|\hat{R}_1\|$. Its Piola image v lies in Y_h^1 (proof of Theorem 6.10). By Lemma 6.8(b) with the affine weights $(x - x_c)_b = (A_F(\hat{x} - \hat{x}_c))_b$, $\int_F \partial_\nu v \otimes (x - x_c) = h_T^{-2} A_F \hat{M} A_F^\top$, and $\partial_n = -\partial_\nu$ (the sign is absorbed by $v \mapsto -v$). So $|S : \mathcal{M}_F^1(v)| = h_T^{-2} \text{tr}(A_F^\top S A_F \hat{M}) = h_T^{-2} |A_F^\top S A_F|_F \geq h_T^{-2} \sigma_{\min}(A_F)^2 |S|_F$. Normalise with Lemma 6.8(c). $\square$

Computation: notes/v7/td/kappa_cert.py 4; leak_k3_image.py; output kappa_k4.log; leak_k3_image.log.

At $k = 3$ the same map has rank 1 and its image is spanned by the skew matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ (exact), so $S : \mathcal{M}_F^1(v) = 0$ for every symmetric S : there is no single-macro leak witness of degree 3, as there is none in two dimensions [29].

Theorem 7.13 (Lower bound with curvature weight). *Let $\mathcal{T}_h$ be an Alfeld refinement, $k \geq 4$, $\kappa_L := \min_F \kappa_L(T_F)$ and $\Lambda_0 = 2 + C_{PT} C_I$ (Corollary 6.4). For every $\mu > 0$ for which δ_R exists (in particular every $\gamma \geq \gamma_0$),*

$$\|\nabla(X - \tilde{U})\|_{\Omega_h} \geq c_L G_\flat h_\Gamma^{1/2} - C h_\Gamma^{3/2}, \quad c_L := \left(\frac{4}{\sqrt{3}}\right)^{1/2} \frac{\kappa_L c_0^{3/2}}{\Lambda_0}.$$

c_L is explicit (Section 9): $c_L = 5.20 \cdot 10^{-3}$ when all macro-tetrahedra at Γ_h are regular and $c_0 = 1$. [COMPUTER-ASSISTED, EXACT]

Proof. Take v_F from Lemma 7.12 with $S = S_F$, signed so that $-S_F : \mathcal{M}_F^1(v_F) \geq \kappa_L |S_F|_F \ell_F^{3/2}$, and $v := \sum_F q_F |S_F|_F v_F \in Y_h^1$. The supports are distinct macro-tetrahedra (Lemma 3.10), so $\|\nabla v\|^2 = \Sigma := \sum_F q_F^2 |S_F|_F^2$, and on F only v_F is seen. By Lemma 7.5, $\partial_n v \perp n_F$ (Lemma 6.1) and $\int_F \partial_n v_F = 0$, the constant τ_F drops out and

$$\begin{aligned} \langle \tilde{U}, \partial_n v \rangle &= \sum_F q_F |S_F|_F (-q_F S_F : \mathcal{M}_F^1(v_F)) + \sum_F |q_F| |S_F|_F O(h_\Gamma^2) \|\partial_n v_F\|_{L^1(F)} \\ &\geq \kappa_L c_0^{3/2} h_\Gamma^{3/2} \Sigma - C h_\Gamma^{3/2} \Sigma^{1/2}, \end{aligned}$$

using $\|\partial_n v_F\|_{L^1(F)} \leq C \ell_F^{1/2}$ and $\#\mathcal{F}_h^\Gamma \leq C h_\Gamma^{-2}$. By Lemma 7.11, Lemma 6.3 and $|a_h(W, v)| \leq 2\|\nabla W\| \|\nabla v\|$, $\Lambda_0 \|\nabla W\| \Sigma^{1/2} \geq \langle \tilde{U}, \partial_n v \rangle - C h_\Gamma^2 \Sigma^{1/2}$. Finally $|F| \leq \frac{\sqrt{3}}{4} h_\Gamma^2$ gives $\Sigma \geq \frac{4}{\sqrt{3}} h_\Gamma^{-2} \sum_F |F| q_F^2 |S_F|_F^2 = \frac{4}{\sqrt{3}} h_\Gamma^{-2} (G_\mathfrak{h}^2 + O(h_\Gamma))$, since the $\pi(F)$ tile Γ with Jacobian $1 + O(h_\Gamma^2)$. If $G_\mathfrak{h}^2 \leq C h_\Gamma$ the claim is trivial after enlarging C . $\square$

Corollary 7.14 (The rate of the leak limit is exactly $\frac{1}{2}$). *On Alfeld refinements with $k \geq 4$, if $G_\mathfrak{h} > 0$, then for all $\gamma \in [\gamma_0, \infty)$ and $h_\Gamma \leq h_1$*

$$\frac{1}{2} c_L G_\mathfrak{h} h_\Gamma^{1/2} \leq \left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\| \leq C (h_\Gamma^{1/2} + E_U^0).$$

The two sides have the same order whenever $E_U^0 = O(h_\Gamma^{1/2})$, for example if $U \in H^{3/2}(\Omega)$ and $h \leq C h_\Gamma$.

The weight is the second fundamental form: on flat parts of Γ there is no $h_\Gamma^{1/2}$ term. In two dimensions the weight is the curvature, and the rate is again $\frac{1}{2}$ [29].

7.4. The geometric part, uniformly in γ , and the H^1 constant. Put $\varepsilon_1 := h^k + h^{k+1} h_\Gamma^{-1/2}$.

Lemma 7.15. *Let $z \in W_h$ be the field of Lemma 5.7, $\delta_G \in Z_h$ with $N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v)$ on Z_h , and $u_h^L := z - \delta_G \in W_h$. For every $\gamma \geq \gamma_0$, $\|\nabla(\tilde{u} - u_h^L)\| \leq C(h_\Gamma^{3/2} + \varepsilon_1)$.*

Proof. (1) The construction of Lemma 5.7 gives $\|\nabla(\tilde{u} - z)\| \leq C\varepsilon_1$ (the γ -dependence of $\varepsilon_h^{(3)}$ comes only from the penalty weight) and $\|\tilde{u} - z\|_{\Gamma_h} \leq C(h_\Gamma^{k+1} + h^{k+1})$; also $\|\tilde{u} - u_h^L\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$ (proof of Theorem 5.12). (2) For $v \in Y_h$, $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$, so $a_h(\tilde{u} - u_h^L, v) = -\langle u_h^L, \partial_n v \rangle - (F, v)$. (3) Let $m_\Sigma := \int_\Sigma g_I \cdot \nu$, $a_\Sigma := -m_\Sigma / (c_b |\Gamma_h|)$ ($|a_\Sigma| \leq C h^{k+1}$) and $W_h^D := \{w \in W_h : w|_{\Gamma_h} = a_\Sigma b\}$. The trace $(z - a_\Sigma b)|_{\Gamma_h}$ has zero flux and norm $\leq C(h_\Gamma^2 + h^{k+1})$; subtracting its lift (Lemma 7.2) gives $z_D \in W_h^D$ with $\|\nabla(\tilde{u} - z_D)\| \leq C(\varepsilon_1 + h_\Gamma^{3/2})$. Let $u^D \in W_h^D$ solve $a_h(\tilde{u} - u^D, v) = -(F, v)$ on Y_h ; comparing with z_D , $\|\nabla(\tilde{u} - u^D)\| \leq C(\varepsilon_1 + h_\Gamma^{3/2})$. (4) $w := u_h^L - u^D \in Z_h$ satisfies $a_h(w, v) = \langle u_h^L, \partial_n v \rangle$ on Y_h and has trace $u_h^L - a_\Sigma b$; with its lift L_w and $y := w - L_w \in Y_h$, $\|\nabla y\| \leq C h_\Gamma^{-1/2} \|u_h^L\|_{\Gamma_h} + 2\|\nabla L_w\|$, so $\|\nabla w\| \leq C h_\Gamma^{-1/2} (\|u_h^L\|_{\Gamma_h} + h^{k+1})$. (5) $\|u_h^L\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h_\Gamma/\gamma)^{1/2} \|\tilde{u} - u_h^L\| \leq C(h_\Gamma^2 + h_\Gamma^{1/2} \gamma^{-1/2} \varepsilon_h^{(3)})$, and $\gamma^{-1/2} \varepsilon_h^{(3)} \leq C\varepsilon_1$. $\square$

Corollary 7.16 (GS in H^1 , uniformly in γ). *For every $\gamma \geq \gamma_0$,*

$$\left\| \nabla \left(\tilde{u} - u_h - \frac{h}{\mu} \tilde{U} \right) \right\|_{\Omega_h} \leq C \frac{h}{\mu} (h_\Gamma^{1/2} + E_U^0) + C(h_\Gamma^{3/2} + \varepsilon_1), \quad \|\nabla(\tilde{u} - u_h)\| \leq C \left(\frac{h}{\mu} + h_\Gamma^{3/2} + \varepsilon_1 \right).$$

Hence, if $G > 0$, $\|\nabla(\tilde{u} - u_h)\| \mu / (hG) \rightarrow K := \|\nabla U\|_{L^2(\Omega)} / G$ whenever $h/\mu \rightarrow 0$ and $\frac{\mu}{h} (h_\Gamma^{3/2} + \varepsilon_1) \rightarrow 0$; for the geometric part the condition is $\gamma h_\Gamma^{1/2} \rightarrow 0$. The second bound improves Theorem 5.12, whose right-hand side contains $(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}$.

Proof. $\tilde{u} - u_h = (\tilde{u} - u_h^L) + \delta_R$ with $\delta_R = \frac{h}{\mu} X$; Lemma 7.15 and Theorem 7.10. $\|\nabla \tilde{U}\|_{\Omega_h} \rightarrow \|\nabla U\|_{\Omega}$ since $|\Omega_h \Delta \Omega| \leq Ch_\Gamma^2$. $\square$

So the H^1 constant of the leak law of Corollary 5.14 is the Dirichlet energy of the Stokes flow driven by the leak, as in two dimensions [29]. In test P of Section 11, $\tilde{u} = 0$ and $\mathcal{G} \equiv 0$, so $u_h = -\delta_R$ and Theorem 7.10 applies for every γ .

Remark 7.17 (General data: the lower bound needs small γ). For GS with $\tilde{u} \neq 0$, the identity on Y_h for $W' := \frac{\mu}{h}(\tilde{u} - u_h) - \tilde{U}$ acquires the term $-\frac{\mu}{h}[\langle \tilde{u}, \partial_n v \rangle + (F, v)]$. With the witness of Theorem 7.13, $\langle \tilde{u}, \partial_n v \rangle = \ell_W(v) + O(h_\Gamma^{5/2})\|\nabla v\|$ with ℓ_W the quadratic edge moments of Lemma 6.5, of size $h_\Gamma^{3/2}\|\nabla v\|$, so the extra term is $O(\gamma h_\Gamma^{1/2})\|\nabla v\|$, the same order as the leak term. In two dimensions it is $O(h_\Gamma^2)$, because the sag is even and the witness odd on each edge. In three dimensions the witness cannot be chosen with vanishing quadratic moments on one Alfeld macro: the joint map (first moments, quadratic edge moments) on $\hat{\mathcal{Y}}_4^1$ has rank 6, not 10 (exact; on barycentric Worsey–Farin macros, rank 9). So all we claim is $\|\nabla W'\| \geq (c_L G_b - C\gamma\|\mathfrak{h}\|W\|_{L^\infty})h_\Gamma^{1/2} - C(1 + \gamma)h_\Gamma^{3/2}$, which is informative only for small γ . The limit statement of Corollary 7.16 is unaffected.

Computation: `notes/v7/td/leak_joint.py` `alfeld 4 / wf 4`; output `leak_joint.log`.

Corollary 7.18 (The constant converges at rate one, with a first-order three-dimensional term). Let $K_h := \|\nabla X\|_{\Omega_h}/G$ and $d_h := \|\nabla(X - \tilde{U})\|/\|\nabla U\|$. With $\sigma_U := \partial_n \tilde{U} - \tilde{P}n$ on Γ_h and $\Theta_\zeta := -\langle \sigma_U, \zeta \rangle_{\Gamma_h}$,

$$\begin{aligned} |K_h^2 - K^2(1 + d_h^2) - 2\Theta_\zeta/G^2| &\leq C(h_\Gamma^2 + h_\Gamma\gamma^{-1/2}(1 + h_\Gamma^{-1/2}E_U^0)), \\ \Theta_\zeta &= \sum_F |F| q_F \sigma_U(x_F^*) \cdot \tau_F + O(h_\Gamma^2) = O(h_\Gamma). \end{aligned}$$

Hence $|K_h - K| \leq C(h_\Gamma + (E_U^0)^2)$ uniformly in γ , as in two dimensions; but the two-dimensional identity $K_h^2 = K^2(1 + d_h^2) + O(h_\Gamma^2)$ acquires the first-order term $2\Theta_\zeta/G^2$, which on the sphere is $O(h_\Gamma^2)$ if and only if the weighted centroid–circumcentre offsets $\sum_F |F| q_F \sigma_U \cdot (x_c - o_F)$ are.

Proof. $\|\nabla X\|^2 = \|\nabla \tilde{U}\|_{\Omega_h}^2 + 2(\nabla \tilde{U}, \nabla W) + \|\nabla W\|^2$ and $(\nabla \tilde{U}, \nabla W) = (F_U, W) + \langle \sigma_U, W \rangle$ (Green; $\text{div } W = 0$, $W|_\Sigma = 0$, zero flux). On Γ_h , $W = D + (\Pi_T \tilde{U} - \tilde{U}) - \Pi_T \zeta$. Now $\langle \sigma_U, \Pi_T \zeta \rangle = \langle \sigma_U, \zeta \rangle + \langle \Pi_T \sigma_U - \sigma_U, \zeta \rangle$ with $\|\Pi_T \sigma_U - \sigma_U\| \leq Ch_\Gamma$ (facewise smooth, jumps $O(h_\Gamma)$ across edges), and $\langle \sigma_U, \zeta \rangle = -\Theta_\zeta$. By Lemma 7.5, $\zeta|_F = -q_F(n(x^*) - n_F) + O(h_\Gamma^2)$, whose tangential face mean is $-q_F \tau_F$, while the affine part $S_F(x - x_c)$ integrates to zero against constants. Finally $|\langle \sigma_U, \Pi_T \tilde{U} - \tilde{U} \rangle| \leq Ch_\Gamma^2$ and $|\langle \sigma_U, D \rangle| \leq C\|D\|_{\Gamma_h}$ (Theorem 7.10). $\square$

Remark 7.19 (What $\partial_n v \cdot n_F = -\text{div}_F v_F$ changes). (1) The rate-1 mechanism of Theorem 5.12 and Lemma 7.4 survive: integration by parts on faces produces edge terms weighted by $|t_E \times (n_F - n_{F'})| = O(h_\Gamma)$, just as vertex terms were weighted by the tangent jump in two dimensions. (2) The upper rate $h_\Gamma^{1/2}$ is dimension-independent: a trace defect of amplitude h_Γ oscillating on the scale h_Γ has $H^{1/2}$ norm of order $h_\Gamma^{1/2}$ on a surface of any dimension. (3) The weight of the lower bound changes from the curvature to $|\mathfrak{h}|_F$, because $\partial_n v$ is tangential on F and pairs with the tangential defect $-qS_F(x - x_c)$ through a 2×2 moment matrix; only its symmetric part is seen, and the $k = 3$ single-macro image is skew. (4) The face means of Remark 7.6 change no rate, only constants.

7.5. Drag, lift and torque. Let $\mathcal{RM} := \{\psi(x) = a + b \times x\}$; then $D(\psi) = 0$, $\nabla\psi$ is skew, $\operatorname{div} \psi = 0$ and $\int_{\Gamma_h} \psi \cdot n = \int_{\Gamma} \psi \cdot n = 0$. Put $J_\psi := \int_{\Gamma} \sigma(u, p)n \cdot \psi$ with $\sigma(u, p) = D(u) - pI$ (drag and lift for $b = 0$, torque for $a = 0$). With $\Phi_\psi := \frac{1}{2}a \times x - \frac{1}{2}|x|^2b$ ($\operatorname{curl} \Phi_\psi = \psi$) and a cutoff χ_0 equal to 1 on $\mathcal{T}_{r/2}$ and 0 near Σ , let $V_\psi := \operatorname{curl}(\chi_0 \Phi_\psi)$, smooth, divergence-free and equal to ψ near Γ . Let Λ be the union of all tetrahedra meeting Γ_h . For $h \leq h_*$, $I_h V_\psi = \psi$ on Λ .

Lemma 7.20. *For $h_\Gamma \leq h_*$, $\Omega_h \triangle \Omega \subset \Lambda$.*

Proof. Points of $\Omega_h \triangle \Omega$ lie within Ch_Γ^2 of Γ_h . Near Γ_h the height above Γ_h in the frame (Gr) is affine up to $O(h_\Gamma^2)$ on sets of diameter Ch_Γ ; on a tetrahedron without vertices on Γ_h it is at least $ch_\Gamma - Ch_\Gamma^2 > Ch_\Gamma^2$, because an interior vertex z has $\operatorname{dist}(z, \partial\Omega_h) \geq ch_z$ and $h_z \geq ch_\Gamma$ near Γ_h . $\square$

Definition 7.21 (Discrete drag). For $w \in V_h^R$ let $\omega_h(w) := a_h(u_h, w) - (p_h, \operatorname{div} w) - (\tilde{f}, w)$, and $J_h^N(\psi) := \omega_h(I_h V_\psi)$. By the method equation, $J_h^N(\psi) = \int_{\Gamma_h} t_h \cdot \psi + \langle u_h, \partial_n \psi \rangle$ with the Nitsche flux $t_h = \partial_n u_h - \theta(p_h - \bar{p}_\Gamma(p_h))n - \frac{\mu}{h}u_h$ ($\theta = 0$ for GS, 1 for CNS*), and $J_h^N(\psi) = \omega_h(w)$ for every $w \in V_h^R$ equal to ψ on Λ . Let $v_0 \in V_h^0$ with $\operatorname{div} v_0 = \operatorname{div} I_h V_\psi$ ((IS3); $\|\nabla v_0\| \leq Ch^k$), $v_\psi := I_h V_\psi - v_0 \in Z_h$, and $\Theta_f(\psi) := (\tilde{f}, \psi)_{\Omega_h \setminus \Omega} - (f, \psi)_{\Omega \setminus \Omega_h} = O(h_\Gamma^2)$.

Lemma 7.22 (Error identity). *For GS and CNS*, with $e_u = \tilde{u} - u_h$,*

$$J_h^N(\psi) - J_\psi + \Theta_f(\psi) = -a_h(e_u, v_\psi) + \rho_0, \quad \rho_0 := \langle u_h, \partial_n v_0 \rangle + (F, v_0),$$

$$|\rho_0| \leq Ch^k h_\Gamma^{-1/2} \|u_h\|_{\Gamma_h} + Ch_\Gamma^2 h^k.$$

Proof. For $v_0 \in V_h^0$ both methods give $\omega_h(v_0) = \langle u_h, \partial_n v_0 \rangle$. For $w \in V_h^R$, $\tilde{\omega}(w) := a_h(\tilde{u}, w) - (\tilde{p}, \operatorname{div} w) - (\tilde{f}, w) = \langle \sigma(\tilde{u}, \tilde{p})n, w \rangle - (F, w)$. For $w = I_h V_\psi$, which equals ψ on $\Lambda \supset \Omega_h \triangle \Omega$ (Lemma 7.20), the divergence theorem on the two signed regions between Γ_h and Γ , $\operatorname{div}(\sigma\psi) = (F - \tilde{f}) \cdot \psi$ there (σ symmetric, $\nabla\psi$ skew, $F = 0$ on Ω), and $\int \psi \cdot n = 0$ give $\tilde{\omega}(I_h V_\psi) = J_\psi - \Theta_f(\psi)$; and $\tilde{\omega}(v_0) = -(F, v_0)$. Since $\operatorname{div} v_\psi = 0$, $\omega_h(v_\psi) - \tilde{\omega}(v_\psi) = -a_h(e_u, v_\psi)$. $\square$

Lemma 7.23 (Reciprocity). *Let (Z_ψ, R_ψ) be the Stokes flow in Ω with $Z_\psi|_\Gamma = \psi$ and $Z_\psi|_\Sigma = 0$. Then $J_\psi(U) := \int_{\Gamma} \sigma(U, P)n \cdot \psi = \int_{\Gamma} q(R_\psi - \bar{R}_\psi) =: K_\psi$, with $\bar{R}_\psi$ the mean of R_ψ over Γ .*

Proof. Both pairs are smooth near Γ and vanish on Σ ; the weak Green formula ($\langle \sigma(U, P)n, W \rangle_\Gamma = \frac{1}{2}(DU, DW) - (P, \operatorname{div} W)$ for $W \in H^1$ with $W|_\Sigma = 0$, smooth near Γ) gives $\langle \sigma(U, P)n, Z_\psi \rangle_\Gamma = \frac{1}{2}(DU, DZ_\psi) = \langle \sigma(Z_\psi, R_\psi)n, U \rangle_\Gamma$, with no $H^2(\Omega)$ regularity. On Γ , $Z_\psi - \psi$ vanishes and is divergence-free, so $\partial_n(Z_\psi - \psi) \cdot n = 0$, and $\partial_n \psi \cdot n = 0$ by skewness; so $\sigma(Z_\psi, R_\psi)n \cdot n = -R_\psi$, and $U = -qn$ with $\int_{\Gamma} q = 0$ gives the claim. $\square$

Theorem 7.24 (GS: drag, lift and torque). *For every $\gamma \geq \gamma_0$ and $\psi \in \mathcal{RM}$,*

$$J_h^N(\psi) = J_\psi - \frac{h}{\mu} K_\psi - \Theta_f(\psi) + \mathcal{R}_h, \quad |\mathcal{R}_h| \leq C \frac{h}{\mu} (h_\Gamma^{1/2} + E_U^0) + C(h_\Gamma^{3/2} + \varepsilon_1 + h^k h_\Gamma^{-1/2}).$$

For fixed γ , bounded ρ and $E_U^0 = O(h^{1/2})$: $J_h^N - J_\psi = -\frac{h}{\mu} K_\psi + O(h^{3/2})$, so the rate is exactly 1 when $K_\psi \neq 0$. The remainder is uniform in γ .

Proof. Lemma 7.22; $e_u = (\tilde{u} - u_h^L) + \delta_R$ (Lemma 7.15), $a_h(\delta_R, v_\psi) = \frac{h}{\mu} a_h(\tilde{U}, v_\psi) + \frac{h}{\mu} a_h(X - \tilde{U}, v_\psi)$ (Theorem 7.10), and by Green's formula ($\operatorname{div} v_\psi = 0$, $v_\psi|_{\Gamma_h} = \psi$, $v_\psi = \psi - v_0$ on $\Lambda \supset \Omega_h \triangle \Omega$, $\operatorname{div} \sigma(\tilde{U}, \tilde{P}) = -F_U$), $a_h(\tilde{U}, v_\psi) = (F_U, v_\psi) + \langle \sigma(\tilde{U}, \tilde{P})n, \psi \rangle_{\Gamma_h} = J_\psi(U) - (F_U, v_0)$ with $|(F_U, v_0)| \leq Ch_\Gamma^2 h^k$. Then Lemma 7.23, and $\|u_h\|_{\Gamma_h} \leq C$. $\square$

For CNS* we need the weighted normal slip. Let $w_\phi := I_h(\chi_0(\phi \circ \pi)(n \circ \pi))$ for $\phi \in C^{k+1}(\Gamma)$ with $\int_{\Gamma} \phi = 0$.

Lemma 7.25 (Weighted normal slip of CNS*). *Let $\gamma \geq \max(1, \gamma_2)$ and $Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}$. Then*

$$|\langle \phi \circ \pi, e_u \cdot n \rangle_{\Gamma_h}| \leq C_\phi \left[\frac{h}{\mu} (h_\Gamma^{-1/2} Y + h^k + \gamma h_\Gamma^3 + h_\Gamma^2 |\bar{e}_p|) + h_\Gamma \left(\frac{h}{\mu} \right)^{1/2} Y \right],$$

with $\bar{e}_p$ the mean of e_p over Ω_h and $|\bar{e}_p| \leq C$ if $\gamma h_\Gamma \leq 1$.

Proof. The two-dimensional proof [29] with three changes. (i) $|\mathcal{G}(w_\phi)| \leq C_\phi(h_\Gamma + \gamma h_\Gamma^3)$ instead of $C_\phi(h_\Gamma^2 + \gamma h_\Gamma^3)$: the transposed term is $\sum_F |F| \phi \partial_n u \cdot \tau_F + O(h_\Gamma^2)$ (Proposition 5.2, Lemma 7.5), generically $O(h_\Gamma)$; the penalty term is $O(\gamma h_\Gamma^3)$ because $\tilde{u} = d \partial_N u(x^*) + O(d^2)$ is tangential to Γ while w_ϕ is normal; and $|(F, w_\phi)| \leq C h_\Gamma^2$. Since $h_\Gamma \leq h_\Gamma^{-1/2} Y$ for $\gamma \geq 1$, the change is absorbed. (ii) The cancellation for fields that are not divergence-free holds: $\partial_n \delta \cdot n_F = (\operatorname{div} \delta)|_{T_F} - \operatorname{div}_F \delta_F$, then the edge bookkeeping of Lemma 5.10. (iii) $|\bar{e}_p| \leq C$: test (2) with $w_1 := I_h(\chi_0 n \circ \pi)$; every term is bounded when $\gamma h_\Gamma \leq 1$, and $\int_{\Gamma_h} w_1 \cdot n = |\Gamma| + O(h_\Gamma^2)$. $\square$

Assumption 7.26 ((E_Z)). With $\tilde{Z}_\psi$ the collar extension of Lemma 3.12 and $\tilde{\zeta} := V_\psi - \tilde{Z}_\psi$, $E_Z := \inf_{z \in Z_h} \|\tilde{\zeta} - z\| \leq C h_\Gamma^{1/2}$.

(E_Z) holds if $Z_\psi \in H^{3/2}(\Omega)$ and $h \leq C h_\Gamma$: near Γ everything is smooth and $\tilde{\zeta} = O(h_\Gamma^2)$ on Γ_h , so only the H^1 approximation away from Γ matters. It is therefore a regularity statement for the dual Stokes flow at Σ . For Lipschitz Σ it is expected from $H^{3/2}$ regularity of the Stokes system on Lipschitz domains, and for the box from Dauge's H^2 theory on convex polyhedra [7]; we have checked neither (Remark 2.6). Without (E_Z), $E_Z \rightarrow 0$ still, and Theorem 7.27(b) holds with $\gamma^{1/2} h_\Gamma^{3/2} E_Z = o(h_\Gamma^{3/2})$.

Theorem 7.27 (CNS*: drag, lift and torque). *Let $\gamma \geq \max(1, \gamma_2)$, $\gamma h_\Gamma \leq 1$, $\psi \in \mathcal{RM}$, and*

$$\Lambda_h(\psi) := \langle \tilde{u}, \partial_n(\tilde{Z}_\psi - \psi) \rangle_{\Gamma_h} = \sum_F \int_F d(\partial_N u \cdot \partial_n(Z_\psi - \psi))(x^*) dA + O(h_\Gamma^3).$$

Then (a) $|J_h^N - J_\psi + \Theta_f| \leq C(Y + h^k h_\Gamma^{-1/2})$, the rate $\frac{3}{2}$; and (b)

$$|J_h^N(\psi) - J_\psi + \Theta_f(\psi) + \Lambda_h(\psi)| \leq C[\gamma^{-1/2} h_\Gamma^2 + \gamma h_\Gamma^3 + \gamma^{1/2} h_\Gamma^{3/2} E_Z + \varepsilon_h^{(3)} + h^k h_\Gamma^{-1/2}].$$

Modulo (E_Z), $|J_h^N - J_\psi| \leq C(h_\Gamma^2 + \varepsilon_h^{(3)} + h^k h_\Gamma^{-1/2})$: the rate is 2.

Proof. The two-dimensional proof [29], with v_ψ of Definition 7.21 and the extra ρ_0 of Lemma 7.22. The auxiliary problem $N_h(\zeta_h, w) + B^*(\pi_h, w) = a_h(v_\psi, w)$ is the CNS* system for $(\tilde{\zeta}, -\tilde{R}_\psi)$, since $a_h(v_\psi, w) = (-\Delta V_\psi, w) + a_h(v_\psi - V_\psi, w)$ ($D(V_\psi) = 0$ near Γ_h , $\|\nabla(v_\psi - V_\psi)\| \leq C h^k$); Theorem 5.16 gives existence and $\|\tilde{\zeta} - \zeta_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + E_Z + h^k]$. The remaining dimension-dependent inputs are: $|\tilde{\zeta}| \leq C h_\Gamma^2$ and $|(\nabla \tilde{u})^\top n_F| \leq C h_\Gamma$ on Γ_h ; $\tilde{\zeta} \cdot n_F = d \partial_n \zeta(x^*) \cdot n_F + O(h_\Gamma^4) = O(h_\Gamma^3)$ by Lemma 3.13 applied to ζ ; $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq C h_\Gamma^{-1/2} Y$ (proof of Theorem 5.16 and Lemma 4.1); and Lemma 7.25 with $\phi = R_\psi - \bar{R}_\psi$. $\square$

With $d = Q_F + O(\ell^3)$ (Lemma 3.1(a)), $\int_F Q_F = \frac{|F|}{24} \sum_{i < j} w_{ij}^F$. On the unit sphere, $\Theta_f + \Lambda_h \approx -\sum_F \frac{|F|}{24} (\sum_{i < j} \ell_{ij}^2) (f \cdot \psi + \partial_N u \cdot \partial_n(Z_\psi - \psi))(x_F^*)$: the leading drag error of CNS* is the shape derivative of J_ψ under the chordal sag. Whether rate 2 is sharp is not proved.

8. THE PENALTY THRESHOLD SCALES WITH ρ

Sufficiency is Lemma 4.2: $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ implies coercivity of N_h on V_h^R . This section shows that a threshold proportional to ρ is also necessary, on the divergence-free space Z_h on which the methods actually need coercivity, under conditions on the cells at a smallest boundary face.

Scaling. For $v \in Z_h$ put $A := a_h(v, v)$, $B := \langle \partial_n v, v \rangle$ and $C := \|v\|_{\Gamma_h}^2$, so that $N_h(v, v) = A - 2B + (\mu/h)C$. Under dilation by ℓ , A and B scale like ℓ and C like ℓ^2 . For a field of length scale ℓ ,

$$N_h(v, v) = \frac{C}{\ell} \left[\frac{\mu\ell}{h} - \frac{\ell(2B - A)}{C} \right], \quad (3)$$

and the quotient $\ell(2B - A)/C$ is dilation invariant. So N_h is indefinite on Z_h as soon as some $v \in Z_h$ has $\ell(2B - A)/C > \mu\ell/h$.

8.1. A witness in one Alfeld macro-element. Let $\mathcal{T}_h$ be an Alfeld refinement, $T = \text{conv}(P_0, \dots, P_3)$ a macro-tetrahedron with $F = \text{conv}(P_1, P_2, P_3) \in \mathcal{F}_h^\Gamma$ and $T \cap \Sigma = \emptyset$, and

$$W_k(T) := \{v \in C^0(T)^3 : v|_S \in P_k^3 \ \forall S \in T^A, \ \text{div } v = 0, \ v = 0 \text{ on } \partial T \setminus F\},$$

with A_T, B_T, C_T the forms above restricted to T and F , and

$$\lambda_T := \text{diam } F \cdot \sup_{v \in W_k(T), C_T(v) > 0} \frac{2B_T - A_T}{C_T}.$$

Lemma 8.1 (Piola invariance). *Let $\Phi(\hat{x}) = J\hat{x} + b$ map $\hat{T}$ onto T with $\hat{F} \mapsto F$. Then $\hat{v} \mapsto (\det J)^{-1} J \hat{v} \circ \Phi^{-1}$ is a bijection $W_k(\hat{T}) \rightarrow W_k(T)$. Hence $\dim W_k(T)$ does not depend on the shape. Pulled back to $W_k(\hat{T})$, A_T and B_T are rational functions of J , and C_T is a rational function of J times $|J^{-T} \hat{v}|$, the square root of a rational function; all three are continuous where $\det J \neq 0$. So a strict inequality $(2B_T - A_T - cC_T)(v) > 0$ at a fixed pulled-back witness persists on an open set of shapes.*

Proof. Φ maps barycentre to barycentre, hence $\hat{T}^A$ onto T^A ; continuity, degree and vanishing on faces are preserved, and $\text{div } v = (\det J)^{-1} (\widehat{\text{div}} \hat{v}) \circ \Phi^{-1}$. In B_T the normalisation of n cancels against Nanson's factor; in C_T it does not. $\square$

Theorem 8.2 (Single-cell necessity). *If $\mu \text{diam } F/h < \lambda_T$, then N_h is indefinite on Z_h , for every $k \geq 3$. In particular, if the macro-tetrahedron on a smallest face F_* of Γ_h has $\lambda_T \geq \lambda_0 > 0$, coercivity on Z_h requires $\mu \geq \lambda_0 \rho$. Exact rational certificates give, for $k = 3$:*

shape (apex)	$\dim W_3$	$\sup \frac{2B-A}{C}$	λ_T	exact certificate
near-regular, $(\frac{1}{2}, \frac{7}{24}, \frac{13}{16})$	8	32.2481	32.50	$2B - A - \frac{28060}{871}C > 0$
right corner [†] , $(0, 0, 1)$	8	22.9834	32.50	$2B - A - \frac{9850}{429}C > 0$
flat, height $\frac{1}{2}$	8	61.4449	61.92	$2B - A - \frac{51071}{832}C > 0$
flat, height $\frac{1}{4}$	8	134.356	135.40	$2B - A - \frac{133014}{991}C > 0$
skew, $(\frac{3}{2}, \frac{1}{3}, \frac{3}{4})$	8	32.3856	32.64	$2B - A - \frac{14656}{453}C > 0$
tall, height 2	8	-10.381	-10.46	$A - 2B - 10C \succ 0$
tall, height 3	8	-57.454	-57.90	$A - 2B - 57C \succ 0$

The base is $(0, 0, 0), (1, 0, 0), (\frac{1}{2}, \frac{7}{8}, 0)$ except for [†], where it is $(0, 0, 0), (1, 0, 0), (0, 1, 0)$; flat and tall cells have their apex above $(\frac{1}{2}, \frac{7}{24}, 0)$. By Lemma 8.1, each positive certificate also holds on an open set of shapes around the tabulated one. [COMPUTER-ASSISTED, EXACT]

Proof. A field $v \in W_k(T)$ extended by zero is continuous (it vanishes on the faces shared with other cells), piecewise P_k , zero on Σ and divergence free, so $v \in Z_h$, and its forms are A_T, B_T, C_T . Apply (3) with $\ell = \text{diam } F$ to the maximiser, and $h/\text{diam } F_* = \rho$. $\square$

Computation: `notes/v6/td_pen/alfeld_tet.py`, `verify_alfeld.py`, `neg_cert.py`; output `alfeld_k3.log`, `verify_*.log`, `neg_cert.log`.

The null space is computed exactly over $\mathbb{Q}$; the suprema are found by floating-point bisection on the 8-dimensional pencil, and the *certificates* are rational witnesses checked in exact arithmetic. An independent quadrature check reproduces the near-regular quotient 32.24809277

against 32.24809278 from the exact pipeline. For tall cells, the exact LDL^T factorisation shows that $A - 2B - 10C$ (height 2) is positive definite on $W_3(T)$, so no single-cell witness of degree 3 exists (higher degrees were not tested); a height scan puts the change of sign at height ≈ 1.69 times the base size (the regular tetrahedron has height 0.816). This is the three-dimensional analogue of the tall-cell limitation in two dimensions [29].

8.2. A continuum witness.

Half-space constant. Take the fluid in $\{z > 0\}$, $n = (0, 0, -1)$ and $\partial_n = -\partial_z$. Let $g(t) = t(1-t)^6$ on $[0, 1]$ (zero for $t \geq 1$), $\phi(s) = (1-s^2)^6$ on $[-1, 1]$, and

$$\mathcal{F} := Y\phi(x/X)\phi(y/X)g(z/Y), \quad w := \text{curl}(\mathcal{F}e_y) = (-\partial_z\mathcal{F}, 0, \partial_x\mathcal{F}).$$

Then $\text{div } w = 0$ and $w \cdot n = 0$ on $z = 0$; at $z = 0$, $w_1 = -\phi\phi$ and $\partial_n w \cdot w = 12\phi^2\phi^2/Y > 0$. Exact integration gives, for $X = 2Y$,

$$\hat{\lambda} := \frac{Y(2B - A)}{C} = \frac{2327591}{151008} = 15.41369\dots,$$

with 15.53940, 15.58222, 15.62292 for $X/Y = 3, 4, 8$, and $\hat{\lambda} \rightarrow -2g'(0)g''(0) - \int_0^1 |g''|^2 = \frac{172}{11} = 15.636\dots$ as $X/Y \rightarrow \infty$ (the one-dimensional limit). [EXACT]

Computation: `notes/v6/td_pen/continuum3d.py`; output `continuum3d.log`.

(LQ) Let $F_* \in \mathcal{F}_h^\Gamma$ have minimal diameter, $x_0 \in F_*$, and $Y := C_q \text{diam } F_*/\kappa$. Every tetrahedron of $\mathcal{T}_h$ meeting $B(x_0, 4Y)$ has diameter at most $C_q \text{diam } F_* = \kappa Y$, and $B(x_0, 4Y) \cap \Sigma = \emptyset$.

Theorem 8.3 (Continuum-witness necessity). *Assume (IS3) with constant β (on Alfeld refinements, $k \geq 3$, Lemma 4.6). There are $\kappa_0, Y_0 > 0$, depending on C_q, β, k, Γ and shape regularity, such that under (LQ) with $\kappa \leq \kappa_0$ and $Y \leq Y_0$, N_h is indefinite on Z_h whenever $\mu < 15h/Y$. Hence the coercivity threshold on Z_h satisfies $\mu^* \geq (15\kappa_0/C_q)\rho$. No condition on the shapes of the boundary cells beyond shape regularity is needed.*

Proof. Constants C depend on the profile, k, Γ and shape regularity.

- (1) *Field.* Use the frame (Gr) at $x_0^* := \pi(x_0)$, with z pointing into Ω , $X = 2Y$, and the field w above; for $z < 0$ extend g polynomially and multiply by a cutoff equal to 1 on $z \geq -Y/2$. Then $w \in W^{4,\infty}$ with $|D^j w| \leq C_j Y^{-j}$, $\text{div } w = 0$ and $\text{supp } w \subset B(x_0, 4Y)$. Near x_0 , Γ_h is a graph $z = \gamma_h(x, y)$ with $|\gamma_h| \leq CY^2$ and $|\nabla \gamma_h| \leq CY$ (Corollary 3.5 and (Gr)).
- (2) *Comparison with the half-space.* On $\text{supp } w$, Ω_h and $\{z > 0\}$ differ in a set of volume $\leq CY^4$ where $|\nabla w|^2 \leq CY^{-2}$, and moving the integrands of B and C from $z = 0$ to $z = \gamma_h$ (with n_F in place of $(0, 0, -1)$) changes them by $O(1)$ and $O(Y)$ pointwise. So $|A_{\Omega_h}(w) - Y\hat{A}| + |B_{\Gamma_h}(w) - Y\hat{B}| \leq CY^2$ and $|C_{\Gamma_h}(w) - Y^2\hat{C}| \leq CY^3$, with $\hat{A}, \hat{B}, \hat{C}$ the half-space values at $Y = 1$.
- (3) *Discrete field.* Let I be degree-3 Lagrange interpolation, so $Iw \in V_h$ and $\|w - Iw\|_{L^\infty} \leq C\kappa^4$, $\|\nabla(w - Iw)\|_{L^\infty} \leq C\kappa^3/Y$. Let $b := \sum_{F \subset \Gamma_h \cap B(x_0, Y)} b_F n_F$; then $\int_{\Gamma_h} b \cdot n \geq cY^2$, $|\nabla b| \leq CC_q/(\kappa Y)$ and $\|\nabla b\|^2 \leq CC_q^2 Y/\kappa$. Since $\int_{\Gamma_h} w \cdot n = \int_{\Omega_h} \text{div } w = 0$, $m := \int_{\Gamma_h} Iw \cdot n$ satisfies $|m| \leq CY^2\kappa^4$; with $a := m/\int_{\Gamma_h} b \cdot n$, $|a| \leq C\kappa^4$. (IS3) gives $v_0 \in V_h^0$ with $\text{div } v_0 = \text{div}(Iw - ab)$ and $\|\nabla v_0\| \leq \beta^{-1}\sqrt{3}(\|\nabla(Iw - w)\| + |a|\|\nabla b\|) \leq C\beta^{-1}(\kappa^3 + C_q\kappa^{7/2})Y^{1/2}$. Put $v := Iw - ab - v_0 \in Z_h$.
- (4) *Perturbation.* On Γ_h , $v = Iw - ab$. So $|C(v)^{1/2} - C_{\Gamma_h}(w)^{1/2}| \leq C\kappa^4 Y$ and $|A(v)^{1/2} - A_{\Omega_h}(w)^{1/2}| \leq C(1 + \beta^{-1})C_q\kappa^3 Y^{1/2}$. In B , the terms $\partial_n(Iw - w)$ and $a\partial_n b$ change B by at most $C(1 + C_q)\kappa^3 Y$, and by the inverse trace inequality and $\sum_F |F|/\text{diam } F \leq CC_q Y/\kappa$, $|\langle \partial_n v_0, Iw - ab \rangle| \leq C(C_q Y/\kappa)^{1/2} \|\nabla v_0\| \leq C\beta^{-1}C_q^{3/2}\kappa^{5/2}Y$.

- (5) *Conclusion.* $Y(2B(v) - A(v))/C(v) \geq \hat{\lambda} - C[(1 + \beta^{-1})C_q^{3/2}\kappa^{5/2} + Y]$. Choose κ_0, Y_0 so that the bracket is at most 0.4; then $\mu < 15h/Y$ gives $N_h(v, v) < 0$ by (3). Finally $Y = C_q \text{diam } F_*/\kappa$ and $h/\text{diam } F_* = \rho$. $\square$

Corollary 8.4 (The threshold scales with ρ). *On Alfled refinements with $k \geq 3$, the coercivity threshold μ^* of N_h on Z_h satisfies $c\rho \leq \mu^* \leq 4\kappa C_I^2\rho$ whenever the mesh is locally quasi-uniform over $O(1/\kappa_0)$ layers at a smallest boundary face in the sense of (LQ), or the macro-cell on a smallest boundary face has a certified $\lambda_T > 0$ (Theorem 8.2).*

The constant κ_0 in Theorem 8.3 is not explicit, and the statement without (LQ) or a shape condition is open, as in two dimensions. If commuting projections onto the Alfled velocity space with $W^{1,\infty}$ -local bounds were used in step 3 in place of the interpolate-and-correct construction [14], (IS3) would not be needed; we have not checked those bounds.

In our computations (Section 11) the threshold on Z_h grows with h/h_Γ : $\mu^* = 76.3$ and 148.0 at the two sphere levels for $k = 3$, 193.6 at the first level for $k = 4$.

9. EXPLICIT CONSTANTS

The constants in Theorems 6.10, 6.7 and 7.13 and the single-cell penalty constant can be made explicit on Alfled refinements. The (IS3) constant cannot, at present; we say why. The values below are certified lower bounds for the stated shapes, not estimates of the true constants: the pre-asymptotic sphere data of Section 11 are orders of magnitude larger.

9.1. The witness constants.

Proposition 9.1 (Explicit κ_0 and κ_L). *Let $k \geq 4$, $T = T_F$ a macro-tetrahedron of an Alfled refinement, A its reference map (any labelling with $\det A > 0$), $\ell = \text{diam } F$, h_T the height of T over F , $D = \text{diam } T$, r the inradius, $c_s := r/D$, and θ the smaller of the minimal angles of F and of its projection onto $T_{y_F}\Gamma$ ($\theta \geq \theta_0/2$ for $h_\Gamma \leq h_*$). Then $(M^3)(iii)$ holds with*

$$\kappa_0(T) = \frac{c_w(F) (\det A)^{1/2}}{2h_T^2 \|A\| \|A^{-1}\|^2 \|\hat{R}\| \ell^{1/2}}, \quad c_w(F) := \frac{\sigma_{\min}(H \mapsto (H(u_i, u_i))_{i=1}^3)}{\ell^2} \geq \frac{\sin^4 \theta}{\sqrt{10}},$$

where $\sigma_{\min}$ is taken with the Frobenius norm on symmetric H and u_1, u_2, u_3 are the projected edge vectors, and

$$\|\hat{R}\| = 91.678155469 \quad (\text{certified; the interval has width below the last digit}).$$

In closed form,

$$\kappa_0(T) \geq \frac{\sin^5 \theta c_s^{11/4} (16\pi/\sqrt{3})^{1/4}}{\sqrt{30} \|\hat{R}\|}, \quad \kappa_L(T) \geq \frac{\sin^5 \theta c_s^{13/4} (16\pi/\sqrt{3})^{3/4}}{\sqrt{6} \|\hat{R}_1\|}.$$

Since $\hat{\mathcal{Y}}_4 \subset \hat{\mathcal{Y}}_k$, both hold for every $k \geq 4$. [COMPUTER-ASSISTED, EXACT] for $\|\hat{R}\|$ and $\|\hat{R}_1\|$; the rest is proved.

Proof. The first formula is Proposition 6.9 with $|P_F t| = 1 - O(h_\Gamma^2)$ (the defect goes into C_1) and, by the definition of c_w , $|w^F| \geq c_w \ell^2 |\mathfrak{h}_{y_F}|_F \geq c_w \ell^2 |\mathfrak{h}_{y_F}|$, up to the $O(\ell^3)$ of Lemma 3.1(d). For the bound on c_w let $E = [e_1, e_2]$ be two edges and $w_3 = (e_1 + e_2)^\top H(e_1 + e_2)$. Then $E^\top H E = \begin{pmatrix} w_1 & (w_3 - w_1 - w_2)/2 \\ (w_3 - w_1 - w_2)/2 & w_2 \end{pmatrix}$ has Frobenius norm at most $\sqrt{5/2} |w|$, and $|E^\top H E|_F \geq \sigma_{\min}(E)^2 |H|_F$, so $|H|_F \leq |E^\top H E|_F / \sigma_{\min}(E)^2$; the bound is in the Frobenius norm, as the definition of c_w requires. Finally $\sigma_{\min}(E)^2 \geq (2|F|)^2 / \text{tr}(E^\top E) \geq \frac{1}{2} \ell^2 \sin^4 \theta$, using $|F| \geq \frac{1}{2} \ell^2 \sin^2 \theta$ (sine rule). For the closed forms: $\det A = 2|F|h_T$, $(2|F|)^{1/2} \geq \ell \sin \theta$, $\|A\| \leq \sqrt{3}D$, $\|A^{-1}\| \leq \sqrt{2}/(2r)$ [5, Theorem 3.1.3] (the reference diameter is $\sqrt{2}$), $h_T \leq D$, and $\ell/D \geq (16\pi/\sqrt{3})^{1/2} c_s^{3/2}$ from $\frac{4}{3} \pi r^3 \leq |T| = \frac{1}{3} |F| h_T \leq \frac{1}{3} \frac{\sqrt{3}}{4} \ell^2 D$. For κ_L , $\sigma_{\min}(A_F)^2 \geq \frac{1}{2} \ell^2 \sin^4 \theta$ in the same way. $\square$

Computation: `notes/v7/td/kappa_cert.py 4; kappa_explicit.py; output kappa_k4.log; kappa_explicit.log.`

The value $\|\hat{R}\| = 91.678155469$ is the certified form of the floating-point value 91.7 quoted after Theorem 6.10; it is obtained, like $\|\hat{R}_1\|$, by bracketing the smallest eigenvalue of $\hat{E}\hat{G}^{-1}\hat{E}^\top$ with exact $LDL^\top$ tests.

Table C1. Explicit constants (floating-point evaluation of the proved formulas, best labelling).

macro shape (apex over base)	$c_w(F)$	$\kappa_0(T)$	$\kappa_L(T)$	c_s	θ	closed-form κ_0
regular	0.866	$2.11 \cdot 10^{-3}$	$8.65 \cdot 10^{-2}$	0.204	60°	$2.85 \cdot 10^{-5}$
near-regular $(\frac{1}{2}, \frac{7}{24}, \frac{13}{16})$	0.861	$2.09 \cdot 10^{-3}$	$8.74 \cdot 10^{-2}$	0.203	59.5°	$2.73 \cdot 10^{-5}$
right corner $(0, 0, 1)$	0.331	$1.52 \cdot 10^{-3}$	$1.63 \cdot 10^{-1}$	0.149	45°	$4.39 \cdot 10^{-6}$
flat, height $\frac{1}{2}$	0.861	$1.79 \cdot 10^{-3}$	$1.18 \cdot 10^{-1}$	0.166	59.5°	$1.57 \cdot 10^{-5}$
skew $(\frac{3}{2}, \frac{1}{3}, \frac{3}{4})$	0.861	$1.33 \cdot 10^{-3}$	$7.73 \cdot 10^{-2}$	0.095	59.5°	$3.40 \cdot 10^{-6}$

The bases are those of Theorem 8.2.

The remaining constants of the lower bounds. In Corollary 6.4, $\Lambda_0 = 2 + C_{PT}C_I$, where on the Alfeld sub-tetrahedron $T_F = \text{conv}(B, F)$, of height $h_{\text{sub}} = h_T/4$ over F and diameter D_{sub} ,

$$C_I^2 \leq k(k+2)\ell/h_{\text{sub}}, \quad C_{PT}^2 \leq \frac{D_{\text{sub}}}{h_{\text{sub}}} \left(\frac{3}{\pi^2} + \frac{2}{\pi} \right).$$

The first is the trace-inverse inequality of Warburton and Hesthaven [36] for $\partial_n v \in P_{k-1}$ in three dimensions, restricted to one face. The second follows by integrating $\text{div}((x-B)|w-\bar{w}_{T_F}|^2)$ over T_F and applying the Payne–Weinberger inequality [30] on the convex set T_F . For regular macros and $k=4$: $C_I \leq 10.84$, $C_{PT} \leq 2.15$, $\Lambda_0 \leq 25.28$. Hence

$$c_L = 5.20 \cdot 10^{-3} \text{ in Theorem 7.13,} \quad c = \frac{\kappa_0 c_0^{5/2} c_A^{1/2}}{\Lambda_0} \cdot \frac{1}{2} \left(\frac{3}{4} \right)^{1/2} = 2.37 \cdot 10^{-5} \text{ in Theorem 6.7,}$$

with $c_0 = 1$ and $c_A = \sqrt{3}/4$. For $k=3$ on general surfaces (Theorem 6.26) κ_0 comes from compactness and is not explicit.

9.2. The single-cell penalty constant over a shape class.

Proposition 9.2 ($\lambda_T \geq 20$ on an explicit box of shapes). *Normalise $F = \text{conv}(0, e_1, (a, b, 0))$ and the apex (x, y, z) ; every tetrahedron with a marked face is similar to one of these. For every (a, b, x, y, z) in the box*

$$|(a, b, x, y, z) - (\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{2}, \frac{\sqrt{3}}{6}, \sqrt{2/3})|_\infty \leq 0.1$$

around the regular tetrahedron, $\lambda_T \geq 20$ for every $k \geq 3$. Hence on Alfeld refinements whose macro-tetrahedron at a smallest wall face lies in this class, coercivity of N_h on Z_h requires $\mu \geq 20\rho$ (Theorem 8.2). [COMPUTER-ASSISTED, INTERVAL ARITHMETIC]

Proof. The box is bisected into a tree of 879 boxes with 440 leaves. For each box a rational witness $\hat{v} \in W_3(\hat{T})$ is fixed (the floating-point maximiser at the box centre, rounded to rationals); the pulled-back forms of Lemma 8.1 are evaluated in ball arithmetic (Arb [22], 80 bits) with a mean-value enclosure (forward-mode derivatives in ball arithmetic over the whole box). Every leaf gives $\text{diam } F \cdot (2B_T - A_T)/C_T \geq 20.0004$. The enclosure was cross-checked against the floating-point pipeline at a random shape (40.5657 both ways) and against the near-regular value 32.50 of Theorem 8.2, which lies in the box. $W_3 \subset W_k$ gives $k \geq 3$. $\square$

Computation: `notes/v7/td/pen_cert.py 20 0.1 (and 20 0.05); output pen_cert_d010.log, pen_cert_d005.log.`

Scope of the certificate. The centres and half-widths of the boxes, including the children $m \pm r/2$ of each bisection, were computed in binary floating point. Strictly, the certified region is therefore

the union of the 440 leaf boxes *as stored in floating point*, which can differ from the exact box above by gaps and overlaps of one unit in the last place between siblings. Given the margin $20.0004 - 20$ this is immaterial for the conclusion, but it is not covered by the certificate; a rerun with dyadic rational centres would remove the caveat. A box of half-width 0.15 was not certified within our time budget: the depth-first mean-value scheme stalls, which says nothing about λ_T there. The tall-cell obstruction of Theorem 8.2 (negative λ_T above height about 1.69) shows that no shape-uniform positive bound exists.

9.3. The (IS3) constant. Lemma 4.6 gives $\beta^{-1} \leq C_{\text{loc}} + C_{\Pi} C_B(\Omega_h)(1 + \sqrt{3}C_{\text{loc}})$. It is not explicit, for three reasons.

- (1) C_{loc} : [18, Theorem 3.1] states only that it depends on k and shape regularity. It can be made explicit by Piola transport: for $p \in \dot{P}_{k-1}(K^r)$ put $\hat{p} := (\det A)p \circ \Phi$; the reference right inverse gives $\hat{v}$ with $\|\hat{\nabla} \hat{v}\| \leq \hat{C}_{\text{loc}}(\det A)^{1/2}\|p\|_K$, and the Piola image has $\text{div } v = p$ and $\|\nabla v\|_K \leq \|A\| \|A^{-1}\| \hat{C}_{\text{loc}} \|p\|_K \leq \sqrt{3/2} c_s^{-1} \hat{C}_{\text{loc}} \|p\|_K$. [NUMERICAL] $\hat{C}_{\text{loc}} = 6.18$ ($k = 3$) and 6.41 ($k = 4$), from generalised eigenvalues in floating point; these could be certified by the same LDL^T bracketing, which we have not done.
- (2) C_{Π} (Scott–Zhang plus face-bubble fluxes) is local and explicit in principle, but depends on the vertex patches; we have not derived a value.
- (3) $C_B(\Omega_h) \leq 2C_B(\Omega)$ holds only for $h_{\Gamma} \leq h_*$ with a non-explicit h_* (Lemma 4.3), and $C_B(\Omega)$ for a box minus a ball is not known explicitly. Explicit Bogovskiĭ bounds exist for domains star-shaped with respect to a ball, and $R \setminus \bar{D}$ is not one; decomposition bounds are explicit but very crude.

Table C2. [NUMERICAL] Discrete inf-sup constant of $(V_h^0, \Pi_h \cap L_0^2)$ on icosphere–Alfeld shells: Ω_h lies between the level- l icosphere polyhedron and its dilate by 2, with m prism layers (radii $2^{j/m}$), 3 tetrahedra per prism, Alfeld-refined. β_h^{-2} is the largest eigenvalue of $M^{1/2}(BA^{-1}B^T)^+M^{1/2}$ on L_0^2 (ARPACK, sparse LU of the saddle system).

k	l	m	macro tets	h_{Γ}	$\dim V_h^0$	β_h (3 smallest)	$1/\beta_h$
3	0	1	60	1.052	3252	.1162, .1162, .1213	8.61
3	1	1	240	0.618	12972	.1161, .1205, .1206	8.61
3	0	2	120	1.052	6780	.1058, .1062, .1064	9.45
3	0	3	180	1.052	10308	.0833, .0834, .0836	12.01
4	0	1	60	1.052	7578	.1230, .1231, .1277	8.13
4	0	2	120	1.052	15642	.1152, .1155, .1156	8.68

Computation: `notes/v7/td/is3_ico.py k l m`; output `is3_k3_10.log`, `is3_more.log`.

At fixed layer structure the constant does not change from $l = 0$ to $l = 1$ (8.608 to 8.612). More layers at fixed l make the prisms flatter (smaller c_s) and β_h smaller, consistent with the c_s^{-1} dependence of C_{loc} . No spurious pressure modes were found. These values, $1/\beta_h \approx 8.6$ ($k = 3$) and 8.1 ($k = 4$) on single-layer shells, are evidence for, not a proof of, a moderate constant.

10. STEADY NAVIER–STOKES FLOW

Gjerde and Scott’s application is cylinder drag, a Navier–Stokes computation [16]. Consider

$$-\nu \Delta u + (u \cdot \nabla)u + \nabla p = f, \quad \text{div } u = 0 \text{ in } \Omega, \quad u|_{\Gamma} = 0, \quad u|_{\Sigma} = g,$$

and, with $c_1(w; u, v) := ((w \cdot \nabla)u, v)$ or $c_s(w; u, v) := \frac{1}{2}[(w \cdot \nabla)u, v] - ((w \cdot \nabla)v, u)$ (denoted c), the methods: find $u_h \in V_h$ with $u_h|_{\Sigma} = g_I$ and $p_h \in \Pi_h$ such that for all $v \in V_h^R$, $q \in \Pi_h$,

$$\nu N_h(u_h, v) + c(u_h; u_h, v) - (p_h, \text{div } v) \left[+ \langle p_h - \bar{p}_{\Gamma}(p_h), v \cdot n \rangle \right] = (\tilde{f}, v), \quad (q, \text{div } u_h) = 0.$$

Without the bracket this is GS, with it CNS*. The wall penalty is then $\nu\mu/h$ (see [29] for the conversion to an unscaled penalty).

Assumptions. (NS1) (u, p) is a solution with $u \in C^{k+1}(\overline{\Omega})$, $p \in C^k(\overline{\Omega})$, $\tilde{f}$ is a smooth extension of f , and the assumptions of Section 2 hold. (NS2) $\tilde{u}, \tilde{p}$ are the extensions of Lemma 3.12; the defect $F := \tilde{f} + \nu \Delta \tilde{u} - (\tilde{u} \cdot \nabla) \tilde{u} - \nabla \tilde{p}$ vanishes on Ω , is supported in $\overline{\Omega_h} \setminus \Omega$ and is bounded. Let $\mathfrak{A}(w, v) := \nu N_h(w, v) + c(w; \tilde{u}, v) + c(\tilde{u}; w, v)$ be the linearisation at $\tilde{u}$, $\|\ell\|_{*,X} := \sup_{0 \neq v \in X} \ell(v)/\|v\|$, and B^* as in the proof of Theorem 5.16. The discrete stability conditions are:

- (L) there is $\beta_L > 0$, independent of h , with $\beta_L \|\delta\| \leq \|\mathfrak{A}(\delta, \cdot)\|_{*,Z_h}$ for all $\delta \in Z_h$;
- (L*) there is $\beta_* > 0$, independent of h , such that for every linear functional ℓ on V_h^R the problem $(\delta, \xi) \in Z_h \times \Pi_h$, $\mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell(v)$ for all $v \in V_h^R$, has a unique solution with $\beta_* \|\delta\| \leq \|\ell\|_{*,V_h^R}$.

They are the discrete forms of “ u lies on a nonsingular branch” for GS and CNS*. Let $X_3 := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}$.

Corollary 10.1 (Navier–Stokes in three dimensions). *Assume (NS1)–(NS2) and (IS3); on Alfeld refinements with $k \geq 3$ (IS3) holds by Lemma 4.6.*

- (i) *Small data. Let N be the constant in $|c(w; u, v)| \leq N\|w\|_{H^1}\|u\|_{H^1}\|v\|_{H^1}$ and C_1 that in $\|v\|_{H^1(\Omega_h)} \leq C_1\|v\|$ for $v \in V_h^R$; both are uniform in h . If $\lambda := NC_1^2\|\tilde{u}\|_{H^1(\Omega_h)}/(\nu c_1) \leq \frac{1}{4}$, then (L) holds with $\beta_L = \nu c_1/2$, and (L*) holds for $\gamma \geq \gamma'_2$ with $\beta_* = \nu/C_*$, γ'_2 and C_* independent of ν and h .*
- (ii) *Nonsingular solutions. If u is a nonsingular solution (the linearisation at u is an isomorphism of the divergence-free H_0^1 fields onto their dual), then there is h_0 , independent of μ, γ, ρ , such that (L), and (L*) for $\gamma \geq \gamma'_2$, hold for $h \leq h_0$.*
- (iii) *GS. Under (L), for either convective form and $X_3 \leq c_*$: GS has a locally unique solution, $\|\nabla(\tilde{u} - u_h)\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$, $\|\tilde{u} - u_h\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX_3$, and under the hypotheses of Corollary 5.14*

$$\tilde{u} - u_h = \frac{h}{\nu\mu}v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}),$$

so that $u_h \cdot n \approx (h/\nu\mu)(p - \bar{p})$ on Γ_h with the Navier–Stokes pressure.

- (iv) *CNS*. Under (L*) and $(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)} \leq c_*$, CNS* has a locally unique solution with $\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$, and $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k)$ for bounded γ and $h_\Gamma \geq h^2$.*

Proof. The two-dimensional proofs [29] consist of a fixed-point argument around the linearisation at $\tilde{u}$, the Stokes estimates of Section 5, and, for (ii), a compactness argument. We list every dimension-dependent input and its three-dimensional replacement.

- (1) *Sobolev constants.* $H^1 \subset L^4$ in three dimensions; transferred to Ω_h by the bi-Lipschitz map Φ_h of Lemma 3.11, the constant N in $|c(w; u, v)| \leq N\|w\|_{H^1}\|u\|_{H^1}\|v\|_{H^1}$ is uniform in h . The convective boundary identities and the wall Péclet estimate are dimension-free (divergence theorem; $\|v\|_{\Gamma_h}^2 \leq (h/\mu)\|v\|^2$).
- (2) *Error equation.* The identity is dimension-free. The geometric term $\nu\mathcal{G}$ is bounded by Lemma 5.3 (the Navier–Stokes defect enters only through $\|F\|_\infty$ and its support), and the convective consistency term by $Ch_\Gamma^4\|\nabla v\|$, since $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$.
- (3) *(i).* Coercivity, continuity and the inverse traces are Lemmas 4.1, 4.2; the inf-sup condition is (IS3); the square count and the uniqueness of the constant use the face bubble (Lemma 4.10). The absorption has the same form, so γ'_2 depends on C_I, c_0, β, c_1 as in two dimensions.
- (4) *(ii).* The compactness argument uses: a strip of volume $|S_h| \leq Ch_\Gamma^2$ (Corollary 3.5); trace transfer by Φ_h ; discrete test fields $I_h\varphi - w_h$ with $w_h \in V_h^0$ from (IS3) and $|\langle \delta_j, \partial_n w_h \rangle| \leq (\rho/\mu)^{1/2}C_I\|\nabla w_h\|$; Rellich’s theorem and compact traces on the fixed Lipschitz domain Ω ;

density of smooth solenoidal fields in the divergence-free H_0^1 fields on Lipschitz domains in $\mathbb{R}^3$ [35]; and compactness of $L^2 \hookrightarrow V'$. All hold in three dimensions.

- (5) (iii). Steps of the GS proof: the approximation step uses Lemmas 5.3 and 5.7; the leak step uses $m(v) = \mathcal{R}(v) - \langle v_\varphi, v \rangle$, bounded only in the energy dual, $|m(v)| \leq C_p h_\Gamma (h/\mu)^{1/2} \|v\|$ (proof of Theorem 5.12), which suffices because (L) is stated in the energy dual; the load ℓ is bounded by Lemma 5.10 with $S = w$ and Lemma 5.9, and the convective term by $C(h/\mu) \|\nabla v\|$ since $\|v_\varphi\|_{H^1} \leq C$. This gives $\|r\| \leq C\beta_L^{-1}(h/\mu + h_\Gamma^2 + \varepsilon_h^{(3)}) \leq CX_3$. The leading-term statement follows as Corollary 5.14, with (1).
- (6) (iv). $|B^*(\eta, v)| \leq 2\|\eta\|_{\Gamma_h} (h/\mu)^{1/2} \|v\|$ with $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$; the fixed point is as for GS with β_* ; the H^1 statement uses the flux proviso of Theorem 5.16. $\square$

Remark 10.2. An explicit threshold h_0 in (ii) by a Schatz-type duality argument [31] is possible in two dimensions under Stokes H^2 -regularity [29]. In three dimensions the same argument needs the corresponding regularity on Ω , which for a box minus an obstacle involves the edges and corners of the box [7]; it also needs a divergence-free extension of the H^2 dual solution across Γ , for which the Piola–Hestenes construction of Lemma 3.12 applies with Sobolev coefficients. We have not written this out, and the edges and corners of the box limit the regularity (Remark 2.6). The Oseen leak limit and the drag statements are in Section 10.1; none of them uses this global regularity.

10.1. The Oseen leak limit and drag. Assume (NS1)–(NS2), (IS3), and that u is a nonsingular solution (Corollary 10.1(ii)); write (H_{ns}) for the last assumption. Let $K(w, v) := c(w; \tilde{u}, v) + c(\tilde{u}; w, v)$, so $\mathfrak{A} = \nu N_h + K$. Besides (L) and (L^*) we use two further discrete stability conditions:

- (L_D) there is $\beta_D > 0$, independent of h , with $\beta_D \|\nabla y\| \leq \sup_{0 \neq v \in Y_h} (\nu a_h(y, v) + K(y, v)) / \|\nabla v\|$ for all $y \in Y_h$ (the discrete Oseen–Dirichlet problem);
- $(L^\dagger)$ the condition (L^*) for the adjoint linearisation $(\delta, v) \mapsto \mathfrak{A}(v, \delta)$.

The compactness argument of Corollary 10.1(ii) gives (L), (L^*) and $(L^\dagger)$ for $\gamma \geq \gamma'_2$, and (L_D) , for $h \leq h_0$ with h_0 independent of μ, γ, ρ ; the two-dimensional proofs are in [29], and the dimension-dependent inputs are those listed in the proof of Corollary 10.1. Let (U_O, P_O) be the Oseen leak flow: $-\nu \Delta U_O + (U_O \cdot \nabla)u + (u \cdot \nabla)U_O + \nabla P_O = 0$, $\text{div } U_O = 0$ in Ω , $U_O|_\Gamma = -qn$ with $q = (p - \bar{p})|_\Gamma$ the Navier–Stokes wall pressure, and $U_O|_\Sigma = 0$. It exists by (H_{ns}) after lifting the data with Lemma 5.8, and it is $C^{2,\alpha}$ on a collar of Γ (local bootstrap with the Stokes Schauder estimate; no global regularity is used). $\tilde{U}_O$, E_O^0 and F_O are defined as $\tilde{U}$, E_U^0 and F_U in Section 7. Let $U_\infty := \|\tilde{u}\|_{W^{1,\infty}(\Omega_h)}$ and C_F the constant in $\|w\|_{L^2(\Omega_h)} \leq C_F \|\nabla w\|$ for w vanishing on Σ (uniform in h by Lemma 3.11).

Theorem 10.3 (Oseen and Navier–Stokes versions). *For $h \leq h_0$ and either convective form:*

- (i) Oseen leak limit, all $\gamma \geq \gamma_0$. With $\mathfrak{A}(\delta_R, v) = \mathcal{R}(v)$ on Z_h and $X := \frac{\nu\mu}{h} \delta_R$: $\|\nabla(X - \tilde{U}_O)\| \leq C(h_\Gamma^{1/2} + E_O^0)$.
- (ii) Lower bound (Alfeld, $k \geq 4$, every μ for which δ_R exists). $\|\nabla(X - \tilde{U}_O)\| \geq c_O G_h h_\Gamma^{1/2} - Ch_\Gamma^{3/2}$ with $c_O := \nu \Lambda_0 c_L / (\nu \Lambda_0 + C_F(1 + C_F)U_\infty)$, G_h and c_L as in Theorem 7.13.
- (iii) GS, uniformly in γ . If $X_u := h/\mu + h_\Gamma^{3/2} + \varepsilon_1 \leq c_*$, then GS has a locally unique solution with $\|\nabla(\tilde{u} - u_h)\| \leq CX_u$ and

$$\left\| \nabla \left(\frac{\nu\mu}{h} (\tilde{u} - u_h) - \tilde{U}_O \right) \right\| \leq C(h_\Gamma^{1/2} + E_O^0) + C \frac{\mu}{h} (h_\Gamma^{3/2} + \varepsilon_1 + X_u^2).$$

So the Navier–Stokes H^1 constant $K_{NS} = \|\nabla U_O\|/G$ is attained whenever $\gamma h_\Gamma^{1/2} \rightarrow 0$ and $\frac{\mu}{h} \varepsilon_1 \rightarrow 0$.

- (iv) GS drag. $J_h^N(\psi) = J_\psi - \frac{h}{\nu\mu} K_\psi^\dagger - \Theta_f(\psi) + O(\frac{h}{\mu}(h_\Gamma^{1/2} + E_O^0) + h_\Gamma^{3/2} + \varepsilon_1 + X_u^2 + h^k h_\Gamma^{-1/2})$, where $K_\psi^\dagger = \int_\Gamma q(R_\psi^\dagger - \bar{R}_\psi^\dagger)$ and $(Z_\psi^\dagger, R_\psi^\dagger)$ is the adjoint Oseen flow with data ψ on Γ and 0 on Σ .
- (v) CNS* drag. Theorem 7.27 holds with ν and $Z_\psi^\dagger$ in place of Z_ψ (the term $\Lambda_h^\dagger$), under $\gamma \geq \max(1, \gamma_2')$, $\gamma h_\Gamma \leq 1$ and $Y \leq c_*$; the rate 2 holds modulo the analogue $(E_Z^\dagger)$ of (E_Z) .

Proof. (i) is Theorem 7.10 with a_h replaced by $\nu a_h + K$: the Oseen–Dirichlet field X^D uses (L_D) , the energy step uses (L) , and $|K(w, v)| \leq C_K \|w\| \|\nabla v\| + \frac{1}{4} U_\infty h_\Gamma^2 \|w\|_{\Gamma_h} \|v\|_{\Gamma_h}$ (the convective boundary identity; dimension-free apart from C_F , via $H^1 \subset L^4$ and Φ_h). The lift, projection and cancellation lemmas are Lemmas 7.2–7.4. (ii) Lemma 7.11 becomes $\nu a_h(W, v) + K(W, v) - \nu \langle W, \partial_n v \rangle = \nu \langle \tilde{U}_O, \partial_n v \rangle - (F_O, v)$ on Y_h , with $|K(W, v)| \leq C_F(1 + C_F) U_\infty \|\nabla W\| \|\nabla v\|$ (W divergence-free and zero on Σ); then the proof of Theorem 7.13, since Lemma 7.5 holds for U_O . (iii) Lemma 7.15 with (L_D) in place of Korn’s inequality, plus the contraction of Corollary 10.1(iii). (iv) Lemma 7.22 with ω_h including $c(u_h; u_h, w)$ (dimension-free; $|\tilde{u}| \leq Ch_\Gamma^2$ on $\Omega_h \Delta \Omega$), and the reciprocity of Lemma 7.23 with $((u \cdot \nabla)U, Z) = -((u \cdot \nabla)Z, U)$, again in weak form. (v) The proof of Theorem 7.27 with $(L^\dagger)$ for the auxiliary problem; the dual pair is smooth near Γ by local regularity, and Lemma 7.25 acquires the terms $K(e_u, w_\phi)$ and $c(e_u; e_u, w_\phi)$, bounded as in two dimensions [29]. $\square$

The quantitative threshold h_0 still needs Stokes H^2 -regularity on Ω (Remark 10.2), which is not used anywhere in Theorem 10.3.

11. COMPUTATIONS

All results of this section are numerical. They are a first three-dimensional implementation, run on a shared machine with 7 GB of memory and two cores; the finer levels needed for asymptotic rates are listed in Appendix A with the exact commands.

11.1. Solver and meshes. The solver (`code3d/svn3d.py`) uses continuous P_k velocities, $k = 3, 4$, and discontinuous full- P_{k-1} pressures on every sub-tetrahedron of an Alfeld refinement. The forms are those of Section 2, with h the largest tetrahedron diameter and n pointing into the obstacle on Γ_h ; GS omits and CNS* includes the mean-free traction. The pressure is eliminated by the iterated-penalty method (penalty 10^4 ; the velocity matrix is factorised once with MKL PARDISO, and the CNS* traction enters only the right-hand side), stopped when $\|\operatorname{div} u_h\| < 10^{-13}$ and the momentum residual is below 10^{-11} , or at round-off stagnation. A nonsymmetric LU variant gives the same u_h to 10^{-12} .

Sphere meshes. An equiangular $N \times N$ grid on each face of the box $R = (-2, 2)^3$ is projected radially onto the unit sphere; the projected points are the vertices of a convex inscribed polyhedron P_h with $12N^2$ faces. Radial layers at $r_j = |b|^{j/m}$, $m = N/2$, between ∂P_h and ∂R give prisms, which are split into three tetrahedra by a minimum-index rule and then Alfeld-refined. Every boundary face is a whole face of P_h or lies on ∂R . Shape regularity is uniform ($\min \text{vol} / \text{diam}^3 = 0.0074\text{--}0.0079$ for $N = 2, \dots, 24$), but $h/h_\Gamma = 3.26, 4.14, 4.79, 5.24$ for $N = 2, 4, 6, 8$: the family is not self-similar at small N , which contaminates rates measured in h_Γ .

Test problems. (box) the cube $(-1, 1)^3$ with a smooth manufactured solution and fully consistent Nitsche conditions on all faces; (P) $u = 0$, $f = \nabla p$ with the wall pressure of the next test, so that $\tilde{u} = 0$, $\mathcal{G} \equiv 0$ and the GS output is exactly the traction response; (sph) Stokes flow past the fixed unit sphere with uniform stream $U = (1, 0.4, -0.3)$, with nonconstant wall pressure; (rot) $u = (1 - r^{-3})\omega \times x$, $p = 0$, so that $u = 0$ and p is constant on Γ ($G = 0$).

11.2. Verification. Patch test. A P_k solution is reproduced to $10^{-9}\text{--}10^{-11}$ in H^1 for $k = 3, 4$, with Nitsche and with strong conditions. *Orientation.* On the $N = 2$ sphere mesh the computed normal points into P_h on Γ_h and outward on the box (checked independently).

k	N	vel. dofs	h	H^1 error	rate	L^2 error	rate	$\ \operatorname{div} u_h\ $
3	2	3 189	1.732	$3.015 \cdot 10^{-1}$	—	$2.439 \cdot 10^{-2}$	—	$1.3 \cdot 10^{-13}$
3	3	10 290	1.155	$9.780 \cdot 10^{-2}$	2.78	$5.265 \cdot 10^{-3}$	3.78	$1.7 \cdot 10^{-13}$
3	4	23 871	0.866	$4.298 \cdot 10^{-2}$	2.86	$1.677 \cdot 10^{-3}$	3.98	$2.3 \cdot 10^{-13}$
3	5	46 038	0.693	$2.252 \cdot 10^{-2}$	2.90	$6.812 \cdot 10^{-4}$	4.04	$3.0 \cdot 10^{-13}$
3	6	78 897	0.577	$1.322 \cdot 10^{-2}$	2.92	$3.250 \cdot 10^{-4}$	4.06	$3.9 \cdot 10^{-13}$
4	2	7 083	1.732	$3.820 \cdot 10^{-2}$	—	$3.266 \cdot 10^{-3}$	—	$6.7 \cdot 10^{-13}$
4	3	23 115	1.155	$8.207 \cdot 10^{-3}$	3.79	$4.653 \cdot 10^{-4}$	4.81	$1.0 \cdot 10^{-12}$
4	4	53 907	0.866	$2.649 \cdot 10^{-3}$	3.93	$1.138 \cdot 10^{-4}$	4.90	$1.5 \cdot 10^{-12}$

Table 1. Box test, $\mu = 400$. The rates approach k in H^1 and $k+1$ in L^2 , and u_h is divergence-free to round-off.

Stability. For the pair $(V_h^0, \Pi_h \cap L_0^2)$ with the vector Laplacian, a dense eigenvalue computation finds exactly one zero eigenvalue (the constants) and no spurious modes in every case; the discrete inf-sup constant is 0.142, 0.149, 0.150 on the box for $N = 1, 2, 3$, and 0.069 on the sphere mesh $N = 2$. Uniformity in h is therefore seen only on the box.

Coercivity. The threshold μ_Z^* of N_h on Z_h (from the inertia of the penalised matrix, independent of the penalty 10^5 – 10^7 , validated against a dense eigenvalue computation) is 76.3 ($N = 2$) and 148.0 ($N = 4$) for $k = 3$, and 193.6 ($N = 2$) for $k = 4$. It grows with h/h_Γ , as $\mu_0 \propto \rho$ predicts (Section 8). A local sufficient threshold on V_h , computed cell by cell, is 216, 341, 421, 476 for $N = 2, 4, 6, 8$ ($k = 3$). All sphere runs below use $\mu \geq 200 > \mu_Z^*$.

11.3. The leak law.

μ	N	$\ \nabla(\tilde{u} - u_h)\ $	$\frac{\ \nabla(\tilde{u} - u_h)\ }{(h/\mu)G}$	slip	leak	energy	$\ \operatorname{div} u_h\ $
200	2	$1.34 \cdot 10^{-1}$	2.34	0.913	0.891	1.001	$1.0 \cdot 10^{-15}$
200	4	$9.21 \cdot 10^{-2}$	2.41	0.949	0.943	1.012	$1.4 \cdot 10^{-15}$
400	2	$7.26 \cdot 10^{-2}$	2.54	0.941	0.923	0.992	$8.7 \cdot 10^{-16}$
400	4	$4.78 \cdot 10^{-2}$	2.50	0.970	0.965	1.002	$7.5 \cdot 10^{-16}$
1000	2	$3.30 \cdot 10^{-2}$	2.90	0.963	0.949	0.991	$2.3 \cdot 10^{-16}$
1000	4	$2.03 \cdot 10^{-2}$	2.66	0.984	0.981	1.001	$3.0 \cdot 10^{-16}$

Table 2. Test P, GS, $k = 3$. slip := $\|\tilde{u} - u_h\|_{\Gamma_h} / ((h/\mu)G)$, leak := $\langle u_h \cdot n, p - \bar{p} \rangle_{\Gamma_h} / ((h/\mu)G^2)$, energy := $\|\tilde{u} - u_h\|(\mu/h)^{1/2}/G$, with G computed on Γ_h . $N = 2$ and $N = 4$ have 9 096 and 70 026 velocity unknowns.

The GS error is of order h/μ (the H^1 ratio is roughly constant, near 2.5), and the slip, leak and energy constants approach 1, with $1 - \text{slip}$ decreasing with h/μ : this is Corollary 5.14 in three dimensions. The H^1 ratio is not universal: in test P, $\tilde{u} = 0$ and $u_h = -\delta_R$, so by Theorem 7.10 it converges, for every $\gamma \geq \gamma_0$, to $K = \|\nabla U\|/G$ with U the Stokes flow driven by the leak (Corollary 7.16), a domain-dependent constant; we have not computed K for this domain. CNS* has no leak: at $\mu = 400$ its error on test P is $2.7 \cdot 10^{-3}$ ($N = 2$) and $2.6 \cdot 10^{-4}$ ($N = 4$), at the level of the pressure projection. At $\mu = 200$ the $N = 4$ CNS* solve returns an error of order 1: $\mu = 200$ is below the well-posedness threshold γ_2 of Theorem 5.16 on that mesh, although it exceeds μ_Z^* . We report CNS* only for $\mu \geq 400$.

11.4. Sphere tests.

test	N	h_Γ	GS H^1	CNS* H^1	interp. H^1	GS $\ e\ _{\Gamma_h}$	D : H^1	D : slip / leak
sph	2	0.919	1.269	1.265	1.633	0.445	0.138	1.254 / 0.978
sph	4	0.524	0.338	0.325	0.192	0.110	0.064	0.978 / 0.925
rot	2	0.919	2.209	2.225	2.390	0.867	—	—
rot	4	0.524	0.541	0.549	0.251	0.223	—	—

Table 3. Tests sph and rot, $k = 3$, $\mu = 400$. “interp.” is the H^1 error of the nodal interpolant of $\tilde{u}$; $D := u_h^{\text{GS}} - u_h^{\text{CNS}^*}$, with slip and leak ratios defined as in Table 2. $\|\text{div } u_h\|$ is 10^{-13} – 10^{-10} in all runs.

11.5. Interpretation. Confirmed at these levels: pointwise incompressibility; the optimal rates h^k and h^{k+1} on the box for $k = 3, 4$; a stable pair without spurious pressure modes; the growth of the coercivity threshold with h/h_Γ ; the leak law of GS with constant tending to 1 in slip and energy (test P at three penalties, and D in test sph, with slip 0.98 and leak 0.93 at $N = 4$); and the removal of the leak by CNS* (the leak correlation of CNS* in test sph is -0.38 at $N = 2$ and -0.06 at $N = 4$, against 0.60 and 0.87 for GS).

Not confirmed: the rates. Only two sphere levels fit in memory, and they are pre-asymptotic. The observed h_Γ -rates of 2.4–2.5 for both methods in tests sph and rot are dominated by the P_3 approximation error (the interpolation error drops from 1.63 to 0.19, and at $N = 2$ it is larger than the finite element error) and by the $O(h_\Gamma^2)$ trace mismatch of $\tilde{u}$ on Γ_h . At $N = 4$ the finite element error is already 1.7–2.2 times the interpolation error, so the geometric $h_\Gamma^{3/2}$ part is taking over; measuring its rate needs $N \geq 8$. In test sph the GS leak (D , 0.064 in H^1 at $N = 4$) is still small compared with the geometric error (0.33), so the rate 1 of GS and the rate 3/2 of CNS* in total H^1 error will separate only at much finer levels. The drift of h/h_Γ also affects rates in h_Γ . We therefore make no claim about asymptotic rates on the sphere from these computations; the commands for $N = 6, \dots, 12$ (up to 1.85 million unknowns) are in Appendix A.

12. WHAT IS NOT PROVED

- (1) *Unsplit meshes.* (IS3) on general unsplit tetrahedral meshes is, to our knowledge, open for the Scott–Vogelius pair in three dimensions (Remark 4.8). All results that need existence of u_h are therefore unconditional only on Alfeld refinements; on unsplit meshes they hold under (IS3). The lower bound of Section 6 for $4 \leq k \leq 8$ on unsplit meshes is not proved (Proposition 6.11 needs $k \geq 9$).
- (2) *Worsey–Farin splits.* (IS3) as formulated, with the full space of discontinuous P_{k-1} pressures, is false when the wall faces are subdivided (Proposition 4.9); this is known [17, 10]. So none of our upper bounds, and not the leak analysis of Section 7, is proved on Worsey–Farin meshes. That would need: a pressure space with the singular-edge conditions of [10]; a uniform-in- h inf-sup constant for that pair on the perturbed domains Ω_h , as Lemma 4.6 provides for Alfeld; and a lift lemma onto the constrained trace space. The lower bound holds for every solution that exists, for $k \geq 4$ on barycentric Worsey–Farin splits (Proposition 6.30) and for $k \geq 9$ in general. For $k = 3$ there is no single-macro witness, and the trace analysis of Section 6.5 has not been redone for split faces; the $k = 3$ edge-star ranks are only numerical (Remark 6.29).
- (3) *Explicit constants.* On Alfeld refinements with $k \geq 4$ the witness constants are explicit (Section 9). Not explicit: the constant κ_0 of Theorem 6.26 (degree 3, general surfaces), which comes from compactness (the sphere criterion of Theorem 6.20 is explicit up to $C_E(\theta)$); the constants κ_0, Y_0 of Theorem 8.3; and the (IS3) constant, because C_Π and $C_B(\Omega)$ are not explicit and $\hat{C}_{\text{loc}}$ is only computed in floating point. The single-cell penalty bound $\lambda_T \geq 20$ is certified on one explicit box of shapes (with the floating-point caveat of Section 9), not over larger classes.
- (4) *Penalty necessity in general.* Necessity of $\mu \gtrsim \rho$ without local quasi-uniformity (LQ) or a certified cell at a smallest face is open, as in two dimensions.
- (5) *Leak and drag.* The leak lower bound of Theorem 7.13 is proved only on Alfeld refinements with $k \geq 4$; for $k = 3$ there is no single-macro leak witness, and multi-macro witnesses would be needed. For general data at a fixed penalty the rate- $\frac{1}{2}$ lower bound is open (Remark 7.17). The rate 2 of the CNS* drag (Theorem 7.27) is proved modulo (E_Z) , a regularity statement

- for the dual Stokes flow at Σ , and in the Oseen case modulo $(E_Z^\dagger)$; its sharpness is not proved. The Gjerde–Scott volume functional and pointwise tractions are not treated.
- (6) *Regularity at the outer boundary.* The smoothness of (u, p) on $\bar{\Omega}$ is a hypothesis on the solution: the edges and corners of the box limit generic regularity (Remark 2.6). Stokes H^2 -regularity on Ω , needed for an explicit threshold h_0 in Corollary 10.1(ii) (Remark 10.2), is reduced to a convex-polyhedron theorem attributed to [7] whose statement we have not checked; (E_Z) rests on the same kind of input.
- (7) *Citations checked indirectly.* The wording of [18, Theorem 3.1] was read through a text-extraction tool; [25, Proposition 6.5] only through [11]; the pressure spaces of [17] only through its abstract and [10]; the Bogovskii regularity used in Lemma 5.8 and the boundary Schauder theory for the leak flow in Section 7 are cited, not re-proved.
- (8) *Numerics.* The sphere computations are pre-asymptotic; the $h_\Gamma^{3/2}$ rates of Theorems 5.16 and 6.27, the rate $h_\Gamma^{1/2}$ of the leak limit, and the separation of the GS and CNS* rates have not been observed in three dimensions (Section 11).

APPENDIX A. REPRODUCIBILITY

All code, logs and raw records are in the repository <https://github.com/jaideepsaipadhi/zero-gradient-prize>. Paths are relative to its root.

Exact and computer-assisted certificates.

Result	Command	Output
Theorem 6.10 (rank 8, $k = 4$)	python3notes/v6/td_sharp/alfeld_witness.py4	alfeld_k4.log
$\ \hat{R}\ = 91.7$	python3notes/v6/td_sharp/k4_constant.py	k4_constant.log
Proposition 6.11	python3notes/v6/td_sharp/bubble_variance.py	bubble_variance.log
Proposition 6.12	python3notes/v6/td_sharp/alfeld_witness.py3	alfeld_k3.log
Lemmas 6.14, 6.18	notes/v6/td_k3/trace1.py, ref_K.py; notes/v6/referee8/ref8_lemmaK.py	logs alongside
Theorem 6.16 (reduced model)	notes/v6/td_k3/validate_reduced.py	validate_reduced.log
Theorem 6.19	notes/v6/td_k3/vertex_formula_general.py, fe_vertex_check.py; notes/v6/referee8/ref8_vertex_fe.py	logs alongside
Icosphere criterion	notes/v6/td_k3/icosphere_check.py	icosphere_check.log
Lemmas 6.21, 6.22	notes/v6/td_k3b/ref_nu.py, face_formula.py, flat_algebra.py	logs alongside
Theorem 6.25 checks	notes/v6/td_k3b/degen.py, degen_random.py, degen_gflat.py, fe_cross.py	logs alongside
Theorem 8.2	python3notes/v6/td_pen/alfeld_tet.py3[shape], verify_alfeld.py, neg_cert.py	alfeld_k3.log, neg_cert.log
Half-space constant	python3notes/v6/td_pen/continuum3d.py	continuum3d.log

Added in this version; all in notes/v7/td/, each run at most 3 minutes on one core

Lemma 7.12; $\ \hat{R}\ = 91.678155469$, $\ \hat{R}_1\ $ (Proposition 9.1) $k = 3$ skew leak image	python3notes/v7/td/kappa_cert.py4 (and 3)	kappa_k4.log, kappa_k3.log
Table C1, Λ_0 , c_L , sharpness constant Remark 7.17 (joint ranks 6, 9)	python3notes/v7/td/leak_k3_image. py python3notes/v7/td/kappa_explicit. py python3notes/v7/td/leak_joint. pyalfeld4; wf 4	leak_k3_image.log kappa_explicit.log leak_joint.log
Proposition 9.2	python3notes/v7/td/pen_cert.py200. 1 (and 20 0.05)	pen_cert_d010.log, pen_cert_d005.log
Table C2	python3notes/v7/td/is3_ico.pyklm	is3_k3_l0.log, is3_more.log
Propositions 4.9, 6.30	python3notes/v7/td/wf_witness. py2 3 4	wf_ranks.log

The exact toolkit for the `notes/v7/td` scripts is `notes/v7/td/refalfeld.py` (Bernstein–Bézier; it loops over the sub-tetrahedra of the split, so it serves Alfeld and Worsey–Farin, with Alfeld results bit-identical to the earlier version) and `notes/v7/td/ratlin.py` (exact linear algebra); `pen_cert.py` needs `python-flint` (Arb). The three scripts `kappa_explicit.py`, `kappa_cert.py` 4 and `wf_witness.py` 4 were rerun independently with identical output (`notes/v7/referee_td_ns/*.rerun.log`). The log `pen_cert_d015.log` is an interrupted earlier run and `pen_cert_l10_d015.log` an uncertified one (Section 9). Each script in the table runs in under five minutes on two cores, except the height scan and the $k = 4$ single-cell pencils (`alfeld_tet.py` 4), estimated at 10–30 minutes per shape and not run.

Three-dimensional solver (code3d/).

- `svn3d.py`: the solver (Section 11); `exact3d.py`: the exact solutions; `run3d.py`: drivers (box, sphere, interpolation); `table3d.py`: tables; `penalty3d.py`: the threshold μ_Z^* (PARDISO inertia); `infsup3d.py`: the discrete inf-sup constant (dense, coarse meshes only); `patch.test.py`.
- `local_*.sh`: exactly the runs of Section 11. Records: `notes/v6/num3d/results.jsonl`; tables: `notes/v6/num3d/tables_local.txt`.
- Requirements: Python 3 with numpy, scipy, sympy and pypardiso (MKL; PYPARDISO_MKL_RT points to `libmkl_rt.so.3`).

Finer levels. The levels that did not fit our machine are scripted in `code3d/run_pod3d.sh`. From the repository root:

```
sh code3d/run_pod3d.sh          # all stages, sequentially
sh code3d/run_pod3d.sh STAGE    # one of: box4 pen k3a k3b k3c pen4 k4
python3 code3d/table3d.py        # tables from results.jsonl
```

Memory was measured locally for $k = 3$ (PARDISO LDL^T of the velocity matrix, peak resident size): sphere $N = 4$ (70k velocity unknowns) 1.0 GB; box $N = 6$ (79k) 1.15 GB. Extrapolated with fill $\propto n^{4/3}$: sphere $N = 6$ (233k) 5–6 GB, $N = 8$ (550k) ≈ 17 GB, $N = 10$ (1.07M) ≈ 40 GB, $N = 12$ (1.85M) ≈ 80 GB; $k = 4$ multiplies the unknowns by about 2.4.

stage	content
box4	$k = 4$ box verification, $N = 2, \dots, 5$
pen	μ_Z^* on the sphere meshes $N = 6, 8$ (run first: confirms $\mu = 400 > \mu_Z^*$)
k3a	tests P, sph, rot; $k = 3$, $\mu = 400$, $N = 6, 8$
k3b	the same at $N = 10, 12$
k3c	test P at $\mu = 1000$, $N = 2, \dots, 8$ (penalty independence of the leak constant)
pen4, k4	μ_Z^* for $k = 4$, then tests P, sph, rot with $k = 4$, $\mu = 1000$, $N = 2, 4, 6$

USE OF GENERATIVE-AI TOOLS DECLARATION

The author used Anthropic’s Claude Opus 5.5 as a research-support tool. It assisted with the following:

- developing and checking the proofs;
- implementing, running and validating the three-dimensional finite element code and the exact-arithmetic certificates;
- independently reviewing the arguments as a referee;
- preparing the manuscript and the accompanying repository.

The author directed the project and retains authorship of the work. Responsibility for all mathematical statements, computations, references and conclusions rests with the author.

DATA AND CODE AVAILABILITY

All code, exact-arithmetic certificates, logs and raw records are available at <https://github.com/jaideepsaipadhi/zero-gradient-prize> (directories `code3d/` and `notes/v6/`); Appendix A lists the command behind each number. The two-dimensional companion paper [29] and its code are in the same repository.

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DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY, WEST LAFAYETTE, IN 47907, USA
 Email address: jpadhi@purdue.edu